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SECTOR IN-DEPTH
26 February 2024

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Default Trends – Global

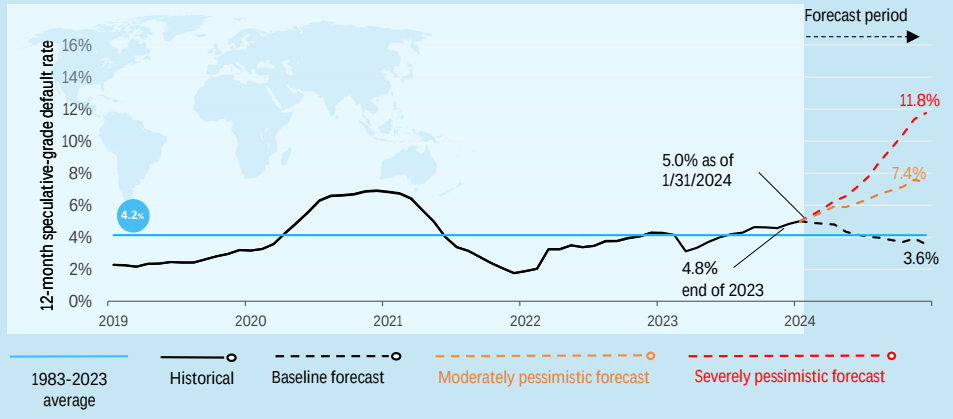
Annual default study: Corporate default rate to moderate in 2024 but remain near its long-term average

Summary

The global trailing 12-month default rate for speculative-grade companies rose to 4.8% in 2023 from 4.3% in 2022. We believe it peaked at 5% in January this year and will likely decline to 3.6% by end 2024, staying above or close to its long-term average of 4.2% for much of the year. This projection is underpinned by our expectation that major central banks will begin lowering policy rates this year and global economic expansion will continue, albeit at a slower pace. The maturity wall in 2024 is manageable. While borrowing conditions are still tight for speculative-grade companies, private/direct lending has provided an alternative source of funding for some. Geopolitics remains a risk, but our baseline does not anticipate an escalation that would materially raise the default rate. In downside scenarios entailing a departure from these baseline assumptions on rates, growth or geopolitics, the default rate could rise further, as shown below.

Exhibit 1

The global speculative-grade default rate will ease in 2024 but remain close to its long-term average



Source: Moody's Investors Service

We republished this report on 21 March 2024 to add "Annual default study" to its title and on 27 March 2024 to correct data in Exhibit 34.

Key highlights of our analysis include:

» **Interest rate hikes, tighter financing conditions and lingering inflation drove up corporate defaults in 2023.** A total of 159 corporate issuers that we rate defaulted in 2023, just surpassing the count of 157 in 2022. Excluding Russian defaults, which accounted for 65 in the 2022 default tally and zero in 2023, the rise would have been substantial at 73%. Last year's default count did not reach the 218 tally observed in 2020, with many of them spurred by the difficult economic conditions brought on by the COVID-19 pandemic.

» **Default trends varied across regions.** Roughly two-thirds of the 2023 defaults were from North America. Within this region, US defaults almost tripled to 105 in 2023 from 36 in 2022. By contrast, the default count fell substantially in Europe to 31 from 89, driven by the decline in Russian defaults. Excluding Russian defaults, the European default count would have risen from 24 to 31. Outside of these two regions, the default tally fell to 12 from 24 in Asia but jumped to nine from four in Latin America.

» **Distressed exchanges (DEs) remained a prevalent approach in debt restructuring.** Roughly 60% of last year's defaults were in the form of distressed exchanges. The share of DEs has been consistently high in recent years as they help preserve shareholders' equity interest and avoid the expensive cost associated with a bankruptcy.

» **Loan issuers were harder hit by interest rate hikes.** The sharp increase in policy rates has put particular pressure on loan issuers whose debt payments typically move up and down with prevailing interest rates. Among last year's 159 defaults, 84 affected loans only, 49 involved bonds only while the remaining 26 resulted in defaults in both.

» **The global speculative-grade default rate will decline gradually in 2024 but remain near its historical average under our baseline scenario.** The rate rose to 4.8% at the end of 2023 from 4.3% a year earlier. We expect it to fall modestly to 3.6% in 2024. In contrast to the 2008-09 global financial crisis and the 2020 pandemic — two recent periods when the default rate rose sharply and then fell quickly — we expect the default rate to stay near its historical average for longer this time against a backdrop of moderating global economic growth in 2024, a gradual pace in interest rate normalization and re-default risk following DEs.

» **We have also developed more pessimistic default scenarios and a more optimistic scenario, given the high degree of uncertainty surrounding the global economy and geopolitical developments.** The global default rate will exceed our baseline expectations if economic and financial conditions are worse than our baseline assumptions. Under our moderately pessimistic scenario, the rate would rise to 7.4% in 2024, higher than the pandemic peak of 6.9%. Under our severely pessimistic scenario, the rate would jump to 11.8%, below the 13.4% peak during the global financial crisis.

» **Default risk will be highest in durable consumer goods for the second year in a row.** Under our baseline forecast, we project there will be 100 corporate defaults across sectors in 2024, or roughly eight per month. We expect durable consumer goods to be the most troubled sector measured by default rate (at 7.1%), while business services will be the riskiest sector measured by default count, with eight.

» **Credit quality weakened for the second consecutive year.** Rating drift, which is calculated by the average upgraded notches per issuer minus that of downgrades, came in at -6.3% of a rating notch for all of Moody's-rated corporate issuers in 2023. The comparable drift was -8.6% in 2022. Construction & building remained the most troubled sector when measured by rating drift. China property developers accounted for most of the downgrades in this sector, reflecting a persistently difficult operating and financing environment.

» **Trading price implied bank loan recovery rate fell noticeably.** The issuer-weighted average recovery rate for first-lien bank loans declined to 59.6% in 2023 from 69.5% in 2022 and the long-term average of 65.0% since 1983. By contrast, the average senior unsecured bond recovery rate increased to 33.4% from 30.9%, both below the long-term average of 37.6% since 1983.

» **Average Default Position (AP) shows that our ratings have been successfully rank ordering credit risk.** Since 1983, the average one-year AP, a measure of rating accuracy, has been high at 91.2%. The metric came in at 91.7% last year, slightly above the long-term average.

For inquiries on the underlying data, please contact ClientServices@moodys.com.

About this report

This report is our 37th annual default study, in which we provide data on defaults, losses and rating transitions for corporate bond, loan and deposit issuers over the past year and for the historical period since 1920. This study covers financial institutions, nonfinancial companies and regulated utilities that have long-term debt ratings. Unless noted otherwise, ratings in this report are senior unsecured ratings (actual or implied), which do not necessarily refer to the rating of a particular debt issue.

The study is based on a proprietary database of ratings and defaults for industrial and transportation companies, utilities and financial institutions. This database covers the credit experiences of almost 30,000 companies that had long-term rated bonds, loans and/or deposits between 1920 and 2023.

Defaults edged up globally with different trends across regions

In 2023, central banks in most major economies maintained hawkish policy stances to fight inflation, increasing borrowing costs for most speculative-grade companies, particularly leveraged loan issuers. Higher funding costs, together with tighter financing conditions following the first quarter's banking stress and the effects of lingering inflation, resulted in slower earnings growth, declining interest coverage and a rise in corporate defaults.

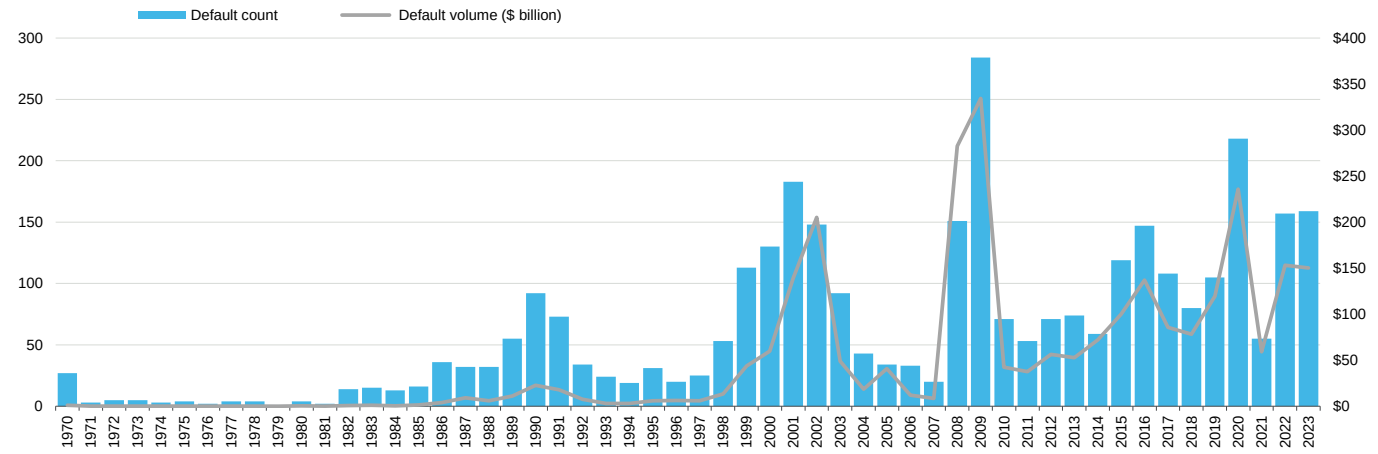
Default count tripled in the US

A total of 159 rated corporate issuers defaulted in 2023, surpassing the year-earlier tally of 157. Approximately 40% of the 2022 defaults were from Russia, primarily stemming from the Russia-Ukraine war and the sanctions against the country. If we exclude such defaults — which are rare in the history of global defaults — the rise in defaults would have been substantial at 73%.[1] Last year's default count was the second highest since the 2008 global financial crisis (GFC). The highest post-GFC level was in 2020 when 218 issuers defaulted amid the difficult economic conditions resulting from COVID-19.

Measured by volume, issuers defaulted on roughly $150 billion of debt in 2023, comprising $77 billion in bonds and $74 billion in loans (see Exhibit 2). Default volume was lower than 2022's $153 billion and 2020's $236 billion. Of the 159 defaults in 2023, one-quarter of them affected at least $1 billion of debt each. The largest default was by Diamond Sports Group, LLC. The US regional sports broadcasting company is the entity through which Sinclair Broadcast Group Inc. executed the acquisition of 21 regional sports networks from Walt Disney Company (A2 stable) in 2019. Diamond Sports entered Chapter 11 in March aiming to renegotiate its broadcast contracts with teams and to restructure its more than $8 billion of total debt stemming from Sinclair's 2019 acquisition.

Exhibit 2

The default count inched up in 2023 from 2022's elevated level driven by Russian defaults



Source: Moody's Investors Service

Default trends varied across regions. Roughly two-thirds of the 2023 defaults were by companies in North America. Within this region, US company defaults almost tripled to 105 in 2023 from 36 in 2022. By contrast, the default count fell substantially in Europe to 31 from 89, resulting from the decline in Russian defaults. Excluding Russian defaults, the European default count would have risen from 24 to 31. Outside of these two regions, the default tally fell to 12 from 24 in Asia but jumped to nine from four in Latin America. In terms of dollar volume, North American companies defaulted on about $101 billion debt, the most across regions. Companies in Asia-Pacific and Europe followed with about $20 billion each.

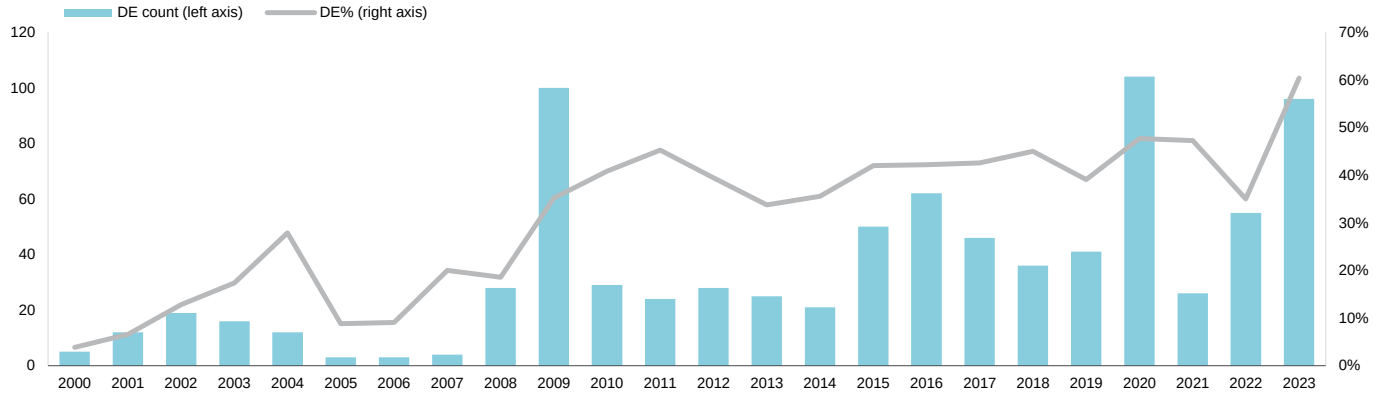
Distressed exchanges (DEs) remained a prevalent approach in debt restructuring.

Measured by initial default type, 96 (or 60%) of last year's defaults were in the form of distressed exchanges, up from 35% in 2022. DEs' contribution would have been higher at 58% in 2022 if we exclude Russian defaults. DEs accounted for a relatively small share in default types before the global financial crisis (see Exhibit 3). Since 2009, however, the share of DEs grew significantly and reached a record high of 60% in 2023. DEs have become more common in debt restructurings because they help preserve shareholders' equity interest. In addition, they save time and avoid expensive costs that will occur in a bankruptcy. Private equity-owned companies are more likely to execute DEs even before the troubled issuers run out of liquidity or are due to refinance, which in part explained the extensive increase in the share of DEs in 2023. We note that initial DEs may not reduce debt enough to repair the capital structure to more sustainable levels and re-default risk is higher post DEs than after bankruptcies.

Of the 63 non-DE defaults in 2023, missed debt payments triggered 35 (or 22%) of last year's defaults. That left 28 cases (or 18%) where bankruptcy was listed as the initial default type.

Exhibit 3

Distressed exchanges remained elevated in 2023



Source: Moody's Investors Service

Loan issuers were harder hit from rate hikes

The sharp increase in policy rates that began two years ago has put particular pressure on loan issuers whose debt payments typically move up and down with prevailing interest rates. In comparison, bonds usually bear fixed rates and will likely not feel the pain of rate hikes until they need refinancing. This led to more defaults in loans than bonds in 2023 as rate hikes materialized. Specifically, 84 of last year's defaults affected loans only, 49 involved bonds only and the remaining 26 resulted in defaults in both. In addition to rate hikes, a few other factors also contributed to the surge in loan defaults last year. Leveraged loans grew rapidly during the easy money era, becoming a mainstay for financially weaker borrowers and debt-financed buyouts and bear higher default risk. A good number of leveraged loan issuers are owned by private equity firms which are more apt to execute distressed exchanges, as mentioned above. Furthermore, banks have tightened lending standards, making it more difficult for weak loan issuers to refinance.

Speculative-grade default rate rose for the second year in a row

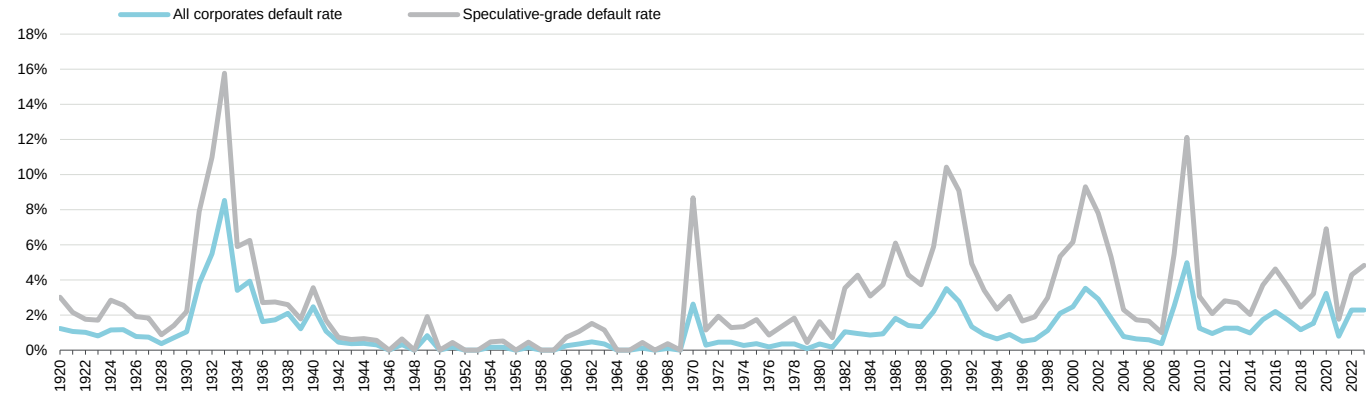
The issuer-weighted annual default rate for Moody's-rated speculative-grade corporate issuers increased to 4.8% at the end of 2023 from 4.3% in 2022 and 1.8% in 2021, but held below the 6.9% level in 2020 when the pandemic broke out (see Exhibit 4). Including investment-grade issuers, the annual default rate for all Moody's-rated corporate issuers held steady at 2.3% from 2022 to 2023. The

2023 year-end figures are higher than the 4.2% average one-year default rate since 1983 for speculative-grade issuers and the 1.7% average for all rated corporate issuers.

Measured on a dollar-volume basis, the speculative-grade bond default rate jumped to 2.1% in 2023 from 1.6% in 2022 and the comparable rate for all rated issuers held steady at 0.4%.[2] In comparison, the dollar-weighted default rates in 2020 were 6.3% for speculative-grade issuers and 1.0% for all rated issuers.

Exhibit 4

The default rate rose for speculative-grade issuers but stabilized for all issuers*



* Issuer-weighted.

Source: Moody's Investors Service

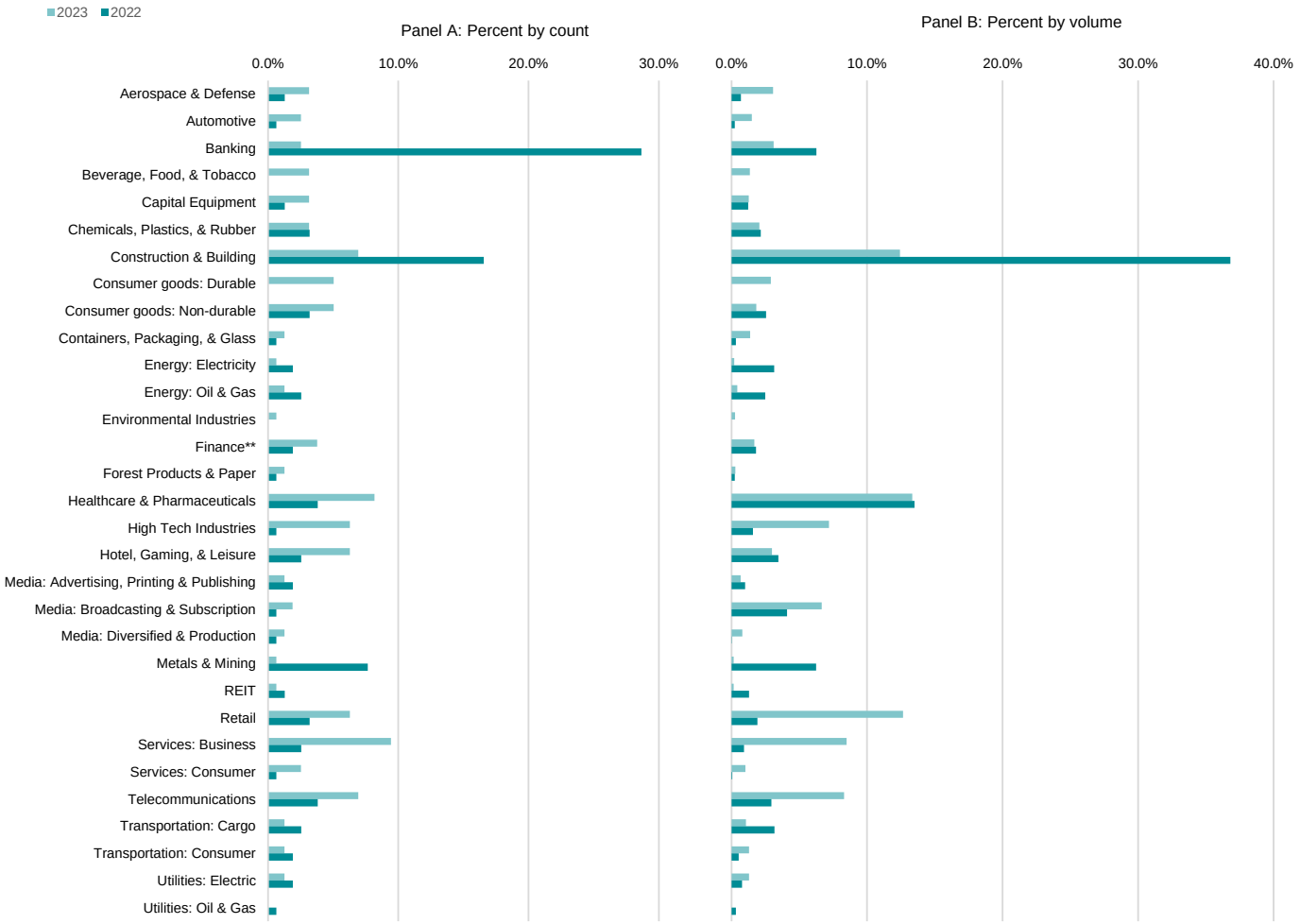
Business services had the most defaults by sector in 2023

The sector had 15 defaults, 13 of which were DEs. All of the defaulters in the sector were highly levered and most also suffered from weak liquidity. The healthcare & pharmaceuticals sector was the second largest contributor of defaults, with 13 (see Exhibit 5). Most of defaulters in this sector also had excessive leverage and weak liquidity. In 2023, the sector continued to contend with a raft of difficulties in the industry, including labor cost inflation and pressure from private and public payers to reduce costs. In addition to higher funding costs due to higher interest rates, refinancing became more difficult as lenders' appetite for highly-leveraged capital structures markedly decreased.

When measured by dollar volume, healthcare & pharmaceuticals defaults stood at the top of the list with 13.3% of the total. The next highest sector by dollar volume was retail, with a 12.7% share of defaults. Although the retail sector only had 10 defaults, six of them had defaulted on more than $1 billion of debt. In 2023, the sector faced increasing financial pressure from years of high inflation, depleted excess consumer savings and higher interest rates.

Exhibit 5

Healthcare & pharmaceuticals had the highest default volume and second highest default count*



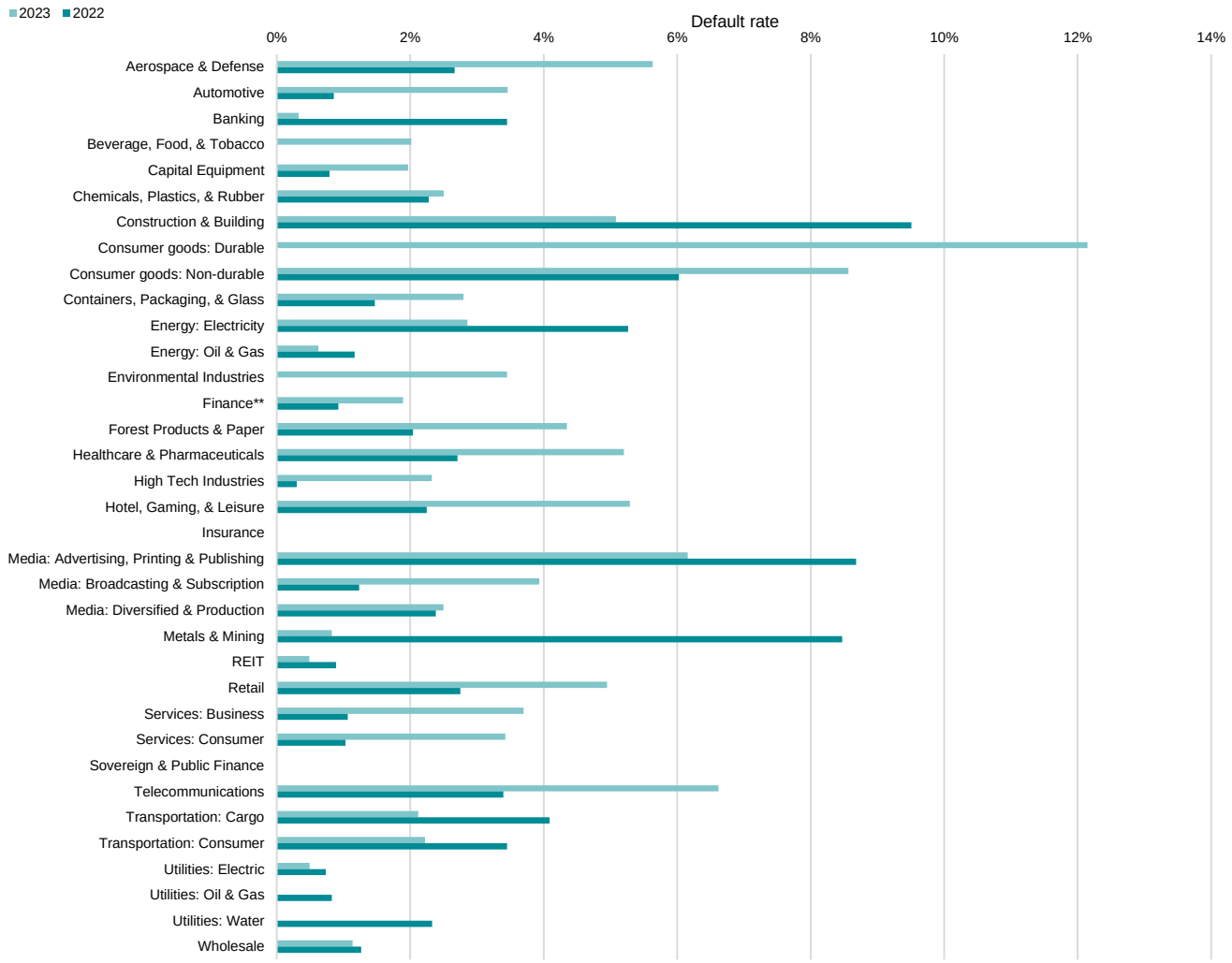
*We omitted those industries that had no defaults in 2022 and 2023.
** Non-bank and non-insurance financial companies.
Source: Moody's Investors Service

Although the business services sector had the most defaults in 2023, its default rate was not the highest because of its relatively large portfolio — i.e. the denominator is big. Durable consumer goods is the sector with the highest default rate, at 12.1%. This sector is a relatively small one dominated by issuers with corporate family rating of single B or below. Defaults rose sharply in this sector last year after product demand weakened because of surging prices, higher interest rates and as consumers shifted spending to travel, entertainment and other experiences after the pandemic. In addition, some financially weak companies that increased debt in recent years for leveraged buyouts and acquisitions found it difficult to service debt with rising funding costs and refinance in a tougher lending environment.

On the other hand, the default rate for the business services sector was lower at 3.7% (see Exhibit 6).

Exhibit 6

Durable consumer goods had the highest one-year default rate in 2023*



*Issuer-weighted and includes investment-grade and speculative-grade issuers.
** Non-bank and non-insurance financial companies.
Companies in the sovereign and public finance sectors are government-related financial and nonfinancial corporate issuers.
Source: Moody's Investors Service

Loan recovery rates fell noticeably

In this section, we discuss recovery rates measured by two methods: one measured by trading price for defaulted debt and the other measured by the actual value that creditors realize at the resolution of a default event, i.e., ultimate recovery rates.

Exhibit 7 shows the trading price implied recovery rates over the past two years, as well as their long-term averages since 1983. On average, recovery rates have generally correlated with the priority of claim in the capital structure for four decades, with a higher priority of claim enjoying a higher average rate of recovery.[3] For example, recovery rates on first-lien bank loans average 65.0% on an issuer-weighted basis and 61.9% on a dollar-volume basis, for the highest recovery rates across security classes. This makes sense given first-lien bank loans' secured nature and seniority in the capital structure.

The exhibit further shows that the issuer-weighted average first-lien loan recovery rate was 59.6% in 2023, down noticeably from 69.5% in 2022. The 2023 first-lien loan recovery rate was also low by historical standards, with the comparable rate averaging 65.0% since 1983. The low recovery rate in last year's first lien loans reflected weaker credit quality and structural deterioration, the latter

resulting from the increase of loan-only structures in recent years. In the worst cases, some first-lien bank loans are becoming bond-like and act as buffers for asset-based loans which are structurally more senior. Moreover, nearly 40% of the loan recovery rates came from re-defaulters, whose debt tends to have lower recovery rates in subsequent defaults. Last but not least, covenant-lite documentation has allowed asset-stripping and priming that tends to diminish recovery prospects for existing secured creditors.

On the bond side, the recovery rate for senior unsecured bonds rose to an average 33.4% in 2023 from 30.9% rate in 2022. This compares to the long-term average of 37.6%.

Measured by dollar volume, the first-lien loan recovery rate declined to an average 53.6% in 2023 from 58.1% in 2022. For senior unsecured bonds, the volume-weighted average recovery rate increased to 30.9% in 2023 from 21.9% in 2022. Some of the average recovery rates in 2023 are based on small samples (see observation counts in Exhibit 8).

Exhibit 7

Average corporate debt recovery rates measured by trading prices

	Issuer-weighted recoveries			Dollar-weighted recoveries		
Priority Position	2023	2022	1983-2023	2023	2022	1983-2023
1st Lien Bank Loan	59.6%	69.5%	65.0%	53.6%	58.1%	61.9%
2nd Lien Bank Loan	40.0%	54.7%	34.4%	44.6%	52.2%	33.9%
Sr. Unsecured Bank Loan	n.a.	n.a.	46.3%	n.a.	n.a.	43.9%
1st Lien Bond	53.5%	58.2%	54.8%	47.0%	52.7%	54.7%
2nd Lien Bond	31.4%	64.7%	43.4%	13.1%	30.2%	37.3%
Sr. Unsecured Bond	33.4%	30.9%	37.6%	30.9%	21.9%	33.4%
Sr. Subordinated Bond	63.5%	n.a.	31.1%	63.5%	n.a.	26.6%
Subordinated Bond	6.7%	18.2%	31.7%	4.3%	10.6%	27.3%
Jr. Subordinated Bond	0.8%	n.a.	21.5%	0.8%	n.a.	11.8%

Source: Moody's Investors Service

Exhibit 8

Recovery observation counts and dollar volume

	Issuer counts			Dollar volume (in $ billions)		
Priority Position	2023	2022	1983-2023	2023	2022	1983-2023
1st Lien Bank Loan	84	32	752	49	29	478
2nd Lien Bank Loan	24	7	150	10	1	36
Sr. Unsecured Bank Loan	0	0	71	0	0	39
1st Lien Bond	30	18	459	20	20	225
2nd Lien Bond	7	2	115	5	1	61
Sr. Unsecured Bond	48	70	1328	41	80	1029
Sr. Subordinated Bond	1	0	517	0	0	118
Subordinated Bond	4	3	423	2	1	86
Jr. Subordinated Bond	1	0	28	1	0	5

Source: Moody's Investors Service

2023 saw more US corporate defaults and subpar ultimate recoveries across the board

Another measure of recovery rates is ultimate recoveries, or the value that creditors realize at the resolution of a default event. Using bankruptcy as an example, the ultimate recovery is the present value of the cash or securities that creditors receive when the debt issuer exits bankruptcy, which typically occurs one to two years following the initial default date. Exhibit 9 shows data on ultimate recovery rates for North American nonfinancial companies included in Moody's Ultimate Recovery Database (URD).[4]

In 2023, the US experienced its highest default tally since the pandemic, a stark contrast to the below-average default rates seen in 2022, which were primarily driven by distressed exchanges (DEs) executed by nonfinancial corporates. Despite DEs constituting approximately two-thirds of all US defaults, bankruptcy filings among our rated debt issuers surged dramatically to 25 in 2023 from eight in 2022.

Approximately half of the companies that filed for bankruptcy in 2023 have exited their court-supervised debt restructurings. However, the average recoveries on defaulted debt were lower than those realized in 2022, as well as long-term historical averages. For example, the 12 companies that filed for and exited bankruptcy protection recovered just 37% at the corporate family level, significantly below the historical average of 50%.

Besides idiosyncratic company-specific issues, some of the reasons behind such poor recovery rates in this rather small data sample can be explained by companies choosing increasingly to equitize most part of their debt structure, with market valuations staying low in their respective sectors and pressured by high interest rates. In addition, three companies in our sample had a single debt class structure comprised solely of first-lien secured loans and fared only a 26% recovery at the enterprise level, thereby skewing the overall recovery average of the 2023 sample.

All in all in 2023, of the 12 resolved bankruptcies, nine recovered below the historical average of 50% at a firm-wide level. However, three distressed exchanges executed by public debt issuers on average fared 85% at a family level, lifting the overall recovery average for 15 defaults from 2023 to around 46%.

Exhibit 9

Average debt ultimate recovery rates, 1987-2023

	Emergence year			Default year		
Debt type	2023	2022	1987-2023	2023	2022	1987-2023
Revolvers*	75.6%	94.5%	86.5%	81.2%	82.7%	86.5%
Term loans**	54.6%	73.6%	71.1%	49.1%	67.6%	71.1%
Senior secured bonds	49.2%	81.5%	61.7%	39.0%	84.5%	61.7%
Senior unsecured bonds	19.3%	40.9%	46.9%	12.4%	46.6%	46.9%
Subordinated bonds	–	–	27.9%	–	–	27.9%

The Moody's Ultimate Recovery Database primarily covers default resolutions of US nonfinancial companies. The emergence year column refers to recovery rates of companies that resolved their defaults that year regardless of when they defaulted. The default year column refers to recovery rates of companies that both defaulted and resolved their defaults in that same year.

* Includes cash revolvers and borrowing base facilities.

** Includes all types of term loans: first-, second-lien and unsecured

Source: Moody's Ultimate Recovery Database

If we look at 2023 family recoveries from another vantage point and include additional four companies that filed for bankruptcy in 2022 but emerged in 2023, the recovery rate still remains subpar at around 46%. Half of the additional bankruptcies filed in 2022 also performed poorly, with names like Phoenix Services International and Cineworld Group fully equitizing their funded debt. Despite not being widespread, three defaulted debt issuers in our sample, namely Serta Simmons, Envision Healthcare and Cineworld, engaged in liability management transactions (LMTs) as part of their DEs. However, these attempts at debt restructuring ultimately led to their subsequent bankruptcy filings in 2023 and 2022. As a consequence of these DEs, primed legacy loans, which were once first-lien secured credit facilities, in some instances, ended up at the bottom of the priority waterfall and were largely wiped out. Depending on which first-lien loan claims creditors held, debt recoveries for this class varied dramatically, ranging from 100% to a mere 4%, as was the case with Serta Simmons. This broad range of recovery rates for first-lien secured facilities, particularly in bankruptcy filings following LMTs, resulted in a substantial 40% standard deviation for loan recoveries.

A majority of loans in our 2023 sample were first-lien secured loans and the 55% loan recovery rate as seen in our exhibit was mainly derived from 43 defaulted first-lien secured term loan tranches — including two tranches which were not impacted in DEs — and four junior lien loans which were originated as second- or third-lien term loans.

In addition, we continue to see a bifurcation in recovery averages of revolvers (at 76%) and term loans (55%) with most defaulted revolvers in our sample being super-senior asset-based facilities and recovering all of the money for their creditors upon default.

For the bonds, senior secured bonds restructuring in 2023 did significantly worse with a 49% recovery rate when compared both to its 2022 average and the historical average of 62%. Of the 18 secured bonds in our sample, the majority diverged from the 2022 trend, stemming from bankruptcies and primarily receiving new equity at settlement. However, this new equity was not valued sufficiently in the current market to cover the debt owed to investors.

Unsurprisingly, a sample of 18 unsecured bonds fared even worse in 2023 defaults with an average recovery rate of just 19 cents on the dollar, compared to 41% recovered in 2022. Meanwhile, in 2022, of the 27 unsecured bonds in our data set, around half were restructured out of court, benefiting debt recoveries.

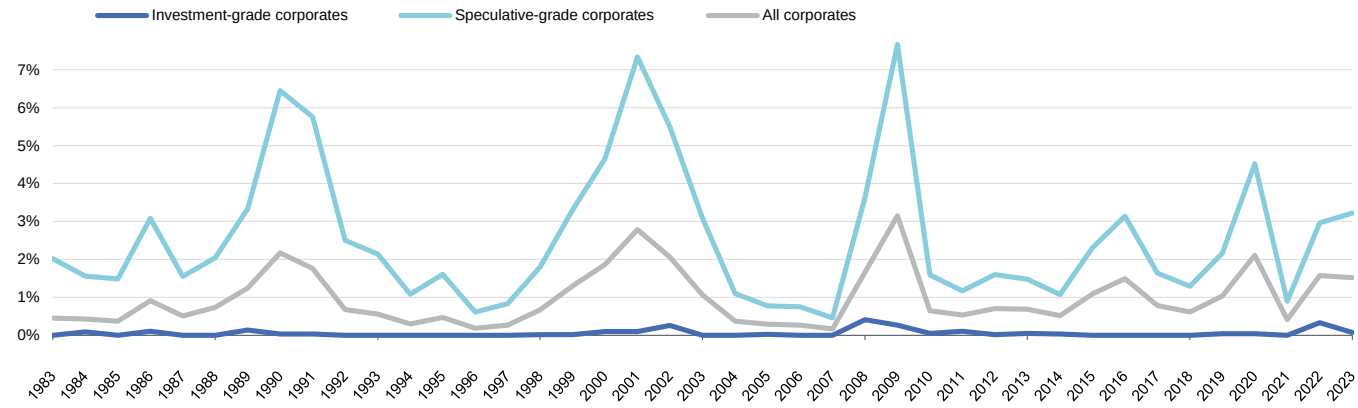
In summary, as we delve further into the distressed exchanges carried out by private equity-owned debt issuers which constituted most our 2023 defaulters — and as businesses continue to emerge from their bankruptcy proceedings — the debt-level recovery rates we have discussed will likely evolve. However, given a growing number of loan-only debt structures among defaulters, we anticipate the recovery rate for this class of debt to remain below its historical average of 70 cents on the dollar

Credit losses edged lower in 2023, resulting from lower loss-given-defaults

Exhibit 10 shows annual average credit loss rates from 1983 through 2023 for all rated corporate issuers. These rates are a function of both the probability of default and loss given default (LGD), or 1-recovery rates. The exhibit indicates that the average credit loss rate among all rated issuers eased to 1.5% in 2023 from 1.6% in 2022. Historically, the average annual credit loss rate for all Moody's-rated corporate issuers has been 1.0% since 1983.

Exhibit 10

Lower losses for investment-grade credits offset by higher losses for speculative-grade issuers



Source: Moody's Investors Service

Credit quality deteriorated for the second consecutive year

Rating downgrades outpaced upgrades in two-thirds of sectors

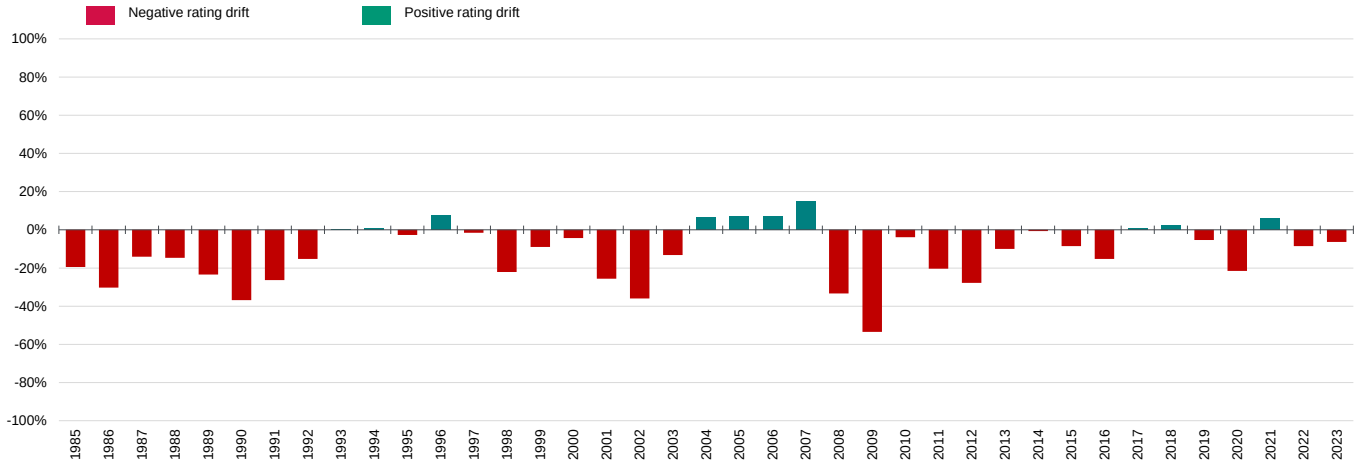
The aggregate credit quality of Moody's-rated issuers deteriorated last year against the backdrop of higher interest rates, tighter financing conditions and stress in sectors such as banking and construction & building. This is reflected in rating drift, which is calculated by the average upgraded notches per issuer minus the average downgraded notches per issuer. By definition, a positive drift

indicates overall improvement in credit quality, while a negative drift signals deterioration. In 2023, rating drift was -6.3% of a notch, the second consecutive year where the drift was in the negative zone that indicates more downgrades than upgrades (see Exhibit 11). In comparison, the drift was -8.6% a year earlier including Russian issuers and -1.8% if Russian issuers were excluded.

Exhibit 11

Rating downgrades dominated rating actions for the second year*

Percent of one notch

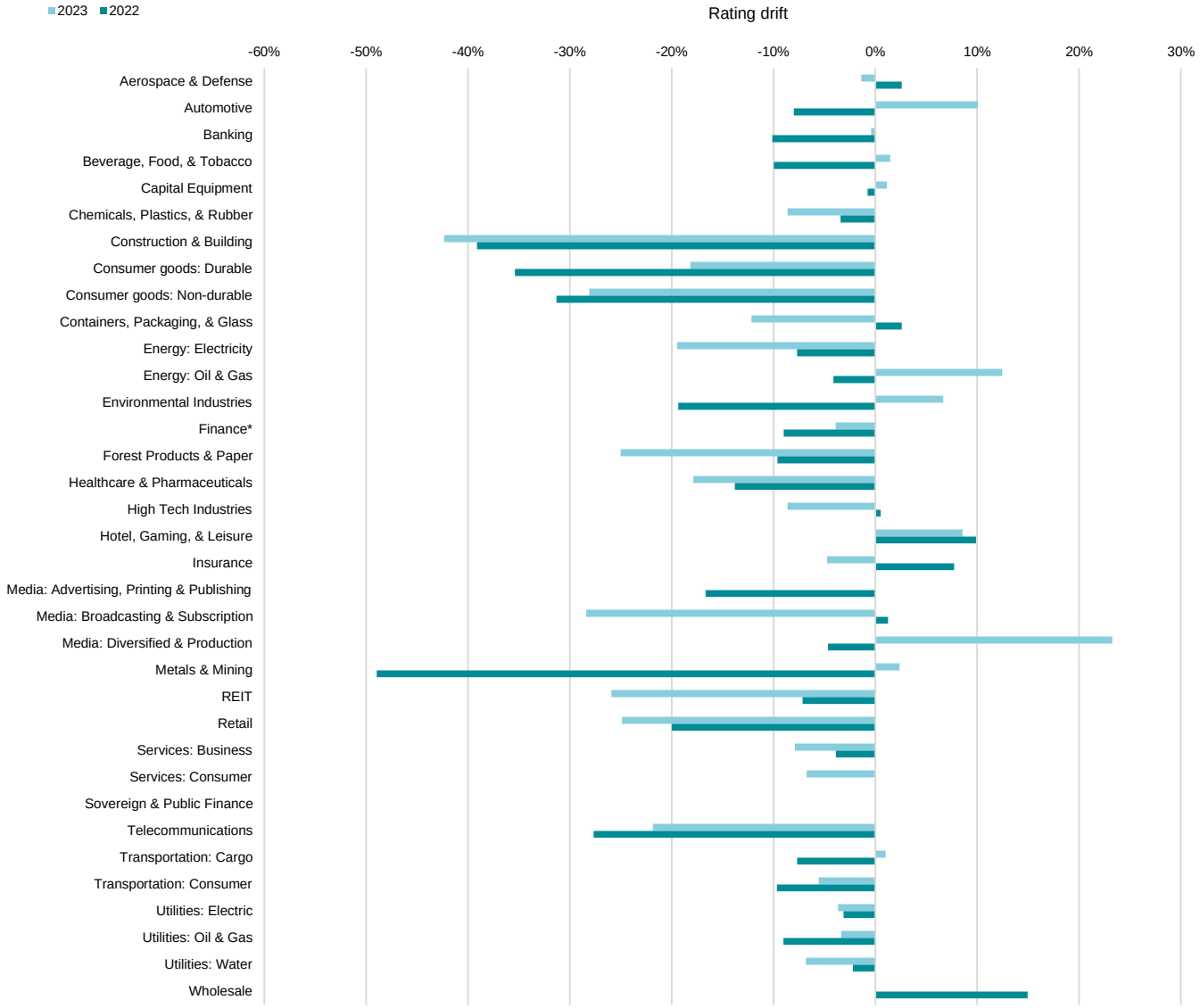


*Rating drift is measured by the average upgraded notches per issuer minus the average downgraded notches per issuer.
Source: Moody's Investors Service

Across sectors, construction and building had the most negative drift at -42.3% (see Exhibit 12). China property developers accounted for most of the downgrades in this sector reflecting a persistently difficult operating and financing environment. Although the government has stepped up policy support, property sales remained sluggish under slowing economic conditions. In addition, offshore funding remained largely closed for financially weak issuers, which led to multiple-notch downgrades for some including Country Garden Holdings Company Limited (Ca negative), a leading integrated property developer.

Exhibit 12

Two-thirds of sectors had negative drift in 2023
(% of rating notch)



* Non-bank and non-insurance financial companies.
Source: Moody's Investors Service

Investment-grade issuers remained more resilient to macro challenges

In 2023, ratings of investment-grade issuers continued to be more stable than those of speculative-grade issuers. This can be seen in Exhibit 13, which presents the one-year rating transition matrix. Each cell of the matrix shows the percentage of issuers that held a given row's rating at the beginning of 2023 and have since migrated to a rating or state (including rating withdrawal and default) indicated in the column heading at the end of 2023.[5]

The exhibit shows that the numbers in the diagonal are greater for investment-grade rating categories than in the speculative-grade universe. Lower triangular elements reflect net upgrades and upper triangular elements reflect net downgrades. Generally speaking, the further one moves away from the diagonal, the smaller the transition rates. This reflects a relatively low probability of issuers experiencing large rating actions during the course of a year.

For investors who are interested in the crossover risk between investment-grade and speculative-grade ratings, the fallen angel rate held steady at 1.2% from 2022 to 2023, below the long-term average of 2.0%. Similarly, the rising star rate also stabilized at 0.9% in the last two years, also below its long-term average of 2.3%. Across sectors, REITs had the most fallen angels. In addition to company-specific reasons, some of the downgrades in this sector reflected our concern of the difficult leasing environment, particularly for office real estate.

Exhibit 13

Ratings continued to be more stable among investment-grade issuers in 2023

Quadrants

II	I (Fallen Angel)
III (Rising Stars)	IV

I: Rating migration from **IG** to **SG**
II: Raing migration within **IG**
III: Rating migration from **SG** to **IG**
IV: Rating migration within **SG**

Rating Downgrade / Rating Upgrade

Probability Heat Map

10 < P | 5 < P < 10 | 1 < P < 5 | 0-1 | 1 < P < 5 | 5 < P < 10 | 10 < P

Upgrade ➤➤ ➤➤ Downgrade ➤➤ ➤➤

Rating Transition Matrix Definition

The matrix summarizes the cumulative changes in credit ratings over a given time horizon, meaning they describe the final position **(➤ TO)** of a credit rating given its initial position **(➤ FROM)**.

➤ TO: **Final rating position** at the end of year (December 31, 2023)

FROM: Initial rating position in the beginning of year (January 1, 2023)		Investment Grade (IG)										Speculative Grade (SG)											
		Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa1	Caa2	Caa3	Ca-C	WR	DEF
Investment Grade (IG)	Aaa	98.0	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Aa1	0.0	88.9	0.0	0.0	0.0	0.0	2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	8.3	0.0
	Aa2	0.0	1.7	94.8	0.9	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.7	0.0
	Aa3	0.0	0.0	1.2	96.4	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.2	0.0
	A1	0.0	0.0	0.0	3.1	90.1	3.1	0.5	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.9	0.0
	A2	0.0	0.0	0.0	0.0	6.4	86.9	2.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.2	0.0
	A3	0.0	0.0	0.0	0.0	0.0	7.0	83.8	5.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.1	0.4
	Baa1	0.0	0.0	0.0	0.0	0.1	0.0	6.5	85.1	4.1	0.6	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.9	0.1
	Baa2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	6.8	82.9	3.8	0.7	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.9	0.1
	Baa3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	8.6	82.0	2.8	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	5.4	0.0
Speculative Grade (SG)	Ba1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	11.1	77.8	3.9	1.9	0.0	0.0	0.5	0.5	0.0	0.0	0.5	2.9	0.0
	Ba2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.0	79.1	5.1	1.7	0.0	0.9	0.4	0.0	0.0	0.0	6.8	0.0
	Ba3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	4.7	7.4	69.8	3.1	2.7	1.9	2.3	0.8	0.4	0.0	4.7	1.9
	B1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	9.5	66.9	5.8	1.8	1.5	1.1	0.0	0.4	8.4	0.7
	B2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.8	1.2	12.1	65.6	10.5	2.0	1.2	0.8	0.0	5.5	0.4
	B3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.0	0.0	3.6	9.9	68.4	9.0	3.0	0.9	0.0	3.9	0.9
	Caa1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.5	0.7	7.3	60.1	13.2	4.2	0.5	10.1	2.8
	Caa2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.8	0.0	4.6	70.3	7.6	2.9	9.7	3.9
	Caa3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.0	7.8	56.2	14.2	9.1	12.3
	Ca-C	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.8	2.5	32.8	12.3	51.6

Data in percentages.
Source: Moody's Investors Service

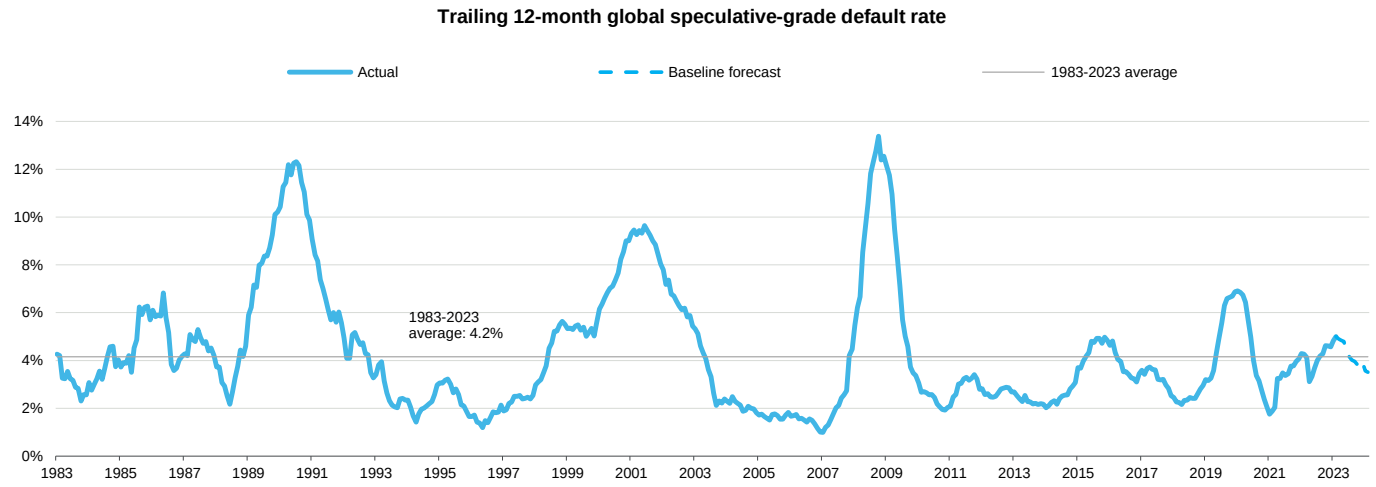
The global speculative-grade default rate will decline gradually in 2024 and remain near its historical average

We expect the default rate to end this year at 3.6%

After finishing at 4.8% at the end of 2023, the global speculative-grade default rate edged higher to 5% for the 12-month ended in January. In the coming months, we expect defaults will gradually normalize, falling from the higher levels that resulted from the pandemic, the Russia-Ukraine war and interest rate hikes. Our Credit Transition Model predicts that January's 5% default rate will mark the peak of the current cycle and that the global speculative-grade default rate will decline modestly to 3.6% by year-end under our baseline scenario (see Exhibit 14). If realized, the global default rate will finish this year at a level close to its historical average of 4.2%.

Exhibit 14

The speculative-grade default rate will decline moderately in 2024



Source: Moody's Investors Service

Our default rate forecast is underpinned by our expectation that major central banks will begin policy rate normalization by lowering rates this year, which will boost borrowers' ability to cover interest expenses, especially in the loan market. The maturity wall in 2024 is manageable as many companies have pushed out near-term maturities in the past few years. In addition, private/direct lending has and will continue to provide an alternative source of debt refinancing for some small and low-rated borrowers when they cannot access the syndicated loan market. In terms of macro conditions, we expect the economy in the US, where most of the defaults typically arise, to continue growing, although at a slower pace this year. While geopolitical developments will prompt market fluctuations and macroeconomic uncertainty, we do not expect them to escalate to levels that would have a materially negative effect on corporate defaults.

Our model's prediction is also in line with what high-yield spreads, strong predictors of defaults, imply. The spreads remain tight and are currently comfortably below the long-term average in both the US and Europe. Looking ahead, we expect them to widen in both regions from the current low base. Specifically, we assume the US high-yield spread will widen from about 323 basis points (bps) currently to the average of 486 bps in the fourth quarter and remain below its long-term average of about 500 bps. Similarly, we assume the European high yield spread to widen from 343 bps to 524 bps, also below its historical average. In terms of the unemployment rate — another macro input in our default forecasting model — we assume the US unemployment rate will rise from the current 3.7% to around 4.5% in the final quarter of 2024, while the European rate will edge slightly higher from 6.6% to 6.7%.[6]

Unlike in 2008-09 global financial crisis and the 2020 pandemic — two recent periods when the spike in defaults was steep but short-lived — we expect the peak default rate will be lower this time and that the rate will not fall rapidly from that peak. That is because this cycle is mainly driven by higher funding costs as opposed to a wide-scale earnings recession. The risks are much more idiosyncratic and centered on credits with overlevered capital structures and that have borrowed heavily in the loan market. Other factors that support

the default rate holding near its historical average in 2024 are moderating global economic growth, the expected gradual interest rate normalization and re-default risk following DEs.

The global default rate will exceed our baseline forecast if risks crystallize

The default rate will exceed our baseline forecasts if certain economic and financial risks crystallize. First, there is still uncertainty surrounding the path of interest rate normalization as inflation's "last mile" is unlikely to be uniform across countries and occasional bumps will complicate policymaking and keep markets on edge. If inflation accelerates severely again, that could force central banks to raise benchmark interest rates, prompting economic recessions. Corporate earnings and cash flow will be negatively affected and default risk will rise.

Another risk is the global economy. If economic growth in the US decelerates more meaningfully than expected and Europe ultimately falls into a severe recession, consumer spending will slow, hurting corporate revenue and debt servicing abilities.

In 2023, stress in the banking sector was evident as Federal Reserve (Fed) tightening negatively impacted some US banks. Banks were more subject to liabilities repricing and the Fed quantitative tightening (QT) added additional pressure to US banks' liabilities. In the US, there are no quantitative limits in bank regulation on interest rate risk. Some banks focused on interest rate risk despite limited regulation; but others did not. Banks that over-focused on net interest income measures during the pandemic ended up with high levels of residential mortgages, fixed rate commercial real estate loans or long-dated fixed rate securities and, in turn, negative economic capital when rates rose and deposits proved less core than anticipated. Uninsured bank creditors worried about significant unrealized economic losses not captured in most US banks' regulatory capital which was a negative shock to confidence sensitive funding that spread. Silicon Valley Bank, First Republic and Signature Bank went into FDIC receivership. As we head into 2024, the US banking sector outlook remains negative with 40% of the rated universe on negative outlook and some banks are still sensitive to confidence shocks, high levels of fixed rate assets and rising asset quality risks related to commercial real estate. Banks are important providers of credit, particularly to small businesses who employ most the US labor force. Renewed or heightened stress in the US banking sector could negatively affect credit availability and, in turn, have negative corporate credit and economic implications.

The geopolitical conflict between Israel and Hamas and the ongoing war between Russia and Ukraine are also worth noting. If the conflicts intensify, the global economy will worsen and inflation could increase. This could also cause a renewed energy shock, resulting in severely negative economic outcomes and an increase in corporate defaults.

Other risks include the potential for China's economic recovery to disappoint, with negative spillovers to global growth and credit conditions. In addition, there is a heavy calendar of elections globally, which could reshape policy priorities.

Amid these uncertainties, we have developed a few scenarios that are more pessimistic than our baseline and one scenario that is more optimistic. Since risks are skewed toward higher numbers, we present default rate forecasts under various pessimistic scenarios in Exhibit 15. In this exhibit, we apply the same rating histories and current ratings in all scenarios so that the only factors driving the different forecasts are economic assumptions. As the exhibit shows, the global default rate could climb above our baseline expectations if economic and financial conditions turn out worse than our baseline assumptions.

Under our moderately pessimistic (downside 10%) scenario, the global default rate would rise to 7.4% in 2024, which is a bit higher than the pandemic peak of 6.9%. Under our severely pessimistic scenario (the downside 4% scenario), the rate would jump to 11.8%, which would be lower than the peak of 13.4% during the 2008-09 global financial crisis.

If global economic and financing conditions are better than expected, the optimistic scenario would unfold. Under this scenario, the global default rate would fall to 3.1% by the end of 2024. This optimistic scenario assumes a smaller widening in the average high-yield spread to 449 bps in the US and 488 bps in Europe. We assume the unemployment rate would improve in both regions, to 3.1% and 6.3% respectively.

Exhibit 15

Default rate forecasts under alternative scenarios

Scenario		Baseline	The downside 25% scenario	The downside 10% scenario	The downside 4% scenario
Scenario description	Probability that the economy will perform better	50%	75%	90%	96%
	Probability that the economy will perform worse	50%	25%	10%	4%
Assumptions for 2024	US unemployment*	4.5%	6.5%	7.6%	8.2%
	US HY spread*	486 bps	666 bps	715 bps	871 bps
Speculative-grade default rate	2023 actual	4.8%	4.8%	4.8%	4.8%
	2024 forecast	3.6%	4.9%	7.4%	11.8%

* Average in Q4 2024.

Sources: Bloomberg, Moody's Analytics and Moody's Investors Service

We expect the durable consumer goods sector to have the highest default rate again

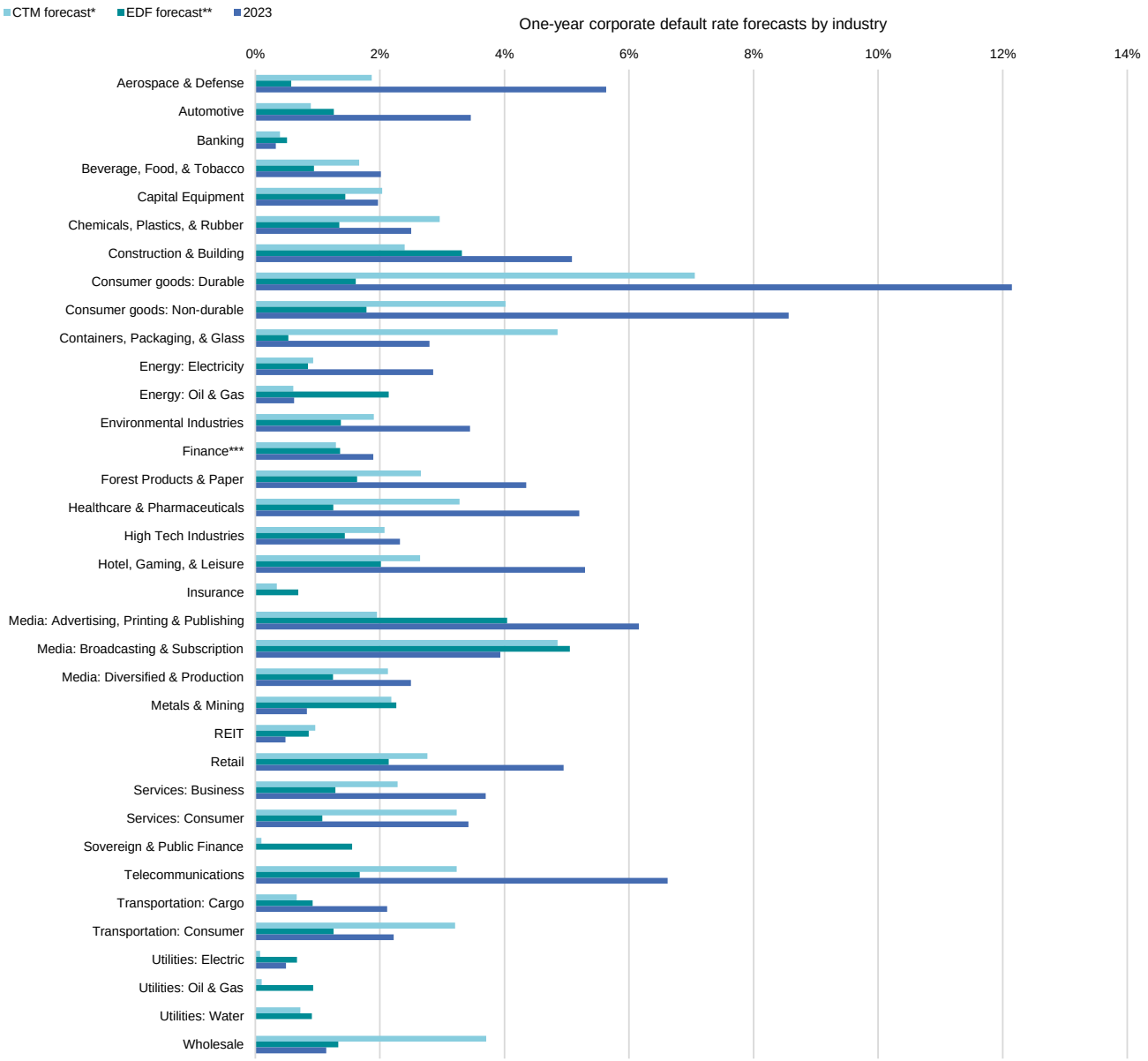
Under our baseline forecast, we project 100 corporate defaults in 2024, or roughly eight per month. Across sectors, we expect that durable consumer goods will have the highest default rate, at 7.1%. We anticipate 2024 will continue to present difficulties for durable goods manufacturers, driven by our expectations that economic growth will slow in the US and remain subdued in the euro area. This will result in weak demand and earnings pressure for durable consumer goods companies. The sector is also dominated with issuers with corporate family rating of single B or below and many are highly leveraged following a debt splurge in recent years, making them vulnerable to higher for longer interest rates, economic shocks and financial market risks.

When measured by default count, we forecast the business services sector will have eight defaults, the most across sectors, because of its relatively larger number of low-rated companies. A good portion of the sector's debt maturing in the coming two years is held from companies with weak credit quality, which presents refinancing risk under tight financing conditions.

Exhibit 16 shows one-year default rate forecasts for all Moody's-rated companies by industry, using the same economic assumption for all industries. The factors driving the different forecast default rates include rating histories, current ratings and the watch/outlook statuses.

Exhibit 16

Moody's CTM forecasts that durable consumer goods will have the highest default rate in 2024



*CTM forecast is based on Moody's CTM.
**EDF forecast is based on Moody's KMV Model.
*** Non-bank and non-insurance financial companies.
Source: Moody's Investors Service

The exhibit also shows the default rate forecast implied by Expected Default Frequency (EDF).[7] The EDF-implied default rate forecast includes rated and unrated issuers. EDF metrics are forward-looking probabilities of default generated by Moody's EDF model, which belongs to a class of credit risk models referred to as structural models. Their basic assumption is that there is a causal, economically motivated reason why companies default. In particular, the EDF model assumes that default is highly likely to occur when the market value of a company's assets is insufficient to cover its liabilities at some future date – in other words, when it is insolvent. EDF-implied ratings may indicate different default outlooks compared with those from Moody's CTM. For example, EDF-implied ratings suggest that the media: broadcasting & subscription sector will have the highest default rate. This indication reflects the volatility in the equity

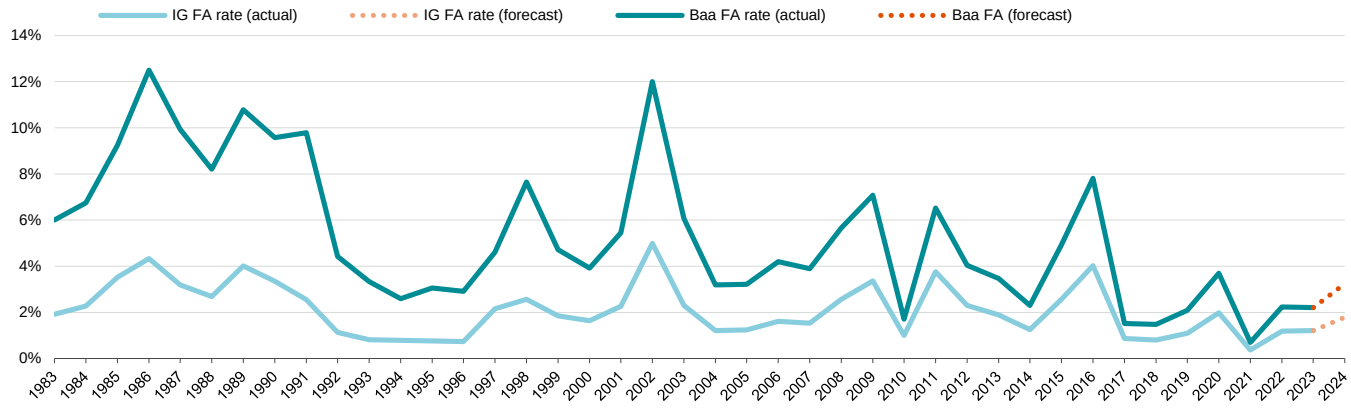
market, which may not be fully captured in rating actions. In addition, the EDF-implied default rate may also be higher in some sectors because that portfolio includes some unrated issuers that may have relatively weak credit quality.

The fallen angel rate will likely rise toward its historical mean

In this section, we use Moody's CTM to predict fallen angel rates based on the same macro assumptions as in our baseline scenario for the default rate forecast. Our model predicts fallen angel rates for Baa-rated companies will rise to 3.2% in 2024 from 2.2% in 2023. Similarly, the model predicts that the fallen angel rate for the entire investment-grade universe will climb to 1.8% from 1.2% under our forecasts (see Exhibit 17). These forecasts, if realized, will remain low by historical standards.

Exhibit 17

Fallen angel (FA) rate will likely rise in 2024



Source: Moody's Investors Service

Loan recovery rate will likely remain low by historical standards

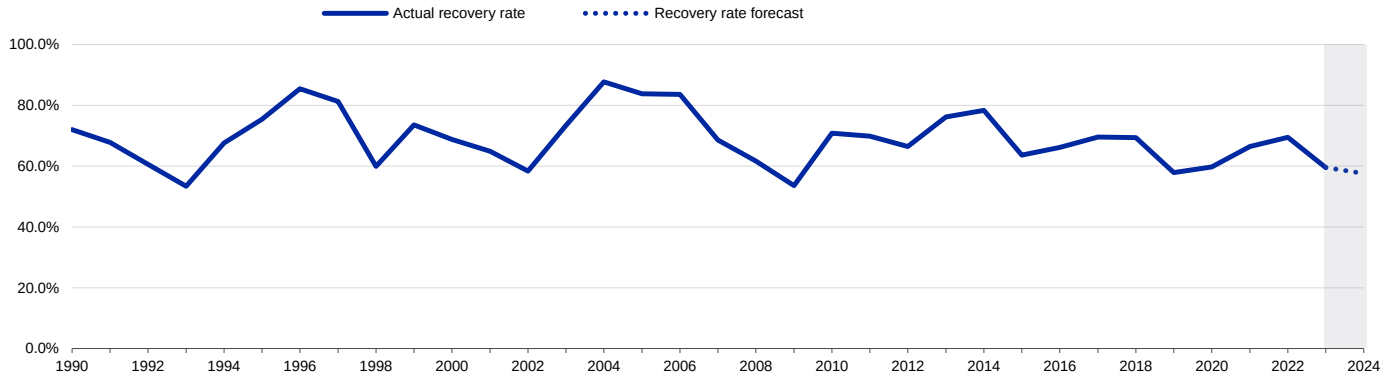
In 2024, we expect leveraged loans will continue to disproportionately default more than high-yield bonds in the US, where most of the defaults come from. The disproportionate defaults will reflect the difference in ratings and the prevalence of senior secured loan-only leveraged buyouts exposed to more expensive floating-rate debt. However, the gap between loan and bond defaults may narrow because the positive effects from policy rate cuts will be reflected faster on loans. In comparison, it will take longer for the effects to materialize on bonds.

In terms of the recovery outlook, our loss given default assessments (LGDAs) at the beginning of this year indicate that recoveries for senior secured loans will remain low this year. Specifically, we expect the average recovery rates of first-lien senior secured bank loans to be 57.6% across all sectors in North America and Europe in the event they all default in 2024 (see Exhibit 18). This is lower than the long-term average recovery rate of 65.0% measured by trading prices. We expect that weak credit quality and structural deterioration will continue to put pressure on loan recoveries in 2024.

We remain concerned about the lack of credit protections in current loan agreements. Less-restrictive credit documentation provides borrowers with a greater set of options that may allow them to delay or avoid a default. However, that also increases the risk of further erosion for first-lien bank lenders because certain cov-lite deals provide greater allowances for collateral transfers to unrestricted subsidiaries, which raises the risk of issuers being more stressed ahead of default, therefore lowering recovery rates. Another trend is that, in recent years, there were asset-stripping and priming financing transactions where companies needed more liquidity and a subset of the existing lender group provided capital while "up-tiering" existing holdings into a super-priority tranche. This had the effect of reducing the recovery for creditors who did not participate in the new money transaction even though their papers were still labeled as first lien. Given the covenant flexibility in recent loan and bond vintages, we expect these kinds of liability management transactions to continue when corporate obligors encounter liquidity difficulties.

Exhibit 18

LGDAs suggest first-lien loan recovery rate will remain below 60% in 2024



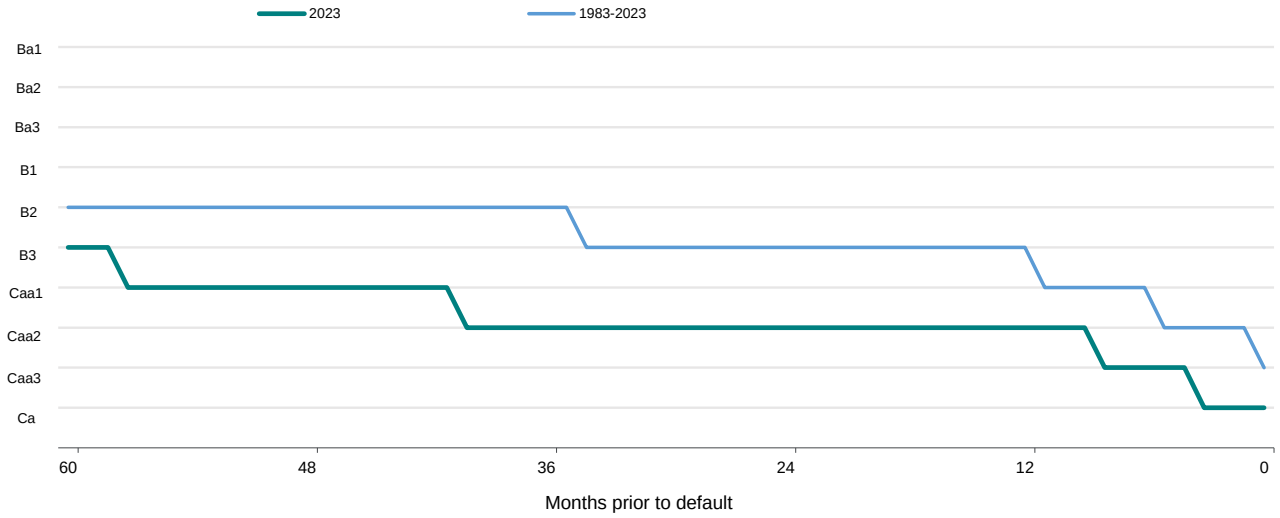
Source: Moody's Investors Service

Rating accuracy

Our ratings have historically proven to be effective predictors of default (see Exhibit 19). The exhibit plots the median ratings of more than 3,000 corporate issuers that defaulted from 1983 to 2023. The exhibit demonstrates that, historically, rated issuers have been downgraded to B2 as early as five years before default. The median rating has been B3 one year before default since 1983. Among last year's defaulters, the median rating one year before default was Caa2, two notches below the rating measured over the entire 1983-2023 period.

Exhibit 19

Median ratings before default, 2023 vs. long-term average



Source: Moody's Investors Service

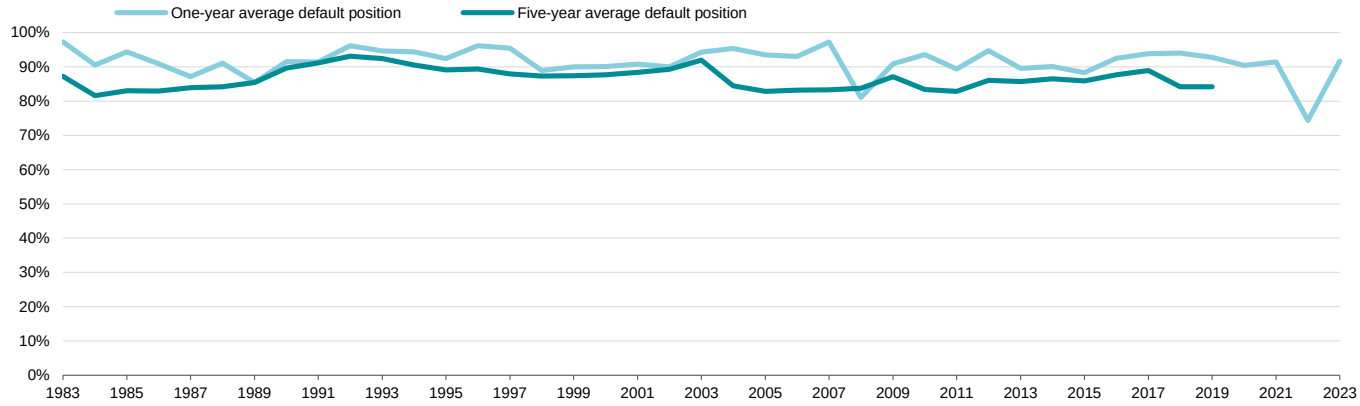
The evolution of median ratings demonstrates that our corporate ratings correlate with subsequent default experiences. To further demonstrate the ability of ratings to separate issuers with low credit risk from those with high credit risk, we use the Average Default Position (AP) of defaults to evaluate ordinal accuracy of our ratings (see Exhibit 20).

AP measures the average position of defaulters, with position defined as the percentage of issuers with higher or equal ratings. A greater AP indicates a more discriminatory rating system because there are more issuers rated higher than the defaulters, or that defaulters are generally found in lower rating categories. The AP of defaults was consistently high during the 1983-2023 period, with an average of 91.2% for the one-year horizon and 86.4% for the five-year horizon. These percentages indicate that our ratings have been

effective in predicting defaults over both the short and long term. In 2023, the one-year AP rebounded sharply to 91.7% from 74.3% in 2002 when high-rated Russian issuers defaulted due to sanction.

Exhibit 20

One- and five-year AP by cohort year, 1983-2023



AP measures how well ratings rank-ordered default risk over the measured horizon.
Source: Moody's Investors Service

Methodology and data sources

Moody's definition of default

Our definition of default is applicable only to debt or debtlike obligations (for example, swap agreements). For details, please refer to Moody's Rating Symbols and Definitions.

Methodology

The methodology used in this study can be found in the Glossary of Moody's Ratings Performance Metrics. The glossary is a technical paper that explains how we calculate default rates, transition rates and rating performance metrics in detail.

Changes in this year's report

We occasionally discover historical defaults, leading to minor revisions of the historical data. As always, the data contained in our most recently published default study supersede the data in previous reports.

Data sources

We base the results of this study on our proprietary database of ratings and defaults for corporate bond and loan issuers. Municipal and sub-sovereign debt issuers, structured finance securities, private placements and issuers with only short-term debt ratings are excluded unless otherwise noted. In total, our data covers the credit experiences of nearly 30,000 corporate issuers that sold long-term public debt at some time between 1920 and 2023. As of January 1, 2023, close to 7,000 corporate issuers held a Moody's long-term bond, loan, deposit or corporate family rating.

Our database of corporate defaults covers more than 5,000 long-term bond and loan defaults by rated and nonrated issuers. Additional data sources, such as Bloomberg's Fixed Income Index data, supplemented our proprietary data in the construction of the aggregate dollar volume-weighted default rates. Defaulted bond pricing data was derived from Bloomberg, Refinitiv, ICE Data Services and TRACE. Most of these market quotes represent an actual bid on the debt instrument, although no trade may have occurred at that price. Over the 1983-2023 period, the data set includes post-default prices for more than 6,000 defaulted instruments issued by over 2,000 defaulting corporations. We make the 1970-2023 credit rating, default and recovery rate data used in this study available through the Default and Recovery Database.

Guide to data tables and charts in appendix

In this section, we briefly describe the interpretation of some of the exhibits contained in this report. Exhibit 19 was derived by mapping our ratings to a linear scale, then taking the median values of the numerically mapped ratings. Exhibit 28 shows average senior unsecured recovery rates by letter rating and year before default. Each cell in the table indicates the average recovery rate on senior unsecured bonds with a specific rating within T years of default. For example, the 36.0% two-year B recovery rate reported in Exhibit 28 indicates the average recovery rate on B- rated issues that default at some time within a two-year period, not recovery rate for issuers rated B exactly two years before default.

Exhibits 38 through 42 show issuer-weighted historical average default rates by rating category over various investment horizons. These data were generated by averaging the multiyear default rates of cohorts formed at monthly intervals. In addition to their being statements of historical fact, these data are also useful proxies for expected default rates. For example, over a five-year period a portfolio of B-rated issuers defaulted at a 20.2% average rate between 1983 and 2023 (see Exhibit 40). For an investor with a five-year exposure to a B-rated debt obligation or counterparty, this estimate also happens to be the best estimate of the expected risk of default for a B-rated exposure based on the available historical data, particularly over long investment horizons.

Exhibit 46 shows average cumulative volume-weighted default rates by rating category. Whereas issuer-based default rates weight each issuer equally, these data weight each issuer by the total volume of defaulted bonds, so larger defaults receive relatively more weight. Average default rates based on bond volume affected are less suitable estimates of expected default risk. One reason is that issuer default volumes vary considerably over time. On average, a leveraged corporate issuer defaults on approximately $300 million of bonds. However, that total has been as high as $30 billion (WorldCom). Issuer-based default rates receive particular emphasis in the rating process because the expected likelihood of default of a debt issuer holding a given rating is likely to be the same regardless of differences in the nominal sizes of the exposures.

Exhibit 47 shows the cumulative issuer-weighted historical default rates of cohorts formed between the years 1970 and 2023 (January 1 of each year). These data are a subset of the data used to calculate the issuer-weighted averages shown in Exhibits 38 through 40 (which, again, are based on cohorts formed at monthly time intervals). The default rates in Exhibit 47 may be useful for scenario analysis. For example, if one believed that future default rates for a given pool of issuers will behave as they did in, say, 1997, then one can use the January 1, 1997 cohort cumulative default rates as proxies for expected default rates.

Appendix

Exhibit 21

Moody's-rated 2023 corporate defaults*

Company	Domicile	Default month	Broad default type	Bond ($mil)	Loan ($mil)	in Jan 2023 cohort?
4L Holdings Corporation	United States	2023.05	Distressed exchange		75	yes
Accuride Corporation	United States	2023.07	Distressed exchange		438	yes
Agung Podomoro Land Tbk (P.T.)	Indonesia	2023.07	Distressed exchange	168		yes
Air Methods Corporation	United States	2023.10	Bankruptcy	500	1,375	yes
Akorn Operating Company LLC	United States	2023.02	Bankruptcy		370	yes
Akumin Inc.	United States	2023.10	Bankruptcy	850	55	yes
Altisource S.a.r.l.	Luxembourg	2023.02	Distressed exchange		247	yes
AMC Entertainment Holdings, Inc.	United States	2023.02	Distressed exchange	109		yes
AMC Entertainment Holdings, Inc.	United States	2023.12	Distressed exchange	342		no
Americanas S.A.	Brazil	2023.01	Payment default	3,000		yes
Amphora Intermediate II Ltd	United Kingdom	2023.12	Payment default		380	yes
AmSurg, LLC	United States	2023.05	Bankruptcy		2,777	yes
Anchor Glass Container Corporation	United States	2023.07	Payment default		727	yes
API Holdings III Corp.	United States	2023.02	Distressed exchange		90	yes
API Holdings III Corp.	United States	2023.11	Distressed exchange		390	no
Aston Midco Limited	Jersey	2023.09	Distressed exchange		92	yes
At Home Group Inc.	United States	2023.05	Distressed exchange	447		yes
Atento Luxco 1	Brazil	2023.09	Payment default	500		yes
Audacy, Inc.	United States	2023.10	Distressed exchange	540		yes
Avaya Holdings Corp.	United States	2023.02	Bankruptcy	221	0	yes
Avaya Inc.	United States	2023.02	Bankruptcy	1,250	1,886	yes
Azul S.A.	Brazil	2023.07	Distressed exchange	862		yes
Bed Bath & Beyond Inc.	United States	2023.04	Bankruptcy	1,030	100	yes
Bering III S.a r.l.	Luxembourg	2023.12	Distressed exchange		393	yes
BIM Land Joint Stock Company	Vietnam	2023.08	Distressed exchange	99		yes
Bioplan USA, Inc.	United States	2023.03	Distressed exchange		386	yes
CareerBuilder, LLC	United States	2023.04	Distressed exchange		176	yes
Carvana Co.	United States	2023.08	Distressed exchange	5,520		yes
Casa Systems, Inc.	United States	2023.06	Distressed exchange		219	yes
Casino Guichard-Perrachon SA	France	2023.08	Payment default	2,196		yes
CatLuxe S.a.r.l.	Spain	2023.04	Distressed exchange		381	yes
Central China Real Estate Limited	China	2023.04	Distressed exchange	1,882		yes
Checkers Holdings, Inc. (OLD)	United States	2023.05	Payment default		305	yes
China SCE Group Holdings Limited	China	2023.10	Payment default		387	yes
CIBT Global, Inc.	United States	2023.05	Distressed exchange		590	yes
ClubCorp Holdings, Inc.	United States	2023.10	Distressed exchange	315		yes
CNG Holdings, Inc.	United States	2023.08	Distressed exchange	228		yes
Codere Luxembourg 2 S.a.r.l.	Spain	2023.04	Payment default	532		yes
ColourOz MidCo	Luxembourg	2023.03	Payment default		1,253	yes
Comet Bidco Limited	United Kingdom	2023.06	Distressed exchange		894	yes
Cooper-Standard Automotive Inc.	United States	2023.01	Distressed exchange	357		yes
Country Garden Holdings Company Limited	China	2023.09	Distressed exchange	4,531		yes
Covis Parent SCA	Luxembourg	2023.06	Distressed exchange		512	yes
Covis Parent SCA	Luxembourg	2023.12	Payment default		495	no
CSTN Merger Sub, Inc.	United States	2023.08	Distressed exchange	450		yes
Curo Group Holdings Corp.	United States	2023.05	Distressed exchange	682		yes
Cyxtera DC Holdings, Inc.	United States	2023.06	Bankruptcy		953	yes

Source: Moody's Investors Service

Company	Domicile	Default month	Broad default type	Bond ($mil)	Loan ($mil)	in Jan 2023 cohort?
Dawn Acquisitions LLC	United States	2023.07	Distressed exchange		166	yes
Diamond Sports Group, LLC	United States	2023.03	Bankruptcy	4,825	3,930	yes
Diebold Nixdorf, Inc. (Old)	United States	2023.06	Bankruptcy	1,511	1,024	no
Digicel Group Holdings Limited	Jamaica	2023.05	Payment default	635		yes
Digicel International Finance Limited	Saint Lucia	2023.03	Payment default	1,492		yes
Digital Media Solutions, LLC	United States	2023.08	Distressed exchange		271	yes
Eagle Intermediate Global Holding B.V.	United States	2023.05	Distressed exchange	275		yes
Elevate Textiles, Inc. (Old)	United States	2023.06	Distressed exchange		644	yes
Envision Healthcare Corporation	United States	2023.05	Bankruptcy	939	4,161	yes
Equinox Holdings, Inc.	United States	2023.02	Distressed exchange		76	yes
Equinox Holdings, Inc.	United States	2023.11	Distressed exchange		66	no
Exela Intermediate LLC	United States	2023.02	Distressed exchange	13		yes
Exela Intermediate LLC	United States	2023.07	Distressed exchange	1,271		no
FDS Holdings BV	Netherlands	2023.09	Distressed exchange		158	yes
FINThrive Software Intermediate Hldgs, Inc.	United States	2023.09	Distressed exchange		46	yes
First Republic Bank	United States	2023.05	Bankruptcy	800		yes
Flamingo Group International Limited	United Kingdom	2023.12	Distressed exchange		269	yes
Foodco Bondco, S.A.U.	Spain	2023.02	Payment default	358		yes
Forming Machining Industries Holdings, LLC	United States	2023.03	Distressed exchange		50	yes
FR BR Holdings, L.L.C.	United States	2023.12	Payment default		460	yes
Frigoglass SAIC	Greece	2023.02	Payment default	334		yes
FXI Holdings, Inc.	United States	2023.05	Distressed exchange	500		yes
Genesis Care Finance Pty Ltd	Australia	2023.06	Bankruptcy		1,629	yes
Gol Linhas Aereas Inteligentes S.A.	Brazil	2023.03	Distressed exchange	1,077		yes
Greenland Holding Group Company Limited	China	2023.07	Payment default	1,350		yes
Haya Holdco 2 plc	United Kingdom	2023.09	Distressed exchange	374		yes
HealthChannels Intermediate Holdco, LLC	United States	2023.09	Distressed exchange		19	yes
Ideal Standard International S.A.	Belgium	2023.09	Distressed exchange	344		yes
Instant Brands Holdings Inc.	United States	2023.06	Bankruptcy		391	yes
JP Intermediate B, LLC	United States	2023.07	Distressed exchange		366	yes
K&N Parent, Inc. (Old)	United States	2023.02	Distressed exchange		285	yes
Keter Group B.V.	Netherlands	2023.08	Payment default		112	yes
KNB Holdings Corporation	United States	2023.02	Bankruptcy		441	yes
Lannett Company, Inc.	United States	2023.05	Payment default	436	223	yes
Learfield Communications, LLC	United States	2023.09	Distressed exchange		1,064	yes
LendingTree, Inc.	United States	2023.12	Distressed exchange	290		yes
LifeScan Global Corporation	United States	2023.05	Distressed exchange		275	yes
Ligado Networks LLC	United States	2023.11	Payment default	4,180		yes
Light Energia S.A.	Brazil	2023.04	Payment default	301		yes
Light S.A.	Brazil	2023.04	Payment default	0	0	yes
Light Servicos De Eletricidade S.A.	Brazil	2023.04	Payment default	1,732	203	yes
Lippo Malls Indonesia Retail Trust	Indonesia	2023.10	Distressed exchange	82	145	yes
Loyalty Ventures Inc.	United States	2023.03	Bankruptcy		636	yes
Lucky Bucks, LLC	United States	2023.03	Payment default		568	yes
Mallinckrodt International Finance S.A.	United States	2023.07	Payment default	978	1,758	yes
Matrix Parent, Inc.	United States	2023.12	Payment default		595	yes
Maverick Gaming LLC	United States	2023.03	Distressed exchange		45	yes
Monitronics International, Inc.	United States	2023.05	Bankruptcy		1,068	yes
Moran Foods LLC	United States	2023.10	Distressed exchange		452	no
MVK Intermediate Holdings, LLC	United States	2023.04	Distressed exchange		362	yes
MVK Intermediate Holdings, LLC	United States	2023.10	Bankruptcy		496	no
National CineMedia, LLC	United States	2023.03	Payment default	630	523	yes
NEP Europe Finco B.V.	United States	2023.12	Distressed exchange		531	yes

Source: Moody's Investors Service

Company	Domicile	Default month	Broad default type	Bond ($mil)	Loan ($mil)	in Jan 2023 cohort?
NEP/NCP Holdco, Inc	United States	2023.12	Distressed exchange		1,708	yes
New Constellis Borrower LLC	United States	2023.03	Distressed exchange		259	yes
Outerstuff LLC	United States	2023.12	Payment default		14	no
Paper Industries Holding S.a r.l.	Luxembourg	2023.12	Distressed exchange		124	yes
Paper Industries Intermediate Fin. S.a r.l.	Luxembourg	2023.12	Distressed exchange	276	0	yes
Party City Holdings Inc.	United States	2023.01	Bankruptcy	1,027	104	yes
Polaris Newco, LLC	United States	2023.05	Distressed exchange		2,500	yes
Praesidiad Group Limited	United Kingdom	2023.11	Distressed exchange		441	yes
Pretium PKG Holdings, Inc.	United States	2023.10	Distressed exchange		1,341	yes
QualTek LLC	United States	2023.03	Distressed exchange	131	555	yes
Quorum Health Corporation	United States	2023.09	Distressed exchange		617	yes
Rackspace Technology Global, Inc.	United States	2023.07	Distressed exchange	222		yes
Resolute Investment Managers, Inc.	United States	2023.12	Distressed exchange		627	yes
Rising Tide Holdings, Inc.	United States	2023.03	Distressed exchange		514	yes
Rising Tide Holdings, Inc.	United States	2023.09	Distressed exchange		521	no
Rite Aid Corporation	United States	2023.10	Bankruptcy	1,596	2,208	yes
Riverbed Technology LLC	United States	2023.02	Payment default		900	yes
RobertShaw US Holding Corp.	United States	2023.05	Distressed exchange		620	yes
Rodan & Fields, LLC	United States	2023.04	Distressed exchange		800	yes
Screenvision, LLC	United States	2023.12	Distressed exchange		22	no
Serta Simmons Bedding, LLC	United States	2023.01	Bankruptcy		1,889	yes
SHO Holding I Corporation	United States	2023.05	Distressed exchange		125	yes
Shutterfly, LLC	United States	2023.06	Distressed exchange	1,050	1,383	yes
Sierra Enterprises LLC	United States	2023.01	Payment default		385	yes
Signature Bank	United States	2023.03	Bankruptcy	575		yes
Silicon Valley Bank	United States	2023.03	Bankruptcy		0	yes
Sino-Ocean Group Holding Limited	China	2023.08	Payment default	3,320		yes
Skillz Inc.	United States	2023.04	Distressed exchange	160		yes
SoLocal Group S.A.	France	2023.07	Payment default	220		yes
SP PF Buyer LLC	United States	2023.07	Distressed exchange		435	yes
Strategic Materials Holding Corp.	United States	2023.08	Payment default		378	yes
SVB Financial Group	United States	2023.03	Bankruptcy	3,300		yes
Tacora Resources Inc.	Canada	2023.05	Distressed exchange	225		yes
Takko Fashion S.a r.l.	Germany	2023.05	Distressed exchange	562	88	yes
Technicolor Creative Studios SA	France	2023.06	Distressed exchange		706	yes
Tele Columbus AG	Germany	2023.12	Payment default	709		yes
Telesat Canada	Canada	2023.05	Distressed exchange	459	105	yes
Toro Private Holdings II, Limited	United Kingdom	2023.03	Distressed exchange		2,100	yes
Toro Private Holdings II, Limited	United Kingdom	2023.12	Distressed exchange		3,826	no
TortoiseEcofin Borrower LLC	United States	2023.10	Distressed exchange		300	yes
Tullow Oil plc	United Kingdom	2023.06	Distressed exchange	167		yes
U.S. Renal Care, Inc	United States	2023.08	Distressed exchange	344	1,758	yes
U.S. TelePacific Corp.	United States	2023.06	Distressed exchange		639	yes
URS Holdco, Inc.	United States	2023.04	Distressed exchange		335	yes
Venator Materials LLC	United Kingdom	2023.05	Bankruptcy	600	351	yes
Vericast Corp.	United States	2023.12	Distressed exchange		785	yes
Vue International Bidco plc	United Kingdom	2023.01	Distressed exchange	285	260	yes
Wahoo Fitness Acquisition L.L.C.	United States	2023.04	Payment default		247	yes
Wanda Commercial Properties (HK) Co. Limited	China	2023.12	Distressed exchange	600		yes
Wellful Inc.	United States	2023.12	Distressed exchange		125	yes
Werner FinCo LP	United States	2023.06	Distressed exchange	251		yes
Wesco Aircraft Holdings, Inc.	United States	2023.06	Bankruptcy	3,821		yes
Western Global Airlines, Inc.	United States	2023.08	Bankruptcy	400	48	yes

Source: Moody's Investors Service

Company	Domicile	Default month	Broad default type	Bond ($mil)	Loan ($mil)		in Jan 2023 cohort?
Wheel Pros, Inc.	United States	2023.09	Distressed exchange		1,154	yes	
Wittur Holding GmbH	Germany	2023.03	Payment default		262	yes	
Yak Access, LLC	United States	2023.03	Distressed exchange		876	yes	
Yellow Corporation	United States	2023.08	Bankruptcy	2	1,245	yes	
Zensun Enterprises Limited	China	2023.09	Distressed exchange	273		yes	
Zensun Group Limited	China	2023.09	Distressed exchange	0		yes	

* This list includes companies that initially defaulted in 2023 and they must have rated bonds, loans and/or deposits within one year before default. Only issuers included in the January 1 cohort are included in the annual default rates in this report. In addition to the 159 companies listed above, nine issuers that initially defaulted in prior years experienced follow-on defaults in 2023 on $4,824 million of bonds and $406 million of loans. We include the amount associated with these follow-on defaults in the 2023 default volume.
Source: Moody's Investors Service

Exhibit 22

Annual Moody's-rated global corporate issuer default counts, 1920-2023

Year	IG	SG	All Ratings	Year	IG	SG	All Ratings	Year	IG	SG	All Ratings
1920	8	25	33	1955	0	1	1	1990	1	88	89
1921	7	25	32	1956	0	0	0	1991	1	68	69
1922	10	23	33	1957	0	1	1	1992	0	33	33
1923	5	22	27	1958	0	0	0	1993	0	23	23
1924	3	37	40	1959	0	0	0	1994	0	19	19
1925	7	34	41	1960	0	2	2	1995	0	29	29
1926	4	18	22	1961	0	3	3	1996	0	18	18
1927	1	15	16	1962	0	4	4	1997	0	25	25
1928	0	8	8	1963	0	3	3	1998	1	50	51
1929	3	12	15	1964	0	0	0	1999	1	106	107
1930	2	21	23	1965	0	0	0	2000	4	125	129
1931	6	78	84	1966	0	1	1	2001	4	179	183
1932	10	108	118	1967	0	0	0	2002	12	136	148
1933	9	189	198	1968	0	1	1	2003	0	91	91
1934	5	60	65	1969	0	0	0	2004	0	39	39
1935	9	51	60	1970	2	25	27	2005	2	31	33
1936	3	19	22	1971	0	3	3	2006	0	31	31
1937	4	18	22	1972	0	5	5	2007	0	20	20
1938	9	17	26	1973	2	3	5	2008	21	119	140
1939	2	13	15	1974	0	3	3	2009	14	258	272
1940	2	22	24	1975	0	4	4	2010	3	62	65
1941	0	10	10	1976	0	2	2	2011	6	44	50
1942	0	4	4	1977	1	3	4	2012	1	68	69
1943	0	3	3	1978	0	4	4	2013	3	68	71
1944	0	3	3	1979	0	1	1	2014	2	56	58
1945	0	2	2	1980	0	4	4	2015	0	109	109
1946	0	0	0	1981	0	2	2	2016	0	140	140
1947	0	2	2	1982	2	11	13	2017	0	106	106
1948	0	0	0	1983	0	14	14	2018	0	74	74
1949	0	5	5	1984	2	11	13	2019	2	96	98
1950	0	0	0	1985	0	16	16	2020	2	205	207
1951	0	1	1	1986	3	33	36	2021	0	51	51
1952	0	0	0	1987	0	31	31	2022	17	135	152
1953	0	0	0	1988	0	32	32	2023	4	143	147
1954	0	1	1	1989	4	50	54				

* Only includes companies in January 1 cohort.
Source: Moody's Investors Service

Exhibit 23

Annual rated global corporate bond and loan default volumes, 1970-2023 (in $ million)

	Investment-grade			Speculative-grade			All Ratings		
Year	Bond	Loan	Total	Bond	Loan	Total	Bond	Loan	Total
1970	154	0	154	756	0	756	910	0	910
1971	0	0	0	132	0	132	132	0	132
1972	0	0	0	215	0	215	215	0	215
1973	17	0	17	94	0	94	112	0	112
1974	0	0	0	69	0	69	69	0	69
1975	0	0	0	176	0	176	176	0	176
1976	0	0	0	34	0	34	34	0	34
1977	68	0	68	179	0	179	247	0	247
1978	0	0	0	112	0	112	112	0	112
1979	0	0	0	18	0	18	18	0	18
1980	0	0	0	302	0	302	302	0	302
1981	0	0	0	47	0	47	47	0	47
1982	243	0	243	515	0	515	758	0	758
1983	0	0	0	1,187	0	1,187	1,187	0	1,187
1984	215	0	215	386	0	386	601	0	601
1985	0	0	0	1,354	0	1,354	1,354	0	1,354
1986	138	0	138	4,004	0	4,004	4,142	0	4,142
1987	0	0	0	8,941	277	9,217	8,941	277	9,217
1988	0	0	0	5,642	361	6,003	5,642	361	6,003
1989	1,506	0	1,506	9,657	0	9,657	11,162	0	11,162
1990	0	0	0	20,429	1,831	22,260	20,429	1,831	22,260
1991	1,348	0	1,348	15,546	678	16,224	16,894	678	17,572
1992	0	0	0	6,667	956	7,623	6,667	956	7,623
1993	0	0	0	2,617	487	3,103	2,617	487	3,103
1994	0	0	0	2,713	299	3,013	2,713	299	3,013
1995	0	0	0	5,401	372	5,773	5,401	372	5,773
1996	0	0	0	4,106	1,435	5,541	4,106	1,435	5,541
1997	0	0	0	5,024	948	5,972	5,024	948	5,972
1998	399	0	399	9,469	3,299	12,768	9,868	3,299	13,167
1999	461	1,225	1,686	26,217	14,054	40,270	26,678	15,279	41,956
2000	4,115	3,950	8,065	25,516	26,206	51,721	29,631	30,156	59,787
2001	21,195	5,363	26,558	78,505	34,103	112,609	99,701	39,466	139,167
2002	46,350	13,158	59,508	104,454	41,121	145,575	150,804	54,279	205,083
2003	0	870	870	37,626	10,308	47,935	37,626	11,178	48,804
2004	0	0	0	13,152	4,712	17,865	13,152	4,712	17,865
2005	2,155	2,825	4,980	26,923	8,745	35,668	29,078	11,570	40,648
2006	0	0	0	8,795	2,630	11,425	8,795	2,630	11,425
2007	0	0	0	6,614	1,983	8,597	6,614	1,983	8,597
2008	182,073	10,380	192,452	51,362	32,451	83,814	233,435	42,831	276,266
2009	35,704	21,931	57,634	149,189	122,495	271,683	184,892	144,425	329,318
2010	4,047	0	4,047	22,256	14,042	36,298	26,303	14,042	40,344
2011	6,561	1,411	7,972	22,696	4,906	27,602	29,257	6,317	35,574
2012	323	0	323	35,001	18,289	53,290	35,324	18,289	53,613
2013	2,408	0	2,408	28,949	13,805	42,754	31,358	13,805	45,162
2014	939	0	939	40,392	30,370	70,762	41,331	30,370	71,701
2015	0	0	0	75,483	19,794	95,277	75,483	19,794	95,277
2016	0	0	0	90,823	43,648	134,472	90,823	43,648	134,472
2017	0	0	0	57,688	27,397	85,085	57,688	27,397	85,085
2018	0	0	0	49,468	25,892	75,360	49,468	25,892	75,360
2019	17,573	3,855	21,428	55,414	33,437	88,851	72,988	37,292	110,279
2020	756	1,766	2,522	127,067	99,430	226,497	127,823	101,196	229,019
2021	0	0	0	41,054	11,918	52,971	41,054	11,918	52,971
2022	17,522	0	17,522	96,155	33,798	129,953	113,677	33,798	147,475
2023	4,675	0	4,675	67,135	66,230	133,366	71,810	66,230	138,041

* Only include companies in Jan. 1 cohort.
Source: Moody's Investors Service

Exhibit 24

Annual issuer default counts and volume by geographical region, 1986-2023

	Default count					Default Volume (in $mil)				
Year	Africa & Middle East	Asia Pacific	Europe	Latin America	North America	Africa & Middle East	Asia Pacific	Europe	Latin America	North America
1986	0	0	0	0	36	$0	$0	$0	$0	$4,142
1987	0	0	0	0	32	$0	$0	$0	$0	$9,297
1988	0	0	0	0	32	$0	$0	$0	$0	$6,003
1989	0	4	0	0	51	$0	$969	$0	$0	$10,253
1990	0	1	1	0	90	$0	$200	$516	$0	$22,097
1991	0	0	2	0	71	$0	$0	$1,228	$0	$16,753
1992	0	0	0	0	34	$0	$0	$0	$0	$7,673
1993	0	0	0	0	24	$0	$0	$0	$0	$3,104
1994	0	0	1	0	18	$0	$0	$774	$0	$2,238
1995	0	0	0	1	30	$0	$0	$0	$100	$5,774
1996	0	0	0	1	19	$0	$0	$0	$207	$5,930
1997	0	2	0	2	21	$0	$614	$0	$310	$5,048
1998	0	4	5	0	44	$0	$1,444	$707	$0	$11,241
1999	0	8	9	8	88	$0	$4,458	$3,950	$2,018	$33,299
2000	0	4	4	4	118	$0	$4,453	$866	$455	$54,080
2001	0	11	15	7	150	$0	$10,088	$10,540	$2,034	$116,505
2002	2	2	25	17	102	$150	$1,842	$47,555	$10,370	$145,167
2003	0	3	7	12	70	$0	$625	$2,927	$4,933	$40,444
2004	1	0	5	0	37	$183	$0	$2,311	$366	$15,775
2005	0	0	1	1	32	$0	$0	$207	$100	$40,445
2006	0	0	7	2	24	$0	$0	$1,601	$167	$10,149
2007	0	0	4	0	16	$0	$0	$2,230	$0	$6,367
2008	0	3	52	3	93	$0	$625	$53,374	$705	$227,926
2009	1	16	31	11	225	$650	$6,989	$28,413	$3,031	$295,225
2010	0	3	10	4	54	$0	$6,913	$9,713	$487	$25,384
2011	0	0	16	1	36	$0	$0	$10,102	$358	$27,149
2012	1	1	13	5	51	$48	$1,805	$17,441	$3,077	$33,780
2013	0	2	26	10	36	$0	$546	$24,263	$4,880	$23,091
2014	1	5	16	5	32	$350	$4,000	$9,974	$4,539	$53,092
2015	1	9	30	10	69	$456	$4,114	$15,950	$4,314	$75,157
2016	2	2	21	12	110	$945	$1,806	$16,248	$26,135	$91,785
2017	1	7	30	3	67	$553	$2,336	$20,684	$19,478	$42,504
2018	0	5	13	5	57	$0	$4,161	$7,950	$8,559	$57,417
2019	3	5	13	3	81	$0	$2,549	$7,418	$7,800	$101,792
2020	3	17	40	13	145	$2,845	$13,053	$41,597	$9,413	$169,042
2021	0	15	10	6	24	$0	$30,869	$4,397	$4,103	$19,959
2022	1	24	89	4	39	$0	$55,953	$46,175	$1,000	$49,944
2023	0	12	31	9	107	$0	$19,551	$20,399	$9,802	$100,518

* Includes defaults outside of January 1 cohort.
Source: Moody's Investors Service

Exhibit 25

2023 defaulted corporate bond and loan recoveries*

		Loan		Bond					
Company	Broad default type	1st Lien	2nd Lien	1st Lien	2nd Lien	Sr. Unsec.	Sr. Sub	Sub	Jr. Sub
4L Holdings Corporation	distressed exchange	90.0%							
Accuride Corporation	distressed exchange	82.6%							
Agung Podomoro Land Tbk (P.T.)	distressed exchange					58.0%			
Air Methods Corporation	bankruptcy	14.8%				1.3%			
Akorn Operating Company LLC	bankruptcy	25.3%							
Akumin Inc.	bankruptcy			78.3%					
Altisource S.a.r.l.	distressed exchange	81.4%							
AMC Entertainment Holdings, Inc. (1)	distressed exchange				59.0%				
AMC Entertainment Holdings, Inc. (2)	distressed exchange				85.3%		63.5%		
Americanas S.A.	payment default					16.3%			
Amphora Intermediate II Ltd	payment default	41.0%							
AmSurg, LLC	bankruptcy	95.0%	55.5%						
Anchor Glass Container Corporation	payment default	81.0%	41.1%						
Aston Midco Limited	distressed exchange	81.7%							
At Home Group Inc.	distressed exchange					96.0%			
Atento Luxco 1	payment default			0.0%					
Audacy, Inc.	distressed exchange				1.0%				
Avaya Holdings Corp.	bankruptcy					7.3%			
Avaya Inc.	bankruptcy	24.5%		25.5%					
Azul S.A.	distressed exchange					82.4%			
Bed Bath & Beyond Inc.	bankruptcy					2.6%			
Bering III S.a r.l.	distressed exchange	82.4%							
BIM Land Joint Stock Company	distressed exchange					49.5%			
Bioplan USA, Inc.	distressed exchange	65.4%	42.1%						
CareerBuilder, LLC	distressed exchange	67.0%							
Carvana Co.	distressed exchange					71.5%			
Casa Systems, Inc.	distressed exchange	90.9%							
Casino Guichard-Perrachon SA	payment default					2.0%			0.8%
CATALINA MARKETING CORPORATION	payment default	75.8%							
CatLuxe S.a.r.l.	distressed exchange	47.5%							
Central China Real Estate Limited	distressed exchange					14.8%			
Checkers Holdings, Inc.	payment default	42.1%							
CIBT Global, Inc.	distressed exchange	72.3%	54.3%						
ClubCorp Holdings, Inc.	distressed exchange					84.8%			
CNG Holdings, Inc.	distressed exchange			85.8%					
Codere Luxembourg 2 S.a.r.l.	payment default			85.3%					
ColourOz MidCo	payment default	67.3%	30.0%						
Comet Bidco Limited	distressed exchange	91.4%							
Cooper-Standard Automotive Inc.	distressed exchange					45.4%			
Country Garden Holdings Company Limited	distressed exchange					7.9%			
Covis Midco 2 S.a r.l. (1)	distressed exchange	57.8%	7.5%						
Covis Midco 2 S.a r.l. (2)	payment default	38.8%							
CSTN Merger Sub, Inc.	distressed exchange			87.1%					
Curo Group Holdings Corp.	distressed exchange			37.6%					
Cyxtera DC Holdings, Inc.	bankruptcy	50.9%							
David's Bridal, Inc.	bankruptcy	35.0%							
Dawn Acquisitions LLC	distressed exchange	76.9%							

		Loan		Bond					
Company	Broad default type	1st Lien	2nd Lien	1st Lien	2nd Lien	Sr. Unsec.	Sr. Sub	Sub	Jr. Sub
Diamond Sports Group, LLC	bankruptcy	93.4%	5.8%		5.5%	1.5%			
Diebold Nixdorf, Inc. (Old)	bankruptcy	17.4%		18.0%	0.8%	2.0%			
Digicel Group Holdings Limited	payment default					43.5%		5.0%	
Digicel International Finance Limited	payment default					46.0%		18.0%	
Digital Media Solutions, LLC	distressed exchange	74.7%							
DTEK ENERGY B.V. (1)	distressed exchange					30.5%			
DTEK ENERGY B.V. (2)	distressed exchange					56.4%			
Eagle Intermediate Global Holding B.V.	distressed exchange			80.0%					
Elevate Textiles, Inc.	distressed exchange	47.5%	6.9%						
Envision Healthcare Corporation	bankruptcy	5.1%				1.4%			
Equinox Holdings, Inc.	distressed exchange	77.8%							
Exela Intermediate LLC	distressed exchange			13.8%					
FDS Holdings BV	distressed exchange	67.5%							
FINThrive Software Intermediate Hldgs, Inc.	distressed exchange		64.1%						
First Republic Bank	bankruptcy							1.0%	
Foodco Bondco, S.A.U.	payment default			42.9%					
Forming Machining Industries Holdings, LLC	distressed exchange	72.5%							
Frigoglass SAIC	payment default			27.3%					
FXI Holdings, Inc.	distressed exchange			90.8%					
Genesis Care Finance Pty Ltd	bankruptcy	12.9%							
Gol Linhas Aereas Inteligentes S.A.	distressed exchange					46.3%			
Greenland Holding Group Company Limited	payment default					15.9%			
Haya Holdco 2 plc	distressed exchange			38.2%					
HealthChannels Intermediate Holdco, LLC	distressed exchange	74.7%							
Ideal Standard International S.A.	distressed exchange			61.8%					
Instant Brands Holdings Inc.	bankruptcy	22.1%							
JP Intermediate B, LLC	distressed exchange	46.9%							
K&N Parent, Inc.	distressed exchange	42.3%							
Kaisa Group Holdings Ltd	payment default					14.0%			
Keter Group B.V.	payment default	83.0%							
Lannett Company, Inc.	payment default			7.3%		1.0%			
Learfield Communications, LLC	distressed exchange	77.0%	62.8%						
LendingTree, Inc.	distressed exchange					78.8%			
LifeScan Global Corporation	distressed exchange		62.5%						
Ligado Networks LLC	payment default			18.5%					
Light Servicos De Eletricidade S.A.	payment default					38.2%			
Lippo Malls Indonesia Retail Trust	distressed exchange					72.2%			
Logan Group Company Limited	payment default					29.8%			
Loyalty Ventures Inc.	bankruptcy	10.9%							
Lucky Bucks, LLC	payment default	32.1%							
Mallinckrodt International Finance S.A.	payment default	75.7%		86.1%	16.3%				
Matrix Parent, Inc.	payment default	67.9%	35.2%						
Maverick Gaming LLC	distressed exchange	68.5%							
MHP SE	distressed exchange					85.0%			
Missouri TopCo Limited	distressed exchange				52.0%				
MLN US Holdco LLC	distressed exchange	28.0%							
Monitronics International, Inc.	bankruptcy	69.2%							

		Loan		Bond					
Company	Broad default type	1st Lien	2nd Lien	1st Lien	2nd Lien	Sr. Unsec.	Sr. Sub	Sub	Jr. Sub
Moran Foods LLC	distressed exchange	87.5%	72.5%						
MVK Intermediate Holdings, LLC (1)	distressed exchange	68.4%							
MVK Intermediate Holdings, LLC (2)	bankruptcy	24.7%							
National CineMedia, LLC	payment default	33.6%		34.5%		2.4%			
NEP Europe Finco B.V.	distressed exchange	86.4%							
NEP/NCP Holdco, Inc	distressed exchange	95.6%							
New Constellis Borrower LLC	distressed exchange	86.1%	58.2%						
Outerstuff LLC	payment default	87.8%							
Paper Industries Intermediate Fin. S.a r.l.	distressed exchange			77.1%					
Party City Holdings Inc.	bankruptcy			16.2%		0.5%			
Polaris Newco, LLC	distressed exchange		94.8%						
Powerlong Real Estate Holdings Limited	distressed exchange					11.5%			
Praesidiad Group Limited	distressed exchange	53.0%							
Pretium PKG Holdings, Inc.	distressed exchange	75.1%	45.4%						
QualTek LLC	distressed exchange	78.6%							
Quorum Health Corporation	distressed exchange	68.5%							
Rackspace Technology Global, Inc.	distressed exchange					28.0%			
Resolute Investment Managers, Inc.	distressed exchange	64.8%	20.0%						
Rising Tide Holdings, Inc. (1)	distressed exchange	59.0%	27.5%						
Rising Tide Holdings, Inc. (2)	distressed exchange	7.0%	3.0%						
Rite Aid Corporation	bankruptcy			75.1%		5.0%			
Riverbed Technology LLC	payment default	31.7%							
Robertshaw US Holding Corp (NEW)	distressed exchange	54.2%	22.3%						
Rodan & Fields, LLC	distressed exchange	46.5%							
Serta Simmons Bedding, LLC	bankruptcy	55.8%							
Shimao Group Holdings Limited	payment default					15.4%			
SHO Holding I Corporation	distressed exchange		71.0%						
Shutterfly, LLC	distressed exchange	60.9%		46.8%					
Sierra Enterprises LLC	payment default	72.0%							
Signature Bank	bankruptcy							2.8%	
Sino-Ocean Group Holding Limited	payment default					18.3%			
Skillz Inc.	distressed exchange			69.5%					
Solocal Group S.A.	payment default			37.0%					
SP PF Buyer LLC	distressed exchange	78.0%							
Strategic Materials Holding Corp.	payment default	63.6%	3.3%						
SVB Financial Group	bankruptcy					57.4%			
Tacora Resources Inc.	distressed exchange			76.3%					
Takko Fashion S.a r.l.	distressed exchange			72.2%					
Tele Columbus AG	payment default			65.3%					
Telesat Canada	distressed exchange	73.8%		60.6%		38.2%			
Toro Private Holdings II, Limited (1)	distressed exchange	57.3%							
Toro Private Holdings II, Limited (2)	distressed exchange	70.6%							
TortoiseEcofin Borrower LLC	distressed exchange	42.8%							
Tullow Oil plc	distressed exchange					60.3%			
U.S. Renal Care, Inc	distressed exchange	50.8%				39.3%			
U.S. Telepacific Corp.	distressed exchange	40.0%							
URS Holdco, Inc.	distressed exchange	33.3%							

		Loan		Bond					
Company	Broad default type	1st Lien	2nd Lien	1st Lien	2nd Lien	Sr. Unsec.	Sr. Sub	Sub	Jr. Sub
Venator Materials LLC	bankruptcy	55.4%				12.0%			
Vericast Corp.	distressed exchange	93.5%							
Vue International Bidco plc	distressed exchange	55.0%							
Wahoo Fitness Acquisition L.L.C.	payment default	42.4%							
Wanda Commercial Properties (HK) Co. Limited	distressed exchange					78.5%			
Werner FinCo LP	distressed exchange					88.1%			
Wesco Aircraft Holdings, Inc.	bankruptcy			90.0%		7.3%			
Western Global Airlines, Inc.	bankruptcy					0.3%			
WeWork Cos US LLC	bankruptcy					0.5%			
Wheel Pros, Inc.	distressed exchange	66.9%							
Wittur Holding GmbH	payment default		68.0%						
Yak Access, LLC	distressed exchange	45.4%	6.6%						
YRC Worldwide Inc.	bankruptcy	91.5%							
Zensun Enterprises Limited	distressed exchange					37.6%			

*Based on trading prices
Source: Moody's Investors Service

Exhibit 26

Annual defaulted corporate bond and loan recoveries*

	Loan	Bond					
Year	1st Lien	1st Lien	Sr. Unsec.	Sr. Sub.	Sub.	Jr. Sub.	All Bonds
1983		40.0%	52.7%	43.5%	41.1%		44.5%
1984			49.4%	67.9%	44.3%		45.5%
1985		83.6%	60.2%	29.6%	39.7%	48.5%	43.6%
1986		59.2%	49.5%	46.8%	41.4%		47.2%
1987		71.0%	63.8%	46.5%	46.9%		51.3%
1988		55.4%	45.2%	31.4%	33.8%	36.5%	38.5%
1989		46.5%	43.6%	35.7%	26.8%	16.9%	32.5%
1990	72.0%	33.8%	38.2%	25.5%	19.5%	10.7%	25.8%
1991	67.9%	50.2%	36.6%	41.8%	24.4%	7.8%	35.5%
1992	60.6%	62.1%	49.2%	49.4%	38.0%	13.5%	45.9%
1993	53.4%		37.1%	51.9%	44.1%		43.1%
1994	67.6%	69.3%	53.7%	29.6%	38.0%	40.0%	45.6%
1995	75.4%	62.0%	47.6%	34.3%	41.5%		43.3%
1996	85.5%	47.6%	62.8%	43.8%	22.6%		41.5%
1997	81.3%	72.0%	56.1%	44.7%	33.1%	30.6%	47.6%
1998	59.9%	46.8%	39.5%	45.0%	18.2%	62.0%	38.3%
1999	73.6%	39.1%	38.0%	26.9%	35.6%		34.3%
2000	68.8%	40.1%	24.2%	20.8%	31.9%	15.5%	25.4%
2001	64.9%	33.6%	21.2%	19.8%	15.9%	47.0%	21.7%
2002	58.4%	50.8%	29.6%	21.4%	23.4%		29.9%
2003	73.4%	63.0%	41.9%	37.8%	12.3%		40.8%
2004	87.7%	73.3%	52.1%	42.3%	94.0%		58.5%
2005	83.8%	69.2%	54.9%	32.8%	51.3%		56.5%
2006	83.6%	74.6%	55.0%	41.4%	56.1%		55.0%
2007	68.6%	86.1%	53.7%	56.2%			57.4%
2008	61.7%	52.5%	33.5%	23.3%	29.5%		34.1%
2009	53.6%	37.3%	36.7%	23.1%	45.3%	8.5%	33.9%
2010	70.9%	57.6%	48.2%	37.5%	33.7%		51.1%
2011	69.9%	70.5%	43.6%	42.8%	31.9%		46.7%
2012	66.4%	57.6%	43.3%	29.9%	40.3%		44.5%
2013	76.2%	68.8%	45.1%	20.7%	26.4%		46.2%
2014	78.4%	73.6%	47.0%	39.1%	38.8%		48.5%
2015	63.7%	54.8%	37.6%	36.6%	58.5%	14.0%	41.2%
2016	66.2%	49.6%	32.2%	36.7%	24.5%	0.6%	37.2%
2017	69.6%	65.4%	54.3%	38.0%	42.6%	17.5%	56.1%
2018	69.4%	58.8%	47.2%	45.6%			51.4%
2019	57.9%	60.3%	32.3%	64.7%	40.0%		41.2%
2020	59.8%	52.5%	34.6%	14.1%			38.4%
2021	66.4%	65.9%	48.7%	17.0%			52.1%
2022	69.5%	58.2%	30.9%		18.2%		35.8%
2023	59.6%	53.5%	33.4%	63.5%	6.7%	0.8%	40.8%

* Measured by trading prices.
Source: Moody's Investors Service

Exhibit 27

Average defaulted corporate bond and loan recoveries by industry, 1983-2023**

	Recoveries					Observation Counts				
	Loan	Bond				Loan	Bond			
Industry groups	1st Lien	1st Lien	Sr. Unsec.	Sr. Sub.	Sub.	1st Lien	1st Lien	Sr. Unsec.	Sr. Sub.	Sub.
Aerospace & Defense	68.3%	84.8%	45.5%	23.3%	40.8%	7	2	11	9	7
Automotive	68.5%	57.9%	47.6%	23.9%	49.5%	41	4	38	28	5
Banking		7.0%	41.5%	20.0%	30.3%	0	1	59	7	69
Beverage, Food, & Tobacco	75.5%	55.5%	42.0%	27.5%	34.0%	24	16	33	19	10
Capital Equipment	62.7%	60.7%	33.2%	34.2%	39.6%	32	11	29	30	15
Chemicals, Plastics, & Rubber	66.7%	51.0%	28.6%	23.6%	63.6%	14	20	27	12	2
Construction & Building	58.1%	62.6%	36.0%	25.3%	28.5%	29	9	108	24	20
Consumer goods: Durable	61.0%	53.4%	34.0%	28.9%	29.4%	15	7	12	15	10
Consumer goods: Non-durable	62.0%	41.0%	32.7%	33.2%	38.6%	43	7	30	37	11
Containers, Packaging, & Glass	73.0%	63.2%	52.1%	36.5%	42.0%	8	9	12	11	3
Energy: Electricity	82.3%	82.0%	58.8%	78.9%	22.3%	7	14	28	2	4
Energy: Oil & Gas	68.1%	56.3%	36.9%	38.0%	39.1%	66	45	177	28	36
Environmental Industries	61.3%	13.0%	29.7%	27.1%	13.6%	9	1	7	6	3
Finance*	68.7%	51.6%	42.9%	36.8%	28.9%	13	7	41	11	15
Forest Products & Paper	74.8%	52.2%	37.3%	20.5%	1.5%	10	16	36	9	3
Healthcare & Pharmaceuticals	62.8%	72.8%	37.1%	33.0%	35.3%	44	9	33	30	30
High Tech Industries	59.4%	52.7%	35.8%	26.1%	23.2%	27	16	28	22	38
Hotel, Gaming, & Leisure	69.4%	56.3%	37.5%	32.3%	26.0%	43	46	45	37	17
Insurance	86.2%	96.3%	37.9%	46.0%	43.2%	2	1	10	4	6
Media: Advertising, Printing & Publishing	57.7%	49.1%	28.4%	22.0%	50.9%	37	21	20	23	4
Media: Broadcasting & Subscription	66.1%	41.9%	41.1%	39.1%	29.1%	21	14	51	29	13
Media: Diversified & Production	65.5%	20.8%	36.4%	32.9%	45.1%	6	3	8	3	3
Metals & Mining	58.7%	41.4%	34.9%	32.1%	45.8%	18	39	76	10	11
REIT	67.8%	44.0%	42.8%	13.0%	27.0%	6	2	19	3	7
Retail	60.3%	59.4%	35.4%	34.4%	22.2%	72	39	89	46	27
Services: Business	69.2%	63.4%	47.3%	28.9%	41.5%	56	27	29	11	6
Services: Consumer	65.4%	35.2%	57.8%	49.1%		15	2	8	5	0
Sovereign & Public Finance		31.4%	50.8%			0	3	7	0	0
Telecommunications	59.4%	53.6%	25.6%	31.0%	29.5%	44	23	146	20	19
Transportation: Cargo	67.4%	51.6%	41.9%	35.4%	32.5%	21	19	36	12	5
Transportation: Consumer	84.0%	53.2%	35.7%	43.6%	28.6%	4	10	33	3	14
Utilities: Electric	100.8%	60.6%	53.5%		27.1%	1	11	22	0	2
Utilities: Oil & Gas		85.0%	72.6%		57.5%	0	1	3	0	1
Utilities: Water			82.1%			0	0	2	0	0
Wholesale	68.4%	53.8%	37.9%	14.1%	37.0%	17	4	15	9	5

* Non-bank and non-insurance financial companies

** Measured by trading prices.

Source: Moody's Investors Service

Exhibit 28

Average senior unsecured bond recovery rates by year before default, 1983-2023

Rating\Year	1	2	3	4	5
Aaa**		3.3%	3.3%	61.9%	69.6%
Aa	37.2%	39.0%	38.1%	44.0%	43.2%
A	37.7%	46.2%	47.4%	46.8%	46.2%
Baa	38.4%	40.3%	40.9%	41.2%	41.3%
Ba	39.9%	39.9%	39.6%	39.5%	39.3%
B	36.4%	36.0%	36.7%	37.3%	37.9%
Caa-C	38.2%	38.5%	38.3%	38.3%	38.2%
IG	38.2%	41.5%	42.6%	43.0%	43.0%
SG	37.8%	37.7%	37.8%	38.0%	38.2%
All	37.8%	37.9%	38.1%	38.4%	38.6%

* Issuer-weighted, based on post default trading prices.
** The Aaa recovery rates are based on five observations, three of which are Icelandic banks that have an average recovery rate of 3.33%.
Source: Moody's Investors Service

Exhibit 29

Annual credit loss rates by letter rating, 1983-2023

Year	Aaa	Aa	A	Baa	Ba	B	Caa-C	IG	SG	All
1983	0.00%	0.00%	0.00%	0.00%	0.55%	1.05%	21.30%	0.00%	2.02%	0.45%
1984	0.00%	0.00%	0.00%	0.31%	0.26%	2.61%	9.20%	0.09%	1.56%	0.43%
1985	0.00%	0.00%	0.00%	0.00%	0.35%	2.83%	2.49%	0.00%	1.48%	0.37%
1986	0.00%	0.00%	0.00%	0.43%	1.20%	5.20%	8.17%	0.11%	3.08%	0.91%
1987	0.00%	0.00%	0.00%	0.00%	1.10%	1.94%	3.56%	0.00%	1.55%	0.51%
1988	0.00%	0.00%	0.00%	0.00%	0.74%	3.07%	6.85%	0.00%	2.04%	0.73%
1989	0.00%	0.28%	0.00%	0.29%	1.68%	4.25%	11.51%	0.14%	3.33%	1.25%
1990	0.00%	0.00%	0.00%	0.16%	2.18%	8.46%	27.86%	0.04%	6.45%	2.17%
1991	0.00%	0.00%	0.00%	0.16%	2.43%	8.32%	10.12%	0.04%	5.76%	1.76%
1992	0.00%	0.00%	0.00%	0.00%	0.17%	3.88%	7.97%	0.00%	2.51%	0.68%
1993	0.00%	0.00%	0.00%	0.00%	0.39%	2.73%	8.77%	0.00%	2.14%	0.56%
1994	0.00%	0.00%	0.00%	0.00%	0.00%	1.94%	2.46%	0.00%	1.08%	0.30%
1995	0.00%	0.00%	0.00%	0.00%	0.14%	2.00%	6.05%	0.00%	1.61%	0.47%
1996	0.00%	0.00%	0.00%	0.00%	0.00%	0.56%	3.69%	0.00%	0.62%	0.19%
1997	0.00%	0.00%	0.00%	0.00%	0.08%	0.88%	3.99%	0.00%	0.83%	0.27%
1998	0.00%	0.00%	0.00%	0.07%	0.46%	2.23%	5.36%	0.02%	1.80%	0.67%
1999	0.00%	0.00%	0.00%	0.06%	0.86%	3.12%	9.32%	0.02%	3.30%	1.30%
2000	0.00%	0.00%	0.00%	0.26%	1.09%	4.27%	13.45%	0.10%	4.66%	1.87%
2001	0.00%	0.00%	0.12%	0.14%	0.92%	6.88%	22.28%	0.10%	7.34%	2.78%
2002	0.00%	0.00%	0.11%	0.59%	1.25%	3.22%	18.59%	0.26%	5.49%	2.06%
2003	0.00%	0.00%	0.00%	0.00%	0.52%	1.49%	11.89%	0.00%	3.09%	1.07%
2004	0.00%	0.00%	0.00%	0.00%	0.18%	0.33%	5.22%	0.00%	1.10%	0.38%
2005	0.00%	0.00%	0.00%	0.07%	0.00%	0.32%	3.28%	0.03%	0.78%	0.29%
2006	0.00%	0.00%	0.00%	0.00%	0.09%	0.49%	2.47%	0.00%	0.75%	0.26%
2007	0.00%	0.00%	0.00%	0.00%	0.00%	0.05%	2.21%	0.00%	0.46%	0.17%
2008	0.00%	0.33%	0.27%	0.67%	1.55%	2.68%	7.11%	0.41%	3.64%	1.67%
2009	0.00%	0.00%	0.15%	0.59%	1.12%	4.30%	16.70%	0.27%	7.67%	3.15%
2010	0.00%	0.00%	0.09%	0.04%	0.00%	0.20%	4.47%	0.05%	1.59%	0.64%
2011	0.00%	0.11%	0.00%	0.20%	0.09%	0.19%	3.51%	0.11%	1.18%	0.53%
2012	0.00%	0.00%	0.00%	0.04%	0.08%	0.31%	4.54%	0.02%	1.60%	0.71%
2013	0.00%	0.00%	0.05%	0.07%	0.33%	0.55%	3.46%	0.05%	1.48%	0.69%
2014	0.00%	0.00%	0.05%	0.03%	0.08%	0.27%	2.52%	0.03%	1.08%	0.52%
2015	0.00%	0.00%	0.00%	0.00%	0.19%	1.51%	4.13%	0.00%	2.31%	1.09%
2016	0.00%	0.00%	0.00%	0.00%	0.10%	1.14%	6.17%	0.00%	3.14%	1.49%
2017	0.00%	0.00%	0.00%	0.00%	0.24%	0.21%	3.45%	0.00%	1.64%	0.78%
2018	0.00%	0.00%	0.00%	0.00%	0.00%	0.36%	2.73%	0.00%	1.30%	0.62%
2019	0.00%	0.00%	0.00%	0.08%	0.00%	0.81%	4.41%	0.04%	2.16%	1.04%
2020	0.00%	0.00%	0.00%	0.07%	0.09%	2.60%	8.39%	0.04%	4.52%	2.11%
2021	0.00%	0.00%	0.00%	0.00%	0.00%	0.68%	1.52%	0.00%	0.90%	0.41%
2022	0.00%	0.00%	0.00%	0.63%	2.88%	3.55%	2.65%	0.33%	2.96%	1.57%
2023	0.00%	0.00%	0.10%	0.07%	0.49%	0.48%	6.10%	0.08%	3.22%	1.52%

* Based on issuer-weighted annual default rates and senior unsecured bond recoveries measured on an issuer-weighted basis.
Source: Moody's Investors Service

Exhibit 30

2023 one-year letter rating migration rates

From\To	Aaa	Aa	A	Baa	Ba	B	Caa	Ca_C	WR	Def
Aaa	98.00%	2.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	0.00%	96.55%	1.25%	0.00%	0.00%	0.00%	0.00%	0.00%	2.19%	0.00%
A	0.00%	0.98%	93.23%	2.26%	0.00%	0.00%	0.00%	0.00%	3.39%	0.15%
Baa	0.00%	0.00%	2.47%	91.40%	1.72%	0.00%	0.00%	0.00%	4.30%	0.11%
Ba	0.00%	0.00%	0.00%	3.72%	85.12%	3.86%	1.57%	0.14%	4.86%	0.72%
B	0.00%	0.00%	0.00%	0.00%	5.08%	81.41%	6.93%	0.12%	5.77%	0.69%
Caa	0.00%	0.00%	0.00%	0.00%	0.14%	3.14%	78.19%	3.85%	9.76%	4.92%
Ca_C	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	3.28%	32.79%	12.30%	51.64%

Source: Moody's Investors Service

Exhibit 31

Average one-year letter rating migration rates, 1920-2023

From\To	Aaa	Aa	A	Baa	Ba	B	Caa	Ca_C	WR	Def
Aaa	87.17%	7.47%	0.77%	0.18%	0.03%	0.00%	0.00%	0.00%	4.38%	0.00%
Aa	0.98%	84.65%	7.51%	0.68%	0.15%	0.04%	0.01%	0.00%	5.92%	0.06%
A	0.06%	2.57%	86.01%	5.14%	0.56%	0.10%	0.03%	0.01%	5.46%	0.08%
Baa	0.03%	0.20%	3.84%	84.30%	4.02%	0.63%	0.11%	0.02%	6.63%	0.23%
Ba	0.01%	0.06%	0.44%	5.97%	74.73%	6.61%	0.67%	0.08%	10.32%	1.10%
B	0.00%	0.04%	0.14%	0.55%	5.44%	72.34%	6.32%	0.44%	11.74%	2.99%
Caa	0.00%	0.01%	0.02%	0.08%	0.35%	5.86%	70.34%	2.94%	13.71%	6.69%
Ca_C	0.00%	0.01%	0.07%	0.07%	0.38%	2.35%	9.83%	44.85%	16.83%	25.60%

Source: Moody's Investors Service

Exhibit 32

Average one-year letter rating migration rates, 1970-2023

From\To	Aaa	Aa	A	Baa	Ba	B	Caa	Ca_C	WR	Def
Aaa	88.08%	7.51%	0.56%	0.06%	0.02%	0.00%	0.00%	0.00%	3.76%	0.00%
Aa	0.74%	85.80%	8.14%	0.40%	0.06%	0.03%	0.02%	0.00%	4.80%	0.02%
A	0.04%	2.34%	87.69%	4.86%	0.41%	0.09%	0.03%	0.00%	4.49%	0.05%
Baa	0.03%	0.12%	3.65%	86.94%	3.25%	0.57%	0.12%	0.02%	5.14%	0.16%
Ba	0.00%	0.03%	0.36%	5.83%	77.03%	6.73%	0.70%	0.10%	8.36%	0.86%
B	0.01%	0.02%	0.11%	0.39%	4.75%	73.97%	6.71%	0.48%	10.60%	2.96%
Caa	0.00%	0.01%	0.01%	0.06%	0.23%	5.64%	70.74%	2.88%	13.95%	6.47%
Ca_C	0.00%	0.00%	0.03%	0.06%	0.32%	1.88%	10.77%	38.63%	18.81%	29.51%

Source: Moody's Investors Service

Exhibit 33

Average five-year letter rating migration rates, 1970-2023

From\To	Aaa	Aa	A	Baa	Ba	B	Caa	Ca_C	WR	Def
Aaa	54.84%	22.06%	4.66%	0.58%	0.29%	0.03%	0.04%	0.00%	17.44%	0.07%
Aa	1.96%	47.16%	23.56%	3.67%	0.71%	0.23%	0.10%	0.03%	22.35%	0.24%
A	0.15%	7.24%	53.62%	13.94%	2.14%	0.64%	0.12%	0.01%	21.54%	0.59%
Baa	0.12%	0.82%	11.22%	52.35%	7.43%	2.29%	0.47%	0.08%	24.02%	1.21%
Ba	0.03%	0.16%	2.15%	14.42%	28.87%	11.24%	1.99%	0.12%	34.99%	6.04%
B	0.02%	0.06%	0.37%	2.16%	7.71%	23.62%	6.68%	0.58%	43.85%	14.96%
Caa	0.00%	0.01%	0.07%	0.40%	1.31%	7.19%	16.45%	1.30%	51.45%	21.82%
Ca_C	0.00%	0.00%	0.02%	0.80%	1.35%	3.67%	3.23%	2.42%	44.34%	44.17%

Source: Moody's Investors Service

Exhibit 34

2023 one-year alphanumeric rating migration rates

From\To	Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa1	Caa2	Caa3	Ca_C	WR	Def
Aaa	98.00%	2.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa1	0.00%	88.89%	0.00%	0.00%	0.00%	0.00%	2.78%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	8.33%	0.00%
Aa2	0.00%	1.74%	94.78%	0.87%	0.87%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.74%	0.00%
Aa3	0.00%	0.00%	1.20%	96.41%	1.20%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.20%	0.00%
A1	0.00%	0.00%	0.00%	3.13%	90.12%	3.13%	0.48%	0.24%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.89%	0.00%
A2	0.00%	0.00%	0.00%	0.00%	6.44%	86.88%	2.48%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	4.21%	0.00%
A3	0.00%	0.00%	0.00%	0.00%	0.00%	7.03%	83.79%	5.66%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	3.13%	0.39%
Baa1	0.00%	0.00%	0.00%	0.00%	0.15%	0.00%	6.47%	85.15%	4.12%	0.59%	0.00%	0.00%	0.44%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.94%	0.15%
Baa2	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.15%	6.76%	82.94%	3.82%	0.74%	0.59%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	4.85%	0.15%
Baa3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	8.60%	82.00%	2.80%	1.20%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	5.40%	0.00%
Ba1	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.97%	11.11%	77.78%	3.86%	1.93%	0.00%	0.00%	0.48%	0.48%	0.00%	0.00%	0.48%	2.90%	0.00%
Ba2	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	5.98%	79.06%	5.13%	1.71%	0.00%	0.85%	0.43%	0.00%	0.00%	0.00%	6.84%	0.00%
Ba3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.39%	4.65%	7.36%	69.77%	3.10%	2.71%	1.94%	2.33%	0.78%	0.39%	0.00%	4.65%	1.94%
B1	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	4.00%	9.45%	66.91%	5.82%	1.82%	1.45%	1.09%	0.00%	0.36%	8.36%	0.73%
B2	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.78%	1.17%	12.11%	65.63%	10.55%	1.95%	1.17%	0.78%	0.00%	5.47%	0.39%
B3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.60%	0.00%	0.00%	3.58%	9.85%	68.36%	8.96%	2.99%	0.90%	0.00%	3.88%	0.90%
Caa1	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.47%	0.47%	0.71%	7.31%	60.14%	13.21%	4.25%	0.47%	10.14%	2.83%
Caa2	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.13%	0.79%	0.00%	4.61%	70.26%	7.63%	2.89%	9.74%	3.95%
Caa3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.46%	0.00%	0.00%	7.76%	56.16%	14.16%	9.13%	12.33%
Ca_C	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.82%	2.46%	32.79%	12.30%	51.64%

Source: Moody's Investors Service

Exhibit 35

Average one-year alphanumeric rating migration rates, 1983-2023

From\To	Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa1	Caa2	Caa3	Ca_C	WR	Def
Aaa	87.36%	5.11%	2.16%	0.53%	0.28%	0.14%	0.02%	0.05%	0.00%	0.02%	0.01%	0.01%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	4.30%	0.00%
Aa1	1.60%	77.10%	7.79%	5.52%	1.35%	1.00%	0.20%	0.14%	0.07%	0.01%	0.03%	0.00%	0.01%	0.04%	0.02%	0.01%	0.02%	0.02%	0.00%	0.00%	5.08%	0.00%
Aa2	0.94%	4.11%	75.12%	9.59%	3.19%	1.50%	0.40%	0.08%	0.14%	0.06%	0.03%	0.01%	0.00%	0.02%	0.01%	0.02%	0.00%	0.02%	0.00%	0.00%	4.75%	0.00%
Aa3	0.14%	0.98%	4.02%	76.30%	8.53%	3.28%	0.78%	0.23%	0.23%	0.11%	0.02%	0.03%	0.01%	0.01%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	5.26%	0.04%
A1	0.04%	0.08%	0.97%	4.94%	77.66%	7.18%	2.43%	0.55%	0.38%	0.17%	0.15%	0.11%	0.04%	0.05%	0.01%	0.01%	0.01%	0.01%	0.01%	0.00%	5.14%	0.06%
A2	0.05%	0.03%	0.19%	0.93%	5.72%	77.54%	6.98%	2.33%	0.89%	0.33%	0.15%	0.12%	0.14%	0.05%	0.03%	0.01%	0.02%	0.02%	0.00%	0.00%	4.44%	0.04%
A3	0.04%	0.04%	0.08%	0.26%	1.32%	6.10%	76.86%	6.55%	2.33%	0.78%	0.30%	0.13%	0.11%	0.09%	0.04%	0.02%	0.02%	0.01%	0.00%	0.01%	4.86%	0.06%
Baa1	0.01%	0.02%	0.06%	0.09%	0.19%	1.31%	6.17%	77.64%	6.27%	1.97%	0.49%	0.25%	0.18%	0.21%	0.05%	0.02%	0.04%	0.02%	0.00%	0.02%	4.88%	0.09%
Baa2	0.03%	0.03%	0.01%	0.06%	0.14%	0.51%	1.64%	6.54%	77.21%	5.85%	1.15%	0.53%	0.38%	0.27%	0.16%	0.07%	0.08%	0.01%	0.01%	0.01%	5.13%	0.16%
Baa3	0.02%	0.01%	0.02%	0.03%	0.06%	0.14%	0.39%	1.62%	8.67%	74.16%	4.45%	1.86%	0.85%	0.66%	0.22%	0.22%	0.12%	0.06%	0.05%	0.04%	6.09%	0.26%
Ba1	0.01%	0.00%	0.01%	0.01%	0.14%	0.11%	0.20%	0.59%	2.20%	10.01%	65.87%	5.53%	3.81%	1.47%	0.55%	0.48%	0.14%	0.19%	0.04%	0.11%	8.03%	0.49%
Ba2	0.00%	0.00%	0.01%	0.02%	0.06%	0.10%	0.14%	0.31%	0.63%	3.52%	7.98%	65.12%	6.19%	3.63%	1.24%	0.81%	0.33%	0.21%	0.07%	0.12%	8.80%	0.71%
Ba3	0.00%	0.01%	0.01%	0.01%	0.05%	0.14%	0.15%	0.08%	0.38%	0.77%	2.77%	7.11%	64.91%	6.91%	3.01%	1.76%	0.69%	0.36%	0.08%	0.11%	9.41%	1.25%
B1	0.01%	0.01%	0.01%	0.01%	0.04%	0.02%	0.07%	0.08%	0.17%	0.34%	0.62%	2.68%	6.97%	63.74%	6.33%	4.35%	1.31%	0.73%	0.22%	0.24%	10.28%	1.76%
B2	0.00%	0.01%	0.00%	0.01%	0.01%	0.02%	0.08%	0.11%	0.12%	0.22%	0.19%	0.62%	2.03%	7.48%	62.51%	7.96%	3.54%	1.87%	0.42%	0.44%	9.61%	2.74%
B3	0.01%	0.00%	0.02%	0.00%	0.03%	0.03%	0.05%	0.03%	0.04%	0.08%	0.13%	0.23%	0.79%	2.47%	6.48%	60.65%	7.50%	3.35%	1.07%	0.77%	12.05%	4.23%
Caa1	0.00%	0.01%	0.00%	0.00%	0.00%	0.01%	0.00%	0.01%	0.01%	0.03%	0.05%	0.10%	0.20%	0.57%	1.25%	7.55%	59.33%	9.39%	2.82%	1.24%	13.68%	3.78%
Caa2	0.00%	0.00%	0.01%	0.00%	0.01%	0.01%	0.00%	0.00%	0.03%	0.05%	0.03%	0.02%	0.09%	0.25%	0.71%	1.75%	6.37%	60.43%	6.61%	2.81%	14.31%	6.49%
Caa3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.04%	0.02%	0.02%	0.09%	0.18%	0.89%	2.68%	9.92%	47.02%	9.22%	14.10%	15.82%
Ca_C	0.00%	0.00%	0.00%	0.00%	0.00%	0.01%	0.02%	0.00%	0.00%	0.06%	0.11%	0.10%	0.12%	0.07%	0.35%	1.42%	1.61%	3.30%	5.98%	38.16%	19.02%	29.67%

Source: Moody's Investors Service

Exhibit 36

Annual issuer-weighted corporate default rates by letter rating, 1920-2023

Year	Aaa	Aa	A	Baa	Ba	B	Caa-C	IG	SG	All
1920	0.00%	0.00%	0.32%	0.94%	2.15%	4.38%	0.00%	0.43%	3.01%	1.23%
1921	0.00%	0.19%	0.35%	0.65%	0.44%	2.68%	13.33%	0.39%	2.15%	1.07%
1922	0.00%	0.18%	0.17%	1.10%	1.08%	1.71%	7.63%	0.51%	1.76%	1.01%
1923	0.00%	0.00%	0.00%	0.62%	0.93%	2.27%	5.93%	0.24%	1.70%	0.80%
1924	0.00%	0.37%	0.00%	0.13%	2.06%	2.70%	12.84%	0.14%	2.85%	1.15%
1925	0.00%	0.00%	0.14%	0.71%	1.74%	2.59%	14.40%	0.32%	2.56%	1.17%
1926	0.00%	0.40%	0.15%	0.11%	1.39%	2.90%	3.70%	0.19%	1.91%	0.77%
1927	0.00%	0.00%	0.21%	0.00%	1.30%	1.98%	12.84%	0.07%	1.83%	0.74%
1928	0.00%	0.00%	0.00%	0.00%	0.16%	1.32%	10.48%	0.00%	0.88%	0.36%
1929	0.00%	0.29%	0.00%	0.44%	0.82%	0.92%	9.73%	0.24%	1.40%	0.71%
1930	0.00%	0.00%	0.00%	0.40%	0.92%	3.16%	7.72%	0.15%	2.20%	1.04%
1931	0.00%	0.00%	0.27%	1.08%	3.00%	9.52%	31.67%	0.50%	7.90%	3.80%
1932	0.00%	0.67%	1.10%	0.92%	6.10%	13.98%	24.06%	0.86%	10.99%	5.50%
1933	0.00%	0.00%	0.26%	1.77%	11.71%	16.15%	25.92%	0.79%	15.77%	8.53%
1934	0.00%	0.62%	0.31%	0.86%	2.52%	4.22%	16.50%	0.59%	5.89%	3.40%
1935	0.00%	0.00%	1.43%	1.92%	5.12%	4.27%	13.02%	1.29%	6.25%	3.93%
1936	0.00%	0.85%	0.54%	0.33%	1.23%	2.38%	7.80%	0.48%	2.71%	1.63%
1937	0.00%	0.00%	0.51%	1.04%	0.99%	2.67%	9.07%	0.62%	2.74%	1.72%
1938	0.00%	0.85%	1.64%	1.99%	0.98%	1.47%	12.81%	1.55%	2.59%	2.11%
1939	0.00%	0.00%	0.00%	0.99%	0.62%	1.74%	6.07%	0.41%	1.77%	1.22%
1940	0.00%	0.00%	0.00%	1.37%	0.43%	3.29%	11.83%	0.59%	3.55%	2.47%
1941	0.00%	0.00%	0.00%	0.00%	0.97%	0.81%	5.07%	0.00%	1.71%	1.08%
1942	0.00%	0.00%	0.00%	0.00%	0.00%	0.78%	2.00%	0.00%	0.73%	0.45%
1943	0.00%	0.00%	0.00%	0.00%	0.00%	1.35%	0.00%	0.00%	0.61%	0.37%
1944	0.00%	0.00%	0.00%	0.00%	0.00%	0.49%	2.55%	0.00%	0.66%	0.39%
1945	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	3.57%	0.00%	0.56%	0.31%
1946	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1947	0.00%	0.00%	0.00%	0.00%	0.00%	0.71%	2.78%	0.00%	0.63%	0.31%
1948	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1949	0.00%	0.00%	0.00%	0.00%	1.36%	1.02%	8.57%	0.00%	1.92%	0.84%
1950	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1951	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	4.76%	0.00%	0.43%	0.18%
1952	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1953	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1954	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	7.14%	0.00%	0.47%	0.17%
1955	0.00%	0.00%	0.00%	0.00%	0.00%	1.61%	0.00%	0.00%	0.52%	0.17%
1956	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1957	0.00%	0.00%	0.00%	0.00%	0.00%	1.27%	0.00%	0.00%	0.45%	0.14%
1958	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1959	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1960	0.00%	0.00%	0.00%	0.00%	1.25%	0.00%	0.00%	0.00%	0.75%	0.25%
1961	0.00%	0.00%	0.00%	0.00%	0.60%	0.00%	8.70%	0.00%	1.07%	0.35%
1962	0.00%	0.00%	0.00%	0.00%	1.75%	1.47%	0.00%	0.00%	1.52%	0.47%
1963	0.00%	0.00%	0.00%	0.00%	1.16%	1.47%	0.00%	0.00%	1.15%	0.35%
1964	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1965	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1966	0.00%	0.00%	0.00%	0.00%	0.00%	2.44%	0.00%	0.00%	0.44%	0.12%
1967	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1968	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	5.00%	0.00%	0.37%	0.11%
1969	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1970	0.00%	0.00%	0.00%	0.54%	4.24%	19.44%	50.00%	0.27%	8.68%	2.63%
1971	0.00%	0.00%	0.00%	0.00%	0.89%	0.00%	12.50%	0.00%	1.16%	0.29%
1972	0.00%	0.00%	0.00%	0.00%	0.00%	6.90%	37.50%	0.00%	1.92%	0.45%
1973	0.00%	0.00%	0.00%	0.46%	0.00%	3.85%	37.50%	0.23%	1.28%	0.46%

Year	Aaa	Aa	A	Baa	Ba	B	Caa-C	IG	SG	All
1974	0.00%	0.00%	0.00%	0.00%	0.51%	7.16%	0.00%	0.00%	1.33%	0.28%
1975	0.00%	0.00%	0.00%	0.00%	1.03%	6.16%	0.00%	0.00%	1.74%	0.36%
1976	0.00%	0.00%	0.00%	0.00%	1.00%	0.00%	0.00%	0.00%	0.87%	0.18%
1977	0.00%	0.00%	0.00%	0.29%	0.54%	3.23%	33.33%	0.11%	1.36%	0.35%
1978	0.00%	0.00%	0.00%	0.00%	1.12%	5.41%	0.00%	0.00%	1.82%	0.35%
1979	0.00%	0.00%	0.00%	0.00%	0.51%	0.00%	0.00%	0.00%	0.43%	0.09%
1980	0.00%	0.00%	0.00%	0.00%	0.00%	5.00%	33.33%	0.00%	1.63%	0.34%
1981	0.00%	0.00%	0.00%	0.00%	0.00%	4.40%	0.00%	0.00%	0.70%	0.16%
1982	0.00%	0.00%	0.26%	0.33%	2.79%	2.22%	21.43%	0.21%	3.54%	1.04%
1983	0.00%	0.00%	0.00%	0.00%	1.16%	2.22%	45.05%	0.00%	4.27%	0.95%
1984	0.00%	0.00%	0.00%	0.61%	0.52%	5.17%	18.18%	0.17%	3.08%	0.85%
1985	0.00%	0.00%	0.00%	0.00%	0.87%	7.11%	6.25%	0.00%	3.72%	0.94%
1986	0.00%	0.00%	0.00%	0.84%	2.37%	10.30%	16.18%	0.21%	6.11%	1.81%
1987	0.00%	0.00%	0.00%	0.00%	3.04%	5.35%	9.82%	0.00%	4.28%	1.41%
1988	0.00%	0.00%	0.00%	0.00%	1.36%	5.61%	12.50%	0.00%	3.73%	1.34%
1989	0.00%	0.50%	0.00%	0.51%	2.97%	7.53%	20.39%	0.25%	5.90%	2.21%
1990	0.00%	0.00%	0.00%	0.26%	3.53%	13.68%	45.06%	0.06%	10.43%	3.51%
1991	0.00%	0.00%	0.00%	0.25%	3.84%	13.12%	15.97%	0.06%	9.08%	2.78%
1992	0.00%	0.00%	0.00%	0.00%	0.34%	7.64%	15.67%	0.00%	4.93%	1.33%
1993	0.00%	0.00%	0.00%	0.00%	0.62%	4.35%	13.95%	0.00%	3.40%	0.89%
1994	0.00%	0.00%	0.00%	0.00%	0.00%	4.19%	5.31%	0.00%	2.34%	0.65%
1995	0.00%	0.00%	0.00%	0.00%	0.27%	3.83%	11.55%	0.00%	3.07%	0.89%
1996	0.00%	0.00%	0.00%	0.00%	0.00%	1.51%	9.90%	0.00%	1.65%	0.50%
1997	0.00%	0.00%	0.00%	0.00%	0.18%	2.00%	9.09%	0.00%	1.89%	0.61%
1998	0.00%	0.00%	0.00%	0.11%	0.76%	3.69%	8.86%	0.03%	2.98%	1.11%
1999	0.00%	0.00%	0.00%	0.09%	1.38%	5.04%	15.04%	0.03%	5.33%	2.09%
2000	0.00%	0.00%	0.00%	0.35%	1.44%	5.64%	17.74%	0.13%	6.15%	2.46%
2001	0.00%	0.00%	0.15%	0.18%	1.17%	8.72%	28.27%	0.12%	9.31%	3.53%
2002	0.00%	0.00%	0.16%	0.84%	1.77%	4.57%	26.40%	0.37%	7.79%	2.92%
2003	0.00%	0.00%	0.00%	0.00%	0.89%	2.56%	20.45%	0.00%	5.32%	1.83%
2004	0.00%	0.00%	0.00%	0.00%	0.38%	0.68%	10.90%	0.00%	2.30%	0.78%
2005	0.00%	0.00%	0.00%	0.16%	0.00%	0.71%	7.27%	0.06%	1.72%	0.64%
2006	0.00%	0.00%	0.00%	0.00%	0.20%	1.09%	5.50%	0.00%	1.67%	0.59%
2007	0.00%	0.00%	0.00%	0.00%	0.00%	0.10%	4.78%	0.00%	0.99%	0.36%
2008	0.00%	0.50%	0.40%	1.01%	2.33%	4.04%	10.70%	0.62%	5.47%	2.50%
2009	0.00%	0.00%	0.24%	0.92%	1.77%	6.80%	26.39%	0.42%	12.11%	4.98%
2010	0.00%	0.00%	0.17%	0.07%	0.00%	0.39%	8.63%	0.09%	3.07%	1.24%
2011	0.00%	0.19%	0.00%	0.36%	0.16%	0.34%	6.22%	0.19%	2.08%	0.94%
2012	0.00%	0.00%	0.00%	0.07%	0.14%	0.54%	8.01%	0.03%	2.81%	1.24%
2013	0.00%	0.00%	0.09%	0.12%	0.60%	1.01%	6.29%	0.09%	2.69%	1.25%
2014	0.00%	0.00%	0.09%	0.06%	0.15%	0.50%	4.76%	0.06%	2.03%	0.97%
2015	0.00%	0.00%	0.00%	0.00%	0.30%	2.42%	6.62%	0.00%	3.71%	1.75%
2016	0.00%	0.00%	0.00%	0.00%	0.14%	1.68%	9.10%	0.00%	4.63%	2.20%
2017	0.00%	0.00%	0.00%	0.00%	0.53%	0.45%	7.55%	0.00%	3.59%	1.71%
2018	0.00%	0.00%	0.00%	0.00%	0.00%	0.68%	5.17%	0.00%	2.45%	1.17%
2019	0.00%	0.00%	0.00%	0.11%	0.00%	1.20%	6.52%	0.06%	3.20%	1.53%
2020	0.00%	0.00%	0.00%	0.11%	0.13%	3.98%	12.83%	0.06%	6.92%	3.23%
2021	0.00%	0.00%	0.00%	0.00%	0.00%	1.33%	2.96%	0.00%	1.75%	0.80%
2022	0.00%	0.00%	0.00%	0.91%	4.17%	5.14%	3.84%	0.48%	4.29%	2.28%
2023	0.00%	0.00%	0.15%	0.11%	0.73%	0.72%	9.16%	0.11%	4.83%	2.28%
Mean	0.00%	0.05%	0.09%	0.26%	1.02%	3.09%	10.27%	0.14%	2.88%	1.20%
Median	0.00%	0.00%	0.00%	0.00%	0.52%	1.86%	7.67%	0.00%	1.92%	0.84%
Std Dev	0.00%	0.17%	0.25%	0.44%	1.60%	3.68%	10.96%	0.26%	2.90%	1.33%
Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Max	0.00%	0.85%	1.64%	1.99%	11.71%	19.44%	50.00%	1.55%	15.77%	8.53%

Source: Moody's Investors Service

Exhibit 37

Annual issuer-weighted corporate default rates by alphanumeric rating, 1983-2023

Year	Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa1	Caa2	Caa3	Ca-C	IG	SG	All
1983	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	3.08%	1.01%	0.00%	7.56%		45.05%			0.00%	4.27%	0.95%
1984	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.75%	0.00%	1.64%	0.00%	6.49%	0.00%	3.03%		18.18%			0.17%	3.08%	0.85%
1985	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.64%	1.15%	4.53%	5.00%	13.40%		6.25%			0.00%	3.72%	0.94%
1986	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.81%	1.84%	1.27%	1.09%	4.00%	8.66%	6.67%	15.08%		16.18%			0.21%	6.11%	1.81%
1987	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	5.21%	0.90%	3.14%	4.17%	5.45%	7.88%		10.16%		0.00%	0.00%	4.28%	1.41%
1988	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.78%	4.24%	4.33%	9.96%		10.53%		50.00%	0.00%	3.73%	1.34%
1989	0.00%	0.00%	0.00%	1.08%	0.00%	0.00%	0.00%	0.00%	0.74%	0.79%	1.02%	1.79%	4.62%	6.70%	5.30%	13.27%		21.49%		0.00%	0.25%	5.90%	2.21%
1990	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.95%	1.08%	5.84%	3.50%	6.86%	16.72%	24.67%		46.46%		33.33%	0.06%	10.43%	3.51%
1991	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.68%	0.00%	0.00%	1.05%	0.00%	7.22%	7.41%	7.16%	29.50%		16.15%		16.67%	0.06%	9.08%	2.78%
1992	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.76%	1.45%	1.37%	24.00%		17.51%		7.69%	0.00%	4.93%	1.33%
1993	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.03%	0.00%	0.80%	2.83%	1.32%	11.84%		14.97%		9.09%	0.00%	3.40%	0.89%
1994	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.52%	2.68%	9.50%		5.01%		7.14%	0.00%	2.34%	0.65%
1995	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.68%	3.79%	5.67%	0.94%		7.69%		23.07%	0.00%	3.07%	0.89%
1996	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.39%	1.25%	3.87%		10.61%		5.88%	0.00%	1.65%	0.50%
1997	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.47%	0.34%	0.50%	7.02%		7.91%		15.19%	0.00%	1.89%	0.61%
1998	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.29%	0.00%	0.00%	1.19%	1.20%	3.15%	3.50%	4.87%	4.08%	9.99%	37.50%	5.00%	0.03%	2.98%	1.11%
1999	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.30%	0.47%	0.92%	2.70%	3.05%	4.29%	8.32%	10.53%	21.65%	14.98%	19.33%	0.03%	5.33%	2.09%
2000	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.26%	0.00%	0.86%	0.49%	1.37%	2.53%	1.57%	4.03%	11.65%	14.34%	24.39%	17.22%	17.06%	0.13%	6.15%	2.46%
2001	0.00%	0.00%	0.00%	0.00%	0.00%	0.41%	0.00%	0.27%	0.24%	0.00%	0.00%	0.97%	2.64%	3.26%	9.66%	12.98%	25.26%	24.97%	39.02%	34.81%	0.12%	9.31%	3.53%
2002	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.43%	0.73%	0.68%	1.16%	2.04%	1.74%	1.53%	1.93%	4.77%	6.96%	18.03%	20.57%	32.86%	37.88%	0.37%	7.79%	2.92%
2003	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.53%	0.65%	1.41%	0.70%	2.38%	4.93%	8.73%	25.27%	27.17%	28.65%	0.00%	5.32%	1.83%
2004	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.00%	0.00%	0.58%	1.71%	8.09%	8.31%	8.65%	27.66%	0.00%	2.30%	0.78%
2005	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.22%	0.28%	0.00%	0.00%	0.00%	0.00%	0.25%	1.96%	5.09%	5.44%	21.37%	13.82%	0.06%	1.72%	0.64%
2006	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.66%	0.65%	0.32%	2.12%	2.27%	5.72%	13.17%	17.22%	0.00%	1.67%	0.59%
2007	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.40%	0.00%	0.00%	1.40%	9.82%	9.82%	46.15%	0.00%	0.99%	0.36%
2008	0.00%	0.00%	0.00%	1.49%	1.02%	0.23%	0.00%	0.97%	1.37%	0.60%	0.92%	2.88%	3.33%	3.15%	4.08%	4.55%	5.70%	17.90%	35.16%	37.90%	0.62%	5.47%	2.50%
2009	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.69%	1.02%	0.86%	0.90%	1.46%	1.74%	2.12%	3.38%	7.18%	8.82%	12.00%	33.26%	53.60%	66.13%	0.42%	12.11%	4.98%
2010	0.00%	0.00%	0.00%	0.00%	0.31%	0.00%	0.22%	0.00%	0.00%	0.24%	0.00%	0.00%	0.00%	0.88%	0.00%	0.28%	1.74%	7.49%	22.89%	26.98%	0.09%	3.07%	1.24%
2011	0.00%	0.00%	0.00%	0.42%	0.00%	0.00%	0.00%	0.00%	0.20%	0.87%	0.00%	0.56%	0.00%	0.00%	0.46%	0.51%	1.91%	8.39%	16.43%	23.23%	0.19%	2.08%	0.94%
2012	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.22%	0.00%	0.00%	0.00%	0.00%	0.41%	0.00%	0.77%	0.75%	2.28%	11.42%	16.45%	48.62%	0.03%	2.81%	1.24%
2013	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.22%	0.00%	0.16%	0.19%	0.00%	0.00%	1.58%	0.71%	1.64%	0.84%	2.10%	7.04%	10.51%	58.04%	0.09%	2.69%	1.25%
2014	0.00%	0.00%	0.00%	0.00%	0.40%	0.00%	0.00%	0.00%	0.00%	0.18%	0.00%	0.00%	0.36%	0.35%	1.08%	0.23%	2.42%	3.41%	12.00%	28.83%	0.06%	2.03%	0.97%
2015	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.95%	0.00%	0.00%	0.37%	3.32%	3.07%	4.10%	5.04%	15.17%	36.86%	0.00%	3.71%	1.75%
2016	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.41%	1.07%	1.01%	2.62%	3.64%	5.30%	19.65%	48.23%	0.00%	4.63%	2.20%
2017	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.42%	1.12%	0.00%	0.39%	0.85%	1.63%	4.42%	16.37%	34.75%	0.00%	3.59%	1.71%
2018	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.10%	0.88%	1.63%	1.86%	13.91%	30.44%	0.00%	2.45%	1.17%
2019	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.15%	0.19%	0.00%	0.00%	0.00%	0.37%	0.35%	2.49%	1.22%	4.08%	16.45%	44.95%	0.06%	3.20%	1.53%
2020	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.41%	0.40%	0.00%	0.00%	1.77%	4.59%	5.31%	6.69%	9.01%	21.81%	63.53%	0.06%	6.92%	3.23%
2021	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.74%	1.21%	1.91%	0.31%	0.80%	7.20%	13.26%	0.00%	1.75%	0.80%
2022	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.04%	1.93%	4.86%	4.19%	3.60%	3.36%	4.92%	6.80%	2.25%	1.58%	6.39%	26.29%	0.48%	4.29%	2.28%
2023	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.40%	0.15%	0.15%	0.00%	0.00%	0.00%	1.98%	0.76%	0.41%	0.93%	3.05%	4.25%	12.76%	54.97%	0.11%	4.83%	2.28%
Mean	0.00%	0.00%	0.00%	0.07%	0.04%	0.02%	0.05%	0.11%	0.17%	0.33%	0.56%	0.72%	1.48%	2.27%	3.07%	6.78%	5.79%	13.06%	19.94%	28.61%	0.09%	4.32%	1.64%
Median	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.00%	1.45%	1.64%	4.87%	3.35%	9.82%	16.44%	27.66%	0.03%	3.71%	1.33%
Std Dev	0.00%	0.00%	0.00%	0.29%	0.17%	0.07%	0.14%	0.26%	0.33%	0.54%	1.14%	1.22%	1.61%	2.30%	3.27%	6.92%	5.87%	10.56%	11.16%	17.66%	0.14%	2.51%	0.99%
Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.31%	0.80%	6.39%	0.00%	0.00%	0.99%	0.36%
Max	0.00%	0.00%	0.00%	1.49%	1.02%	0.41%	0.69%	1.02%	1.37%	1.93%	5.21%	5.84%	7.22%	8.66%	16.72%	29.50%	25.26%	46.46%	53.60%	66.13%	0.62%	12.11%	4.98%

Source: Moody's Investors Service

Exhibit 38

Average cumulative issuer-weighted global default rates by letter rating, 1920-2023

Rating\Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	0.00%	0.01%	0.02%	0.07%	0.13%	0.20%	0.29%	0.40%	0.52%	0.66%	0.78%	0.87%	0.97%	1.00%	1.03%	1.08%	1.13%	1.17%	1.22%	1.26%
Aa	0.06%	0.17%	0.26%	0.41%	0.62%	0.87%	1.12%	1.35%	1.57%	1.81%	2.07%	2.35%	2.62%	2.88%	3.07%	3.22%	3.36%	3.53%	3.72%	3.89%
A	0.08%	0.23%	0.47%	0.72%	1.01%	1.32%	1.64%	1.96%	2.31%	2.67%	3.04%	3.40%	3.73%	4.07%	4.46%	4.81%	5.09%	5.36%	5.62%	5.89%
Baa	0.24%	0.66%	1.15%	1.69%	2.24%	2.79%	3.32%	3.90%	4.50%	5.09%	5.67%	6.26%	6.87%	7.42%	7.92%	8.43%	8.93%	9.38%	9.78%	10.22%
Ba	1.17%	2.77%	4.54%	6.38%	8.21%	9.96%	11.58%	13.13%	14.64%	16.23%	17.64%	19.04%	20.37%	21.54%	22.68%	23.77%	24.82%	25.79%	26.66%	27.39%
B	3.20%	7.35%	11.59%	15.52%	19.03%	22.12%	24.88%	27.32%	29.56%	31.57%	33.44%	35.22%	36.97%	38.69%	40.24%	41.78%	43.11%	44.28%	45.27%	46.13%
Caa-C	9.07%	16.23%	22.27%	27.41%	31.86%	35.58%	38.65%	41.32%	43.83%	45.96%	48.00%	49.79%	51.33%	52.82%	54.46%	56.08%	57.50%	58.83%	60.13%	61.41%
IG	0.13%	0.38%	0.68%	1.01%	1.37%	1.75%	2.12%	2.50%	2.90%	3.30%	3.70%	4.11%	4.50%	4.87%	5.22%	5.56%	5.86%	6.14%	6.42%	6.70%
SG	3.77%	7.55%	11.11%	14.33%	17.21%	19.72%	21.92%	23.89%	25.73%	27.47%	29.06%	30.57%	32.00%	33.33%	34.59%	35.82%	36.94%	37.96%	38.86%	39.65%
All	1.55%	3.12%	4.57%	5.87%	7.03%	8.04%	8.93%	9.74%	10.50%	11.23%	11.91%	12.56%	13.18%	13.75%	14.28%	14.80%	15.25%	15.68%	16.06%	16.43%

Source: Moody's Investors Service

Exhibit 39

Average cumulative issuer-weighted global default rates by letter rating, 1970-2023

Rating\Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	0.00%	0.01%	0.01%	0.03%	0.08%	0.12%	0.18%	0.23%	0.28%	0.34%	0.40%	0.46%	0.53%	0.56%	0.60%	0.64%	0.68%	0.69%	0.69%	0.69%
Aa	0.02%	0.05%	0.10%	0.18%	0.27%	0.38%	0.48%	0.58%	0.66%	0.73%	0.82%	0.92%	1.03%	1.12%	1.20%	1.27%	1.37%	1.48%	1.66%	1.83%
A	0.05%	0.15%	0.31%	0.47%	0.67%	0.89%	1.13%	1.38%	1.64%	1.91%	2.18%	2.44%	2.71%	2.99%	3.32%	3.64%	3.94%	4.26%	4.55%	4.86%
Baa	0.16%	0.42%	0.73%	1.09%	1.45%	1.82%	2.18%	2.60%	3.07%	3.54%	4.03%	4.55%	5.09%	5.62%	6.13%	6.66%	7.20%	7.68%	8.09%	8.50%
Ba	0.90%	2.42%	4.15%	5.99%	7.79%	9.48%	11.01%	12.48%	13.97%	15.52%	17.01%	18.50%	19.87%	21.15%	22.50%	23.79%	24.94%	25.93%	26.88%	27.56%
B	3.14%	7.51%	12.06%	16.28%	20.12%	23.56%	26.61%	29.32%	31.83%	34.06%	36.09%	38.00%	39.86%	41.74%	43.39%	44.98%	46.34%	47.61%	48.79%	50.01%
Caa-C	9.00%	16.22%	22.50%	27.92%	32.70%	36.66%	39.96%	42.88%	45.56%	47.77%	49.66%	51.01%	52.04%	52.74%	53.61%	54.57%	55.15%	55.70%	56.33%	56.88%
IG	0.09%	0.23%	0.42%	0.64%	0.87%	1.12%	1.36%	1.63%	1.92%	2.20%	2.50%	2.79%	3.10%	3.40%	3.71%	4.02%	4.32%	4.62%	4.90%	5.18%
SG	4.06%	8.19%	12.12%	15.65%	18.79%	21.52%	23.86%	25.95%	27.91%	29.71%	31.36%	32.92%	34.35%	35.72%	37.06%	38.34%	39.45%	40.43%	41.37%	42.15%
All	1.60%	3.19%	4.65%	5.92%	7.02%	7.96%	8.76%	9.47%	10.15%	10.77%	11.35%	11.90%	12.42%	12.91%	13.40%	13.87%	14.31%	14.71%	15.08%	15.43%

Source: Moody's Investors Service

Exhibit 40

Average cumulative issuer-weighted global default rates by letter rating, 1983-2023

Rating\Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	0.00%	0.01%	0.01%	0.03%	0.06%	0.09%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%
Aa	0.02%	0.06%	0.11%	0.19%	0.29%	0.37%	0.46%	0.54%	0.61%	0.69%	0.77%	0.88%	0.98%	1.05%	1.12%	1.19%	1.27%	1.39%	1.56%	1.70%
A	0.05%	0.16%	0.33%	0.50%	0.71%	0.94%	1.18%	1.43%	1.67%	1.92%	2.16%	2.40%	2.64%	2.93%	3.27%	3.60%	3.91%	4.23%	4.50%	4.78%
Baa	0.17%	0.43%	0.74%	1.08%	1.43%	1.78%	2.12%	2.51%	2.91%	3.32%	3.74%	4.19%	4.69%	5.16%	5.60%	6.06%	6.55%	6.96%	7.31%	7.61%
Ba	0.89%	2.45%	4.27%	6.18%	7.96%	9.62%	11.14%	12.56%	13.95%	15.36%	16.66%	17.96%	19.14%	20.25%	21.48%	22.59%	23.56%	24.42%	25.27%	25.78%
B	3.13%	7.52%	12.09%	16.31%	20.16%	23.62%	26.68%	29.39%	31.86%	34.04%	36.01%	37.90%	39.76%	41.64%	43.25%	44.82%	46.20%	47.54%	48.78%	50.07%
Caa-C	8.95%	16.14%	22.43%	27.87%	32.64%	36.60%	39.88%	42.79%	45.49%	47.75%	49.70%	51.08%	52.14%	52.87%	53.78%	54.80%	55.41%	55.99%	56.66%	57.24%
IG	0.09%	0.24%	0.43%	0.65%	0.88%	1.12%	1.35%	1.60%	1.85%	2.10%	2.36%	2.62%	2.89%	3.17%	3.45%	3.74%	4.02%	4.29%	4.54%	4.77%
SG	4.16%	8.40%	12.46%	16.11%	19.33%	22.12%	24.52%	26.64%	28.58%	30.34%	31.92%	33.39%	34.77%	36.08%	37.36%	38.57%	39.61%	40.58%	41.50%	42.26%
All	1.69%	3.38%	4.93%	6.28%	7.44%	8.42%	9.24%	9.97%	10.63%	11.22%	11.77%	12.28%	12.77%	13.25%	13.71%	14.16%	14.57%	14.96%	15.31%	15.62%

Source: Moody's Investors Service

Exhibit 41

Average cumulative issuer-weighted global default rates by alphanumeric rating, 1983-2023

Rating\Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	0.00%	0.01%	0.01%	0.03%	0.06%	0.09%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%	0.12%
Aa1	0.00%	0.00%	0.00%	0.05%	0.09%	0.13%	0.13%	0.13%	0.15%	0.19%	0.24%	0.29%	0.39%	0.50%	0.63%	0.73%	0.81%	0.90%	1.00%	1.00%
Aa2	0.00%	0.01%	0.10%	0.21%	0.31%	0.39%	0.47%	0.55%	0.66%	0.79%	0.89%	1.00%	1.09%	1.14%	1.18%	1.27%	1.44%	1.62%	1.79%	1.97%
Aa3	0.04%	0.11%	0.16%	0.23%	0.36%	0.47%	0.60%	0.70%	0.78%	0.85%	0.94%	1.07%	1.18%	1.25%	1.31%	1.35%	1.38%	1.48%	1.66%	1.84%
A1	0.06%	0.17%	0.35%	0.52%	0.70%	0.88%	1.03%	1.17%	1.29%	1.42%	1.58%	1.75%	1.93%	2.17%	2.42%	2.63%	2.86%	3.06%	3.18%	3.30%
A2	0.04%	0.13%	0.27%	0.45%	0.64%	0.92%	1.21%	1.50%	1.79%	2.10%	2.40%	2.66%	2.92%	3.24%	3.60%	4.01%	4.43%	4.81%	5.10%	5.39%
A3	0.06%	0.18%	0.36%	0.53%	0.78%	1.02%	1.28%	1.58%	1.87%	2.15%	2.41%	2.69%	2.98%	3.29%	3.67%	4.03%	4.29%	4.66%	5.05%	5.48%
Baa1	0.10%	0.26%	0.44%	0.64%	0.83%	1.07%	1.30%	1.55%	1.83%	2.15%	2.53%	2.98%	3.44%	3.89%	4.33%	4.76%	5.19%	5.43%	5.60%	5.82%
Baa2	0.16%	0.41%	0.68%	0.99%	1.26%	1.55%	1.84%	2.17%	2.53%	2.89%	3.33%	3.74%	4.18%	4.59%	5.04%	5.48%	5.97%	6.43%	6.90%	7.26%
Baa3	0.27%	0.66%	1.14%	1.71%	2.33%	2.92%	3.44%	4.08%	4.70%	5.26%	5.74%	6.24%	6.83%	7.43%	7.86%	8.40%	8.97%	9.54%	9.98%	10.32%
Ba1	0.51%	1.51%	2.68%	3.81%	5.01%	6.24%	7.26%	8.12%	8.84%	9.62%	10.44%	11.29%	12.04%	12.72%	13.68%	14.46%	15.12%	15.80%	16.75%	17.45%
Ba2	0.74%	2.00%	3.46%	4.94%	6.46%	7.83%	9.04%	10.21%	11.59%	13.04%	14.30%	15.55%	16.66%	17.78%	19.07%	20.14%	20.94%	21.66%	22.35%	22.45%
Ba3	1.33%	3.61%	6.31%	9.27%	11.80%	14.14%	16.43%	18.66%	20.76%	22.83%	24.71%	26.59%	28.36%	30.04%	31.56%	33.17%	34.77%	36.06%	37.00%	37.71%
B1	1.88%	5.05%	8.53%	11.95%	15.37%	18.44%	21.51%	24.26%	26.55%	28.47%	30.24%	31.82%	33.60%	35.48%	37.11%	38.83%	40.55%	42.34%	44.03%	45.66%
B2	2.91%	7.40%	11.95%	16.20%	19.91%	23.40%	26.35%	28.78%	31.43%	34.01%	36.43%	38.92%	40.95%	42.79%	44.43%	46.17%	47.71%	49.12%	50.30%	51.77%
B3	4.52%	10.02%	15.72%	20.80%	25.29%	29.23%	32.45%	35.46%	38.04%	40.14%	41.93%	43.57%	45.41%	47.46%	49.12%	50.21%	50.79%	51.22%	51.78%	52.19%
Caa	6.93%	13.69%	19.78%	25.17%	30.11%	34.28%	37.66%	40.70%	43.54%	45.99%	48.07%	49.55%	50.71%	51.58%	52.67%	53.95%	54.44%	54.93%	55.84%	56.65%
Ca-C	33.62%	46.61%	55.71%	62.19%	64.97%	66.09%	68.12%	69.46%	70.49%	70.94%	71.64%	72.15%	72.47%	72.47%	72.47%	72.47%	73.14%	73.69%	73.69%	73.69%
IG	0.09%	0.24%	0.43%	0.65%	0.88%	1.12%	1.35%	1.60%	1.85%	2.10%	2.36%	2.62%	2.89%	3.17%	3.45%	3.74%	4.02%	4.29%	4.54%	4.77%
SG	4.16%	8.40%	12.46%	16.11%	19.33%	22.12%	24.52%	26.64%	28.58%	30.34%	31.92%	33.39%	34.77%	36.08%	37.36%	38.57%	39.61%	40.58%	41.50%	42.26%
All	1.69%	3.38%	4.93%	6.28%	7.44%	8.42%	9.24%	9.97%	10.63%	11.22%	11.77%	12.28%	12.77%	13.25%	13.71%	14.16%	14.57%	14.96%	15.31%	15.62%

Source: Moody's Investors Service

Exhibit 42

Average cumulative issuer-weighted global default rates by alphanumeric rating, 1998-2023

Rating\Year	1	2	3	4	5	6	7	8	9	10
Aaa	0.00%	0.02%	0.02%	0.02%	0.02%	0.02%	0.02%	0.02%	0.02%	0.02%
Aa1	0.00%	0.00%	0.00%	0.00%	0.02%	0.07%	0.08%	0.08%	0.11%	0.17%
Aa2	0.00%	0.01%	0.12%	0.24%	0.31%	0.39%	0.49%	0.58%	0.72%	0.83%
Aa3	0.04%	0.11%	0.15%	0.23%	0.36%	0.50%	0.70%	0.86%	0.97%	1.07%
A1	0.08%	0.18%	0.32%	0.48%	0.67%	0.87%	1.06%	1.23%	1.39%	1.55%
A2	0.06%	0.16%	0.33%	0.47%	0.67%	0.97%	1.26%	1.56%	1.92%	2.35%
A3	0.07%	0.20%	0.39%	0.58%	0.86%	1.06%	1.33%	1.64%	1.99%	2.35%
Baa1	0.11%	0.26%	0.43%	0.61%	0.75%	0.97%	1.18%	1.46%	1.78%	2.16%
Baa2	0.18%	0.41%	0.66%	0.88%	1.07%	1.27%	1.46%	1.67%	1.96%	2.22%
Baa3	0.27%	0.65%	1.12%	1.63%	2.18%	2.63%	3.07%	3.68%	4.27%	4.87%
Ba1	0.42%	1.31%	2.23%	3.08%	4.16%	5.23%	6.21%	7.07%	7.84%	8.66%
Ba2	0.71%	1.80%	3.10%	4.41%	5.85%	7.10%	8.21%	9.42%	10.86%	12.46%
Ba3	0.94%	2.52%	4.33%	6.38%	8.06%	9.65%	11.45%	13.50%	15.52%	17.48%
B1	1.19%	3.63%	6.36%	9.23%	11.86%	14.17%	16.58%	18.83%	20.77%	22.59%
B2	2.62%	6.83%	11.26%	15.50%	19.02%	22.29%	25.04%	27.44%	30.19%	32.87%
B3	3.45%	8.48%	14.05%	18.96%	23.16%	26.78%	29.69%	32.56%	35.28%	37.49%
Caa1	4.10%	9.70%	15.19%	20.29%	25.19%	29.40%	32.77%	35.64%	38.48%	41.07%
Caa2	6.36%	12.77%	18.85%	24.38%	29.34%	33.53%	36.97%	40.29%	43.32%	45.38%
Caa3	17.05%	29.15%	37.80%	44.02%	49.16%	53.65%	57.45%	60.50%	62.40%	63.81%
Ca-C	35.39%	49.27%	58.93%	65.63%	68.54%	69.72%	71.82%	73.18%	74.21%	74.66%
IG	0.11%	0.27%	0.46%	0.66%	0.88%	1.10%	1.33%	1.59%	1.88%	2.18%
SG	4.13%	8.33%	12.32%	15.85%	18.89%	21.47%	23.70%	25.74%	27.70%	29.49%
All	1.79%	3.56%	5.18%	6.56%	7.72%	8.68%	9.49%	10.25%	10.97%	11.65%

Source: Moody's Investors Service

Exhibit 43

Average cumulative issuer-weighted global default rates by industry group, 1970-2023

Industry\Year	1	2	3	4	5	6	7	8	9	10
Aerospace & Defense	1.08%	2.19%	3.23%	3.95%	4.49%	4.80%	5.00%	5.16%	5.24%	5.25%
Automotive	2.25%	4.54%	6.70%	8.70%	10.60%	12.35%	13.93%	15.45%	16.85%	17.89%
Banking	0.59%	1.19%	1.77%	2.33%	2.88%	3.39%	3.86%	4.33%	4.82%	5.31%
Beverage, Food, & Tobacco	1.17%	2.29%	3.40%	4.39%	5.19%	5.90%	6.44%	6.84%	7.26%	7.59%
Capital Equipment	1.59%	3.30%	5.00%	6.54%	7.85%	8.88%	9.72%	10.45%	11.13%	11.79%
Chemicals, Plastics, & Rubber	1.04%	2.20%	3.34%	4.37%	5.36%	6.16%	6.71%	7.16%	7.63%	8.09%
Construction & Building	3.12%	6.46%	9.67%	12.56%	15.15%	17.51%	19.62%	21.75%	23.65%	25.14%
Consumer goods: Durable	3.04%	6.51%	10.09%	13.46%	16.77%	19.50%	21.31%	22.92%	24.61%	26.19%
Consumer goods: Non-durable	3.94%	8.19%	12.11%	15.30%	17.93%	20.08%	21.99%	23.66%	24.94%	25.98%
Containers, Packaging, & Glass	2.22%	4.66%	6.74%	8.52%	10.07%	11.41%	12.67%	13.90%	14.82%	15.74%
Energy: Electricity	3.17%	6.35%	9.30%	11.66%	13.96%	16.17%	18.33%	20.39%	22.46%	24.70%
Energy: Oil & Gas	2.75%	5.26%	7.47%	9.33%	10.89%	12.21%	13.38%	14.43%	15.43%	16.32%
Environmental Industries	2.99%	5.75%	7.93%	9.68%	11.03%	12.17%	12.96%	13.83%	14.78%	15.81%
Finance*	1.09%	2.19%	3.24%	4.09%	4.81%	5.44%	5.96%	6.47%	7.02%	7.55%
Forest Products & Paper	2.34%	4.58%	6.77%	8.78%	10.52%	12.06%	13.35%	14.53%	15.45%	16.22%
Healthcare & Pharmaceuticals	1.66%	3.32%	4.82%	6.18%	7.35%	8.45%	9.46%	10.22%	10.98%	11.56%
High Tech Industries	1.57%	3.20%	4.65%	5.98%	7.07%	8.05%	8.99%	9.94%	10.90%	11.83%
Hotel, Gaming, & Leisure	3.91%	7.77%	11.18%	14.25%	16.90%	19.19%	21.17%	23.03%	24.69%	26.28%
Insurance	0.28%	0.58%	0.89%	1.23%	1.56%	1.90%	2.22%	2.53%	2.86%	3.17%
Media: Advertising, Printing &	5.90%	11.71%	16.80%	20.96%	24.75%	28.06%	30.63%	32.57%	34.45%	36.54%
Media: Broadcasting & Subscription	3.24%	6.63%	10.10%	13.08%	15.85%	18.29%	20.34%	22.14%	23.71%	25.11%
Media: Diversified & Production	2.23%	4.40%	6.38%	7.70%	8.63%	9.42%	9.88%	10.26%	10.68%	11.14%
Metals & Mining	3.02%	6.06%	8.84%	11.17%	13.05%	14.71%	16.27%	17.67%	19.02%	20.33%
REIT	0.50%	1.02%	1.47%	1.89%	2.23%	2.56%	2.83%	3.07%	3.35%	3.67%
Retail	3.37%	6.72%	9.98%	12.95%	15.59%	17.84%	19.71%	21.35%	22.81%	24.10%
Services: Business	2.35%	4.77%	7.03%	9.13%	11.01%	12.60%	14.02%	15.21%	16.39%	17.55%
Services: Consumer	2.55%	5.12%	7.51%	9.84%	11.99%	14.09%	15.90%	17.57%	19.15%	20.71%
Sovereign & Public Finance	0.43%	0.86%	1.24%	1.54%	1.76%	1.92%	1.97%	1.97%	1.97%	1.97%
Telecommunications	2.30%	4.53%	6.51%	8.12%	9.39%	10.31%	11.00%	11.48%	11.89%	12.23%
Transportation: Cargo	2.07%	3.86%	5.42%	6.77%	7.80%	8.73%	9.45%	10.07%	10.64%	11.26%
Transportation: Consumer	2.60%	5.08%	7.54%	9.97%	12.13%	13.92%	15.53%	17.14%	18.79%	20.41%
Utilities: Electric	0.17%	0.32%	0.47%	0.61%	0.75%	0.87%	0.98%	1.09%	1.22%	1.33%
Utilities: Oil & Gas	0.15%	0.27%	0.41%	0.55%	0.67%	0.81%	0.93%	1.01%	1.09%	1.17%
Utilities: Water	0.18%	0.37%	0.52%	0.64%	0.76%	0.76%	0.76%	0.76%	0.76%	0.76%
Wholesale	3.26%	6.95%	10.34%	12.82%	14.76%	16.34%	17.18%	17.90%	18.53%	19.16%

* Non-bank and non-insurance financial companies

Source: Moody's Investors Service

Exhibit 44

Annual default rates by broad industry group, 1970-2023

	Aerospace & Defense	Automotive	Banking	Beverage, Food, & Tobacco	Capital Equipment	Chemicals, Plastics, & Rubber	Construction & Building	Consumer goods: Durable	Consumer goods: Non-durable	Containers, Packaging, & Glass	Energy: Electricity	Energy: Oil & Gas	Environmental Industries	Finance*	Forest Products & Paper	Healthcare & Pharmaceuticals	High Tech Industries
1970	0.00%	0.00%		0.00%	0.00%	0.00%	0.00%		0.00%			0.00%			0.00%	0.00%	3.70%
1971	0.00%	0.00%		0.00%	0.00%	0.00%	0.00%		0.00%			0.00%			0.00%	0.00%	0.00%
1972	0.00%	0.00%		0.00%	1.08%	0.00%	0.00%	0.00%	0.00%			0.00%			0.00%	0.00%	0.00%
1973	0.00%	0.00%		0.00%	0.00%	0.00%	0.00%	0.00%	0.00%			0.00%			0.00%	0.00%	0.00%
1974	0.00%	0.00%		0.00%	1.10%	0.00%	0.00%	0.00%	0.00%			0.00%			0.00%	0.00%	0.00%
1975	0.00%	3.03%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%			0.00%		0.00%	0.00%	4.76%	0.00%
1976	0.00%	0.00%	0.00%	0.00%	1.02%	0.00%	0.00%	0.00%	2.63%			0.00%		0.00%	0.00%	0.00%	0.00%
1977	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.70%			0.00%		0.00%	0.00%	0.00%	0.00%
1978	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.70%			1.32%		0.00%	0.00%	0.00%	3.13%
1979	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%			0.00%		0.00%	0.00%	0.00%	2.78%
1980	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	3.23%	0.00%	0.00%			0.00%		0.00%	0.00%	0.00%	2.70%
1981	0.00%	0.00%	0.00%	0.00%	1.18%	0.00%	0.00%	0.00%	0.00%			0.00%		0.00%	0.00%	0.00%	0.00%
1982	0.00%	2.86%	0.00%	0.00%	0.00%	0.00%	3.33%	0.00%	0.00%			0.97%		0.00%	0.00%	0.00%	4.00%
1983	0.00%	7.69%	0.00%	1.45%	0.00%	0.00%	0.00%	0.00%	0.00%		0.00%	3.19%		0.00%	0.00%	0.00%	1.75%
1984	0.00%	2.78%	0.00%	1.43%	0.00%	0.00%	0.00%	0.00%	0.00%		0.00%	3.98%		0.00%	0.00%	2.22%	4.84%
1985	0.00%	0.00%	0.00%	0.00%	1.75%	2.08%	0.00%	0.00%	1.69%		0.00%	3.41%		0.00%	0.00%	0.00%	1.52%
1986	2.86%	0.00%	0.00%	1.19%	0.00%	1.89%	1.69%	5.88%	0.00%		0.00%	8.31%		0.00%	0.00%	1.41%	0.00%
1987	0.00%	1.75%	0.28%	0.00%	2.43%	0.00%	4.44%	0.00%	0.00%	0.00%	0.00%	5.53%		0.00%	0.00%	3.69%	1.11%
1988	2.33%	0.00%	1.45%	0.00%	0.00%	1.67%	2.70%	4.35%	3.64%	0.00%	4.76%	2.13%	0.00%	0.00%	0.00%	3.54%	1.68%
1989	0.00%	7.41%	1.49%	2.20%	0.00%	0.00%	8.77%	0.00%	3.80%	10.00%	0.00%	0.00%	0.00%	4.68%	0.00%	2.52%	1.76%
1990	0.00%	3.85%	1.84%	5.49%	2.61%	0.00%	16.93%	4.76%	9.90%	0.00%	8.33%	1.47%	0.00%	0.00%	0.00%	1.15%	5.68%
1991	2.17%	4.19%	1.40%	0.00%	4.54%	0.00%	7.55%	5.88%	6.09%	0.00%	4.35%	3.06%	0.00%	0.00%	5.41%	4.88%	2.22%
1992	0.00%	0.00%	0.32%	1.20%	3.00%	0.00%	0.00%	0.00%	5.98%	7.69%	0.00%	0.79%	0.00%	0.00%	0.00%	1.45%	3.35%
1993	2.22%	0.00%	0.30%	1.11%	4.93%	0.00%	1.69%	0.00%	2.00%	0.00%	0.00%	2.25%	0.00%	0.00%	0.00%	0.00%	0.00%
1994	0.00%	0.00%	0.00%	1.03%	0.85%	0.00%	0.00%	0.00%	0.00%	0.00%	4.00%	0.00%	0.00%	0.00%	2.17%	1.02%	3.00%
1995	0.00%	0.00%	0.00%	1.89%	0.83%	0.00%	5.02%	0.00%	7.63%	0.00%	0.00%	0.63%	0.00%	1.98%	1.79%	0.98%	1.87%
1996	0.00%	0.00%	0.00%	0.82%	0.72%	0.00%	1.05%	0.00%	4.37%	5.56%	0.00%	0.57%	5.00%	0.00%	0.00%	0.00%	1.03%
1997	0.00%	1.35%	0.00%	0.76%	0.00%	0.00%	1.11%	0.00%	6.43%	4.35%	0.00%	0.00%	0.00%	0.00%	0.00%	0.85%	1.79%
1998	0.00%	1.09%	0.30%	0.68%	0.00%	1.59%	0.00%	0.00%	4.42%	0.00%	0.00%	1.28%	5.56%	1.93%	1.18%	3.03%	0.73%
1999	0.00%	0.92%	0.14%	1.82%	1.40%	1.49%	6.38%	10.99%	4.29%	0.00%	0.00%	6.13%	4.35%	1.82%	1.06%	5.26%	2.67%
2000	1.56%	5.07%	0.00%	2.83%	4.19%	1.45%	1.94%	3.57%	14.34%	4.55%	0.00%	0.39%	15.79%	0.60%	6.94%	4.66%	3.17%

	Aerospace & Defense	Automotive	Banking	Beverage, Food, & Tobacco	Capital Equipment	Chemicals, Plastics, & Rubber	Construction & Building	Consumer goods: Durable	Consumer goods: Non-durable	Containers, Packaging, & Glass	Energy: Electricity	Energy: Oil & Gas	Environmental Industries	Finance*	Forest Products & Paper	Healthcare & Pharmaceuticals	High Tech Industries
2001	4.59%	5.36%	0.07%	2.23%	5.18%	5.68%	4.71%	3.57%	15.50%	11.54%	1.79%	1.17%	11.11%	3.58%	7.53%	1.47%	4.50%
2002	0.00%	1.74%	0.56%	1.62%	2.05%	0.00%	5.04%	3.23%	6.01%	4.88%	8.87%	2.25%	13.33%	0.00%	3.31%	0.70%	2.72%
2003	0.00%	2.60%	0.08%	1.66%	2.52%	4.36%	2.41%	0.00%	2.74%	0.00%	11.85%	0.00%	0.00%	0.62%	1.10%	2.85%	3.47%
2004	0.00%	2.51%	0.00%	1.65%	0.96%	0.00%	2.27%	2.78%	4.49%	2.08%	0.00%	0.34%	0.00%	0.00%	1.18%	0.73%	0.00%
2005	0.00%	5.22%	0.07%	0.00%	0.95%	0.00%	0.00%	4.17%	1.59%	4.40%	4.00%	0.35%	0.00%	0.64%	1.27%	0.00%	0.00%
2006	0.00%	4.89%	0.00%	1.13%	1.49%	0.00%	2.05%	1.92%	3.04%	2.33%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.56%
2007	0.00%	1.69%	0.00%	1.07%	0.49%	0.00%	1.02%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.48%	0.76%	0.00%
2008	1.47%	4.38%	3.05%	1.59%	2.68%	1.91%	5.20%	2.00%	6.18%	10.48%	0.00%	1.59%	5.26%	3.90%	6.40%	0.78%	0.00%
2009	4.39%	19.67%	0.93%	4.33%	10.35%	7.41%	14.56%	21.42%	8.25%	9.04%	1.96%	1.93%	0.00%	5.34%	15.38%	0.74%	6.71%
2010	0.00%	1.16%	0.48%	0.57%	3.53%	0.00%	1.74%	0.00%	3.74%	0.00%	6.12%	0.96%	10.00%	2.55%	3.03%	1.42%	1.49%
2011	0.00%	1.01%	0.55%	0.00%	1.90%	0.00%	2.26%	0.00%	0.00%	5.14%	4.00%	1.27%	0.00%	0.58%	1.41%	0.00%	0.00%
2012	1.23%	0.00%	0.63%	1.64%	2.24%	1.40%	0.00%	0.00%	0.00%	2.22%	5.56%	0.89%	4.35%	1.72%	2.90%	0.68%	1.13%
2013	1.20%	0.93%	1.29%	1.61%	0.57%	0.64%	2.48%	0.00%	2.04%	2.13%	1.85%	1.59%	4.55%	0.00%	0.00%	1.84%	0.56%
2014	1.25%	0.00%	0.74%	0.00%	0.00%	0.58%	0.53%	2.22%	0.00%	0.00%	7.27%	0.49%	0.00%	0.50%	1.52%	0.00%	2.59%
2015	2.59%	0.00%	1.56%	1.95%	0.00%	0.00%	2.32%	2.00%	4.64%	0.00%	0.00%	7.06%	4.17%	0.49%	3.13%	0.00%	0.44%
2016	1.25%	0.83%	0.23%	0.50%	0.83%	1.62%	1.40%	2.33%	1.45%	0.00%	1.96%	14.66%	7.85%	2.66%	2.99%	0.62%	0.78%
2017	2.60%	0.00%	0.84%	0.00%	0.84%	0.98%	0.46%	0.00%	0.00%	0.00%	5.45%	6.85%	10.53%	2.38%	9.29%	0.61%	1.20%
2018	0.00%	0.00%	0.08%	1.15%	0.00%	0.00%	4.08%	5.88%	1.43%	0.00%	7.12%	3.12%	0.00%	1.34%	1.69%	1.06%	0.36%
2019	1.39%	0.90%	0.31%	1.11%	1.99%	0.47%	2.17%	3.92%	4.11%	0.00%	0.00%	5.08%	0.00%	0.81%	2.00%	2.63%	0.70%
2020	4.15%	3.45%	0.39%	2.20%	2.50%	1.43%	3.30%	7.88%	7.82%	1.72%	7.81%	13.10%	0.00%	0.00%	0.00%	4.94%	2.51%
2021	0.00%	0.00%	0.08%	0.00%	1.64%	0.00%	4.29%	3.92%	1.35%	0.00%	8.95%	1.76%	0.00%	0.69%	0.00%	0.92%	0.00%
2022	2.67%	0.85%	3.45%	0.00%	0.79%	2.27%	9.51%	0.00%	6.02%	1.47%	5.26%	1.17%	0.00%	0.92%	2.04%	2.71%	0.30%
2023	5.63%	3.46%	0.33%	2.01%	1.97%	2.50%	5.08%	12.15%	8.56%	2.80%	2.86%	0.62%	3.45%	1.89%	4.35%	5.20%	2.32%

	Hotel, Gaming, & Leisure	Insurance	Media: Advertising, Printing & Publishing	Media: Broadcasting & Subscription	Media: Diversified & Production	Metals & Mining	REIT	Retail	Services: Business	Services: Consumer	Sovereign & Public Finance	Telecommunications	Transportation: Cargo	Transportation: Consumer	Utilities: Electric	Utilities: Oil & Gas	Utilities: Water	Wholesale
1970			0.00%			0.00%		0.00%				0.00%	18.46%	0.00%	0.00%	0.00%		0.00%
1971			0.00%			0.00%		0.00%				0.00%	1.92%	4.76%	0.00%	0.00%		0.00%
1972						0.00%		0.00%	0.00%			0.00%	3.77%	0.00%	0.00%	0.00%		0.00%
1973						0.00%		3.51%	0.00%			0.00%	2.00%	0.00%	0.00%	0.00%		
1974						0.00%		3.54%	0.00%			0.00%	0.00%	0.00%	0.00%	0.00%		
1975						0.00%		1.79%	0.00%			0.00%	0.00%	0.00%	0.00%	0.00%		
1976						0.00%		0.00%	0.00%			0.00%	0.00%	0.00%	0.00%	0.00%		
1977						0.00%		0.00%	0.00%			0.00%	2.25%	0.00%	0.00%	0.00%		
1978						0.00%		1.89%	0.00%			0.00%	0.00%	0.00%	0.00%	0.00%		
1979	0.00%					0.00%		0.00%	0.00%			0.00%	0.00%	0.00%	0.00%	0.00%		
1980	0.00%	0.00%				0.00%		0.00%	0.00%			0.00%	0.00%	5.00%	0.00%	0.00%		
1981	0.00%	0.00%			0.00%	0.00%		0.00%	0.00%			0.00%	1.15%	0.00%	0.00%	0.00%		
1982	0.00%	0.00%			0.00%	2.38%		6.00%	0.00%			1.22%	1.20%	5.56%	0.00%	0.00%		0.00%
1983	0.00%	0.00%		0.00%	0.00%	1.75%		0.00%	0.00%			0.00%	0.00%	19.05%	0.00%	0.00%	0.00%	0.00%
1984	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%		0.00%	5.26%			0.00%	1.19%	0.00%	0.00%	0.00%	0.00%	0.00%
1985	5.26%	3.45%	0.00%	0.00%	0.00%	3.28%		0.00%	0.00%			0.00%	1.18%	0.00%	0.00%	0.00%	0.00%	0.00%
1986	2.38%	0.00%	0.00%	3.85%	5.00%	18.45%		0.00%	0.00%			0.00%	1.14%	8.17%	0.00%	0.00%	0.00%	4.76%
1987	6.49%	0.00%	5.88%	0.00%	0.00%	0.00%	0.00%	1.05%	7.69%			0.00%	1.08%	0.00%	0.00%	1.59%	0.00%	4.35%
1988	2.04%	1.47%	0.00%	4.26%	3.57%	0.00%	0.00%	1.92%	3.45%		0.00%	0.00%	0.00%	0.00%	0.54%	0.00%		0.00%
1989	13.81%	1.35%	0.00%	4.35%	11.27%	0.00%	5.88%	2.54%	3.13%	0.00%	10.53%	0.98%	1.12%	3.57%	0.00%	0.00%		0.00%
1990	24.62%	0.00%	11.76%	6.43%	4.55%	3.81%	0.00%	6.39%	10.81%	11.11%	0.00%	0.00%	0.00%	24.14%	0.00%	0.00%		0.00%
1991	2.70%	1.22%	0.00%	4.94%	4.35%	1.96%	0.00%	11.58%	0.00%		0.00%	0.89%	2.52%	30.00%	0.54%	1.79%		0.00%
1992	5.27%	0.00%	0.00%	10.53%	9.77%	5.92%	6.67%	2.89%	0.00%		0.00%	0.00%	0.00%	0.00%	1.55%	0.00%		0.00%
1993	2.27%	0.00%	0.00%	8.74%	0.00%	1.69%	0.00%	2.89%	2.70%		0.00%	0.00%	0.00%	0.00%	0.00%	0.00%		0.00%
1994	5.26%	0.00%	0.00%	2.38%	0.00%	0.00%	0.00%	2.51%	2.33%		0.00%	0.00%	1.25%	2.94%	0.00%	0.00%	0.00%	3.70%
1995	2.61%	0.81%	0.00%	0.00%	0.00%	0.00%	0.00%	1.47%	2.50%		0.00%	0.00%	0.00%	5.64%	0.00%	0.00%	0.00%	3.57%
1996	2.26%	0.00%	0.00%	0.00%	17.65%	0.00%	0.00%	0.75%	2.22%	0.00%	0.00%	0.68%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1997	2.04%	0.62%	0.00%	2.30%	0.00%	0.00%	0.00%	3.59%	0.00%	0.00%	0.00%	0.53%	1.20%	0.00%	0.00%	0.00%	0.00%	2.70%
1998	2.84%	0.00%	2.33%	2.51%	5.26%	2.89%	1.03%	5.41%	0.00%	0.00%	0.00%	0.94%	1.06%	0.00%	0.00%	0.00%	0.00%	7.14%
1999	6.41%	0.00%	3.77%	3.68%	0.00%	4.14%	0.00%	1.92%	4.33%	7.41%	4.65%	1.97%	7.68%	0.00%	0.00%	1.49%	0.00%	5.60%
2000	6.06%	1.46%	1.54%	1.56%	3.23%	6.81%	0.00%	4.83%	1.98%	10.00%	4.44%	2.99%	4.13%	5.17%	0.00%	0.00%	0.00%	10.06%

	Hotel, Gaming, & Leisure	Insurance	Media: Advertising, Printing & Publishing	Media: Broadcasting & Subscription	Media: Diversified & Production	Metals & Mining	REIT	Retail	Services: Business	Services: Consumer	Sovereign & Public Finance	Telecommunications	Transportation: Cargo	Transportation: Consumer	Utilities: Electric	Utilities: Oil & Gas	Utilities: Water	Wholesale
2001	8.63%	0.00%	3.17%	3.85%	3.33%	13.83%	0.00%	6.53%	4.20%	3.57%	2.38%	10.14%	3.99%	1.69%	0.73%	0.00%	0.00%	12.54%
2002	1.82%	0.44%	3.43%	14.14%	7.85%	7.40%	0.88%	2.76%	2.13%	3.45%	0.00%	14.19%	4.61%	8.51%	0.00%	1.23%	3.57%	8.62%
2003	2.91%	0.78%	1.67%	5.48%	0.00%	4.48%	0.00%	2.87%	1.10%	0.00%	0.00%	8.15%	3.74%	3.42%	0.32%	1.28%	0.00%	9.33%
2004	4.06%	0.37%	0.00%	3.89%	0.00%	1.98%	0.00%	0.71%	2.26%	0.00%	0.00%	1.52%	0.00%	3.28%	0.30%	0.00%	0.00%	2.78%
2005	0.95%	0.00%	0.00%	1.05%	0.00%	0.89%	0.00%	1.57%	0.00%	0.00%	0.00%	0.43%	0.00%	7.81%	0.30%	0.00%	0.00%	2.86%
2006	0.83%	0.67%	0.00%	3.19%	0.00%	0.00%	0.00%	0.74%	0.00%	0.00%	0.00%	0.00%	3.67%	0.00%	0.00%	0.00%	0.00%	2.50%
2007	0.78%	0.00%	3.15%	0.00%	0.00%	0.00%	0.00%	2.23%	0.74%	2.63%	0.00%	0.91%	1.16%	0.00%	0.00%	0.00%	0.00%	0.00%
2008	9.93%	0.00%	7.37%	2.01%	6.90%	0.80%	2.44%	3.58%	0.64%	0.00%	0.00%	2.43%	5.04%	0.00%	0.00%	0.00%	0.00%	0.00%
2009	17.65%	0.32%	35.06%	23.18%	0.00%	10.23%	5.85%	3.72%	3.13%	7.26%	0.00%	5.94%	7.37%	3.10%	0.29%	0.00%	0.00%	10.53%
2010	4.38%	0.37%	14.25%	2.14%	0.00%	0.00%	0.00%	4.06%	1.81%	0.00%	0.00%	1.56%	1.47%	0.00%	0.00%	1.00%	0.00%	2.63%
2011	3.40%	0.37%	8.38%	0.00%	0.00%	0.00%	0.82%	3.26%	1.24%	1.89%	0.00%	0.00%	6.10%	5.35%	0.00%	0.00%	0.00%	0.00%
2012	3.80%	0.37%	15.22%	1.04%	0.00%	2.20%	0.00%	2.14%	1.62%	6.95%	0.00%	1.72%	1.19%	1.85%	0.54%	0.00%	0.00%	0.00%
2013	2.31%	0.38%	19.27%	0.91%	0.00%	1.95%	0.70%	1.35%	1.94%	0.00%	0.00%	1.67%	2.27%	0.00%	0.00%	0.00%	0.00%	0.00%
2014	1.49%	0.00%	8.22%	0.00%	0.00%	5.29%	0.00%	2.35%	2.47%	3.44%	1.11%	1.63%	1.01%	0.00%	0.00%	0.00%	0.00%	0.00%
2015	2.22%	0.72%	2.22%	1.03%	3.33%	6.79%	0.00%	2.11%	2.56%	4.75%	0.00%	0.55%	0.00%	0.00%	0.25%	0.00%	0.00%	1.61%
2016	0.69%	0.00%	16.19%	0.00%	0.00%	9.30%	0.00%	3.67%	1.46%	0.00%	0.00%	1.57%	2.12%	1.52%	0.25%	0.00%	0.00%	5.94%
2017	0.00%	0.00%	0.00%	4.18%	2.94%	2.03%	0.00%	7.08%	3.98%	1.41%	0.00%	1.57%	3.11%	0.00%	0.00%	0.00%	0.00%	2.96%
2018	0.00%	0.00%	6.58%	1.05%	0.00%	1.93%	0.00%	7.38%	2.34%	0.00%	0.00%	1.02%	2.02%	0.00%	0.24%	0.00%	0.00%	4.37%
2019	2.06%	0.00%	3.33%	1.11%	0.00%	3.17%	0.00%	5.56%	2.86%	1.52%	1.08%	3.10%	2.91%	0.00%	0.47%	0.00%	0.00%	2.86%
2020	7.49%	0.00%	14.18%	0.00%	0.00%	2.65%	0.90%	12.51%	7.03%	10.02%	0.00%	3.23%	5.04%	4.07%	0.25%	0.00%	0.00%	5.54%
2021	1.22%	0.00%	4.00%	0.00%	5.13%	0.67%	0.44%	1.18%	1.89%	0.00%	0.00%	1.13%	1.05%	0.00%	0.00%	0.00%	0.00%	1.39%
2022	2.25%	0.00%	8.68%	1.23%	2.38%	8.47%	0.89%	2.75%	1.07%	1.03%	0.00%	3.39%	4.09%	3.45%	0.73%	0.83%	2.33%	1.27%
2023	5.29%	0.00%	6.16%	3.93%	2.50%	0.83%	0.49%	4.95%	3.70%	3.42%	0.00%	6.62%	2.12%	2.22%	0.49%	0.00%	0.00%	1.14%

* Non-bank and non-insurance financial companies.
Source: Moody's Investors Service

Exhibit 45

Annual volume-weighted corporate bond default rates by letter rating, 1994-2023

Year	Aaa	Aa	A	Baa	Ba	B	Caa-C	IG	SG	All
1994	0.00%	0.00%	0.00%	0.00%	0.00%	2.29%	22.76%	0.00%	2.13%	0.43%
1995	0.00%	0.00%	0.00%	0.00%	0.00%	4.32%	12.41%	0.00%	3.08%	0.71%
1996	0.00%	0.00%	0.00%	0.00%	0.00%	2.29%	18.46%	0.00%	2.34%	0.67%
1997	0.00%	0.00%	0.00%	0.00%	0.05%	3.07%	6.42%	0.00%	1.91%	0.49%
1998	0.00%	0.00%	0.00%	0.00%	0.65%	3.22%	11.79%	0.00%	2.87%	0.85%
1999	0.00%	0.00%	0.00%	0.14%	1.15%	5.50%	21.19%	0.03%	5.88%	1.13%
2000	0.00%	0.00%	0.00%	0.62%	1.95%	5.30%	19.99%	0.14%	5.63%	1.30%
2001	0.00%	0.00%	0.72%	0.96%	1.28%	14.34%	50.28%	0.55%	15.24%	2.95%
2002	0.00%	0.00%	2.72%	2.30%	5.25%	16.06%	78.06%	1.86%	21.56%	4.90%
2003	0.00%	0.00%	0.00%	0.00%	3.62%	3.69%	16.17%	0.00%	5.95%	1.04%
2004	0.00%	0.00%	0.00%	0.00%	0.00%	0.26%	9.19%	0.00%	1.88%	0.33%
2005	0.00%	0.00%	0.00%	0.20%	0.00%	0.76%	18.50%	0.07%	3.79%	0.75%
2006	0.00%	0.00%	0.00%	0.00%	0.03%	1.07%	4.00%	0.00%	1.05%	0.21%
2007	0.00%	0.00%	0.00%	0.00%	0.00%	0.11%	4.63%	0.00%	0.82%	0.16%
2008	0.00%	0.82%	2.37%	1.11%	8.27%	1.05%	11.16%	1.47%	5.80%	2.22%
2009	0.00%	0.00%	0.01%	0.75%	3.45%	5.19%	38.84%	0.22%	16.15%	2.74%
2010	0.00%	0.00%	0.15%	0.02%	0.00%	0.13%	6.22%	0.08%	1.70%	0.35%
2011	0.00%	0.11%	0.00%	0.40%	0.00%	0.17%	7.16%	0.15%	1.46%	0.40%
2012	0.00%	0.00%	0.00%	0.02%	0.00%	0.22%	11.19%	0.01%	2.05%	0.40%
2013	0.00%	0.00%	0.06%	0.02%	0.00%	0.87%	4.37%	0.04%	1.09%	0.26%
2014	0.00%	0.00%	0.02%	0.00%	0.11%	0.32%	9.26%	0.01%	1.74%	0.39%
2015	0.00%	0.00%	0.00%	0.00%	0.81%	1.38%	17.04%	0.00%	3.47%	0.79%
2016	0.00%	0.00%	0.00%	0.00%	0.13%	0.94%	17.97%	0.00%	3.27%	0.69%
2017	0.00%	0.00%	0.00%	0.00%	0.33%	0.28%	9.77%	0.00%	1.82%	0.36%
2018	0.00%	0.00%	0.00%	0.00%	0.00%	0.56%	11.85%	0.00%	1.92%	0.35%
2019	0.00%	0.00%	0.00%	0.42%	0.00%	1.38%	16.45%	0.22%	2.82%	0.66%
2020	0.00%	0.00%	0.00%	0.02%	0.00%	3.63%	35.52%	0.01%	6.26%	1.02%
2021	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.43%	0.00%	0.31%	0.06%
2022	0.00%	0.00%	0.00%	0.27%	0.97%	0.35%	8.21%	0.14%	1.56%	0.40%
2023	0.00%	0.00%	0.08%	0.03%	0.00%	0.06%	17.03%	0.05%	2.11%	0.38%
Mean	0.00%	0.03%	0.20%	0.24%	0.94%	2.63%	17.28%	0.17%	4.25%	0.91%
Median	0.00%	0.00%	0.00%	0.01%	0.02%	1.06%	12.13%	0.01%	2.23%	0.58%
Std Dev	0.00%	0.15%	0.65%	0.49%	1.89%	3.83%	15.68%	0.42%	4.91%	1.03%
Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.43%	0.00%	0.31%	0.06%
Max	0.00%	0.82%	2.72%	2.30%	8.27%	16.06%	78.06%	1.86%	21.56%	4.90%

Source: Moody's Investors Service

Exhibit 46

Average volume-weighted corporate bond default rates by letter rating, 1994-2023

Rating\Year	1	2	3	4	5
Aaa	0.0%	0.0%	0.0%	0.0%	0.0%
Aa	0.0%	0.1%	0.1%	0.1%	0.2%
A	0.2%	0.3%	0.5%	0.6%	0.7%
Baa	0.1%	0.2%	0.3%	0.5%	0.7%
Ba	0.5%	1.2%	1.8%	2.6%	3.2%
B	2.0%	4.7%	7.5%	9.8%	11.3%
Caa-C	14.6%	24.6%	31.5%	35.0%	37.1%
IG	0.1%	0.2%	0.4%	0.5%	0.6%
SG	3.2%	6.2%	8.7%	10.5%	11.7%
All	0.7%	1.4%	1.9%	2.4%	2.7%

Source: Moody's Investors Service

Exhibit 47

Cumulative issuer-weighted default rates by annual cohort, 1970-2023

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Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	39	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.70%	2.70%	2.70%
Aa	77	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.45%	1.45%	1.45%	2.94%	2.94%	2.94%	2.94%
A	254	0.00%	0.00%	0.00%	0.41%	0.41%	0.84%	0.84%	0.84%	0.84%	0.84%	1.29%	1.29%	1.29%	1.29%	1.29%	1.29%	1.79%	1.79%	2.84%	2.84%
Baa	372	0.54%	0.54%	0.54%	1.12%	1.42%	1.42%	1.73%	2.37%	3.04%	3.04%	3.40%	3.40%	4.59%	5.00%	5.42%	5.87%	7.25%	8.29%	9.35%	9.92%
Ba	238	4.24%	5.12%	5.57%	6.05%	7.04%	8.07%	8.61%	9.21%	10.46%	10.46%	10.46%	12.02%	14.56%	15.49%	15.49%	17.58%	21.08%	23.61%	23.61%	25.08%
B	36	19.44%	19.44%	22.32%	22.32%	22.32%	22.32%	22.32%	22.32%	22.32%	22.32%	22.32%	22.32%	30.50%	30.50%	30.50%	30.50%	36.29%	36.29%	36.29%	36.29%
Caa-C	16	50.00%	56.25%	75.00%	81.25%	81.25%	81.25%	81.25%	90.63%	90.63%	90.63%	90.63%	90.63%	90.63%	90.63%	90.63%	90.63%	90.63%	90.63%	90.63%	90.63%
IG	742	0.27%	0.27%	0.27%	0.70%	0.84%	0.99%	1.14%	1.44%	1.76%	1.76%	2.09%	2.09%	2.61%	2.97%	3.15%	3.34%	4.30%	4.90%	5.73%	5.95%
SG	290	8.68%	9.76%	11.60%	12.38%	13.19%	14.03%	14.47%	15.44%	16.45%	16.45%	16.45%	17.69%	21.01%	21.73%	21.73%	23.35%	26.97%	28.93%	28.93%	30.10%
All	1032	2.63%	2.93%	3.43%	3.95%	4.27%	4.60%	4.82%	5.28%	5.76%	5.76%	6.01%	6.28%	7.37%	7.79%	7.94%	8.39%	9.79%	10.61%	11.29%	11.65%

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Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	40	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.70%	2.70%	2.70%	2.70%
Aa	74	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.75%
A	282	0.00%	0.00%	0.37%	0.37%	0.74%	0.74%	0.74%	0.74%	0.74%	1.14%	1.14%	1.55%	1.96%	1.96%	1.96%	2.40%	2.40%	3.30%	3.30%	3.30%
Baa	398	0.00%	0.00%	0.53%	0.80%	1.07%	1.37%	1.97%	2.59%	2.59%	2.93%	2.93%	4.06%	4.45%	4.85%	5.28%	6.58%	7.55%	8.55%	9.08%	10.22%
Ba	228	0.89%	1.33%	1.82%	2.81%	3.84%	4.38%	4.99%	6.24%	6.24%	6.24%	7.80%	10.32%	11.25%	11.25%	13.31%	17.94%	20.49%	20.49%	21.96%	21.96%
B	27	0.00%	3.85%	3.85%	3.85%	3.85%	3.85%	3.85%	3.85%	3.85%	3.85%	3.85%	15.87%	15.87%	15.87%	15.87%	23.51%	23.51%	23.51%	23.51%	23.51%
Caa-C	8	12.50%	50.00%	62.50%	62.50%	62.50%	62.50%	81.25%	81.25%	81.25%	81.25%	81.25%	81.25%	81.25%	81.25%	81.25%	81.25%	81.25%	81.25%	81.25%	81.25%
IG	794	0.00%	0.00%	0.39%	0.53%	0.80%	0.94%	1.22%	1.51%	1.51%	1.82%	1.82%	2.47%	2.80%	2.97%	3.15%	3.86%	4.43%	5.20%	5.40%	6.02%
SG	263	1.16%	3.11%	3.94%	4.81%	5.71%	6.17%	7.22%	8.30%	8.30%	8.30%	9.62%	13.17%	13.95%	13.95%	15.67%	20.48%	22.59%	22.59%	23.84%	23.84%
All	1057	0.29%	0.77%	1.27%	1.57%	1.99%	2.20%	2.65%	3.11%	3.11%	3.35%	3.61%	4.80%	5.20%	5.34%	5.78%	7.12%	7.92%	8.56%	8.91%	9.44%

1/1/72

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	41	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.63%	2.63%	2.63%	2.63%	2.63%
Aa	78	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.67%	1.67%
A	303	0.00%	0.34%	0.34%	0.34%	0.34%	0.34%	0.34%	0.34%	0.34%	0.34%	0.72%	1.10%	1.10%	1.10%	1.50%	1.50%	2.34%	2.34%	2.80%	3.27%
Baa	431	0.00%	0.49%	0.74%	1.25%	1.51%	2.06%	2.64%	2.64%	3.27%	3.27%	3.98%	4.34%	5.09%	5.48%	6.68%	7.57%	8.48%	9.48%	11.07%	13.83%
Ba	224	0.00%	0.48%	1.45%	2.47%	3.00%	3.59%	4.81%	4.81%	4.81%	6.29%	9.48%	10.35%	10.35%	13.24%	17.53%	19.85%	19.85%	21.19%	22.62%	28.73%
B	29	6.90%	6.90%	6.90%	6.90%	6.90%	6.90%	6.90%	6.90%	6.90%	6.90%	17.85%	17.85%	17.85%	17.85%	25.32%	25.32%	25.32%	25.32%	25.32%	25.32%
Caa-C	8	37.50%	50.00%	50.00%	50.00%	50.00%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	
IG	853	0.00%	0.36%	0.49%	0.74%	0.87%	1.13%	1.40%	1.40%	1.68%	1.68%	2.14%	2.45%	2.77%	2.93%	3.59%	4.11%	4.83%	5.21%	6.19%	7.40%
SG	261	1.92%	2.73%	3.58%	4.46%	4.91%	5.92%	6.96%	6.96%	6.96%	8.22%	12.23%	12.96%	12.96%	15.41%	19.91%	21.87%	21.87%	23.01%	24.28%	29.70%
All	1114	0.45%	0.92%	1.21%	1.60%	1.80%	2.21%	2.64%	2.64%	2.87%	3.11%	4.22%	4.60%	4.86%	5.40%	6.65%	7.39%	7.99%	8.47%	9.48%	11.22%

1/1/73																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	41	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.63%	2.63%	2.63%	2.63%	2.63%	2.63%
Aa	81	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.59%	1.59%	1.59%
A	309	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.37%	0.74%	0.74%	0.74%	1.13%	1.13%	1.95%	1.95%	2.40%	2.85%	3.31%
Baa	441	0.46%	0.69%	1.17%	1.43%	1.94%	2.48%	2.48%	3.08%	3.08%	3.75%	4.43%	5.14%	5.87%	7.01%	7.85%	8.70%	9.65%	11.65%	13.76%	13.76%
Ba	205	0.00%	1.02%	2.08%	2.63%	3.24%	4.50%	4.50%	4.50%	5.99%	10.01%	10.87%	10.87%	12.77%	17.03%	20.52%	20.52%	21.82%	23.22%	30.45%	31.96%
B	27	3.85%	3.85%	3.85%	3.85%	3.85%	3.85%	3.85%	3.85%	3.85%	15.16%	15.16%	15.16%	15.16%	22.87%	22.87%	22.87%	22.87%	22.87%	22.87%	22.87%
Caa-C	6	37.50%	37.50%	37.50%	37.50%	58.33%	58.33%	58.33%	58.33%	58.33%	58.33%	58.33%	58.33%	58.33%	58.33%	58.33%	58.33%	58.33%	58.33%		
IG	872	0.23%	0.35%	0.59%	0.71%	0.96%	1.22%	1.22%	1.49%	1.49%	1.93%	2.37%	2.68%	2.99%	3.63%	4.13%	4.82%	5.18%	6.31%	7.29%	7.49%
SG	238	1.28%	2.16%	3.09%	3.56%	4.61%	5.69%	5.69%	5.69%	6.97%	11.74%	12.48%	12.48%	14.10%	18.61%	21.58%	21.58%	22.72%	23.97%	30.47%	31.83%
All	1110	0.46%	0.73%	1.11%	1.31%	1.71%	2.13%	2.13%	2.35%	2.58%	3.78%	4.27%	4.53%	5.04%	6.26%	7.12%	7.71%	8.17%	9.31%	11.00%	11.34%
1/1/74																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	44	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Aa	89	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.27%	1.27%	1.27%	1.27%	1.27%	1.27%	1.27%	1.27%	2.74%	2.74%	2.74%	2.74%
A	307	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.37%	0.74%	0.74%	0.74%	1.12%	1.12%	1.93%	1.93%	2.37%	2.82%	3.27%	3.27%
Baa	431	0.00%	0.48%	0.73%	1.24%	1.79%	1.79%	2.38%	2.38%	3.05%	3.73%	4.44%	5.17%	5.56%	6.40%	7.27%	8.22%	9.72%	11.83%	11.83%	11.83%
Ba	197	0.51%	1.60%	2.17%	2.79%	4.09%	4.09%	4.09%	4.85%	8.92%	9.80%	9.80%	11.70%	17.93%	21.39%	21.39%	22.72%	25.53%	32.82%	34.35%	36.17%
B	29	7.16%	7.16%	7.16%	7.16%	7.16%	7.16%	7.16%	11.80%	22.18%	22.18%	22.18%	22.18%	29.96%	29.96%	29.96%	29.96%	29.96%	29.96%	29.96%	29.96%
Caa-C	3	0.00%	0.00%	0.00%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%			
IG	871	0.00%	0.24%	0.36%	0.60%	0.86%	0.86%	1.12%	1.12%	1.70%	2.14%	2.44%	2.75%	3.06%	3.56%	4.24%	4.60%	5.52%	6.48%	6.67%	6.67%
SG	229	1.33%	2.27%	2.76%	3.82%	4.93%	4.93%	4.93%	6.23%	11.03%	11.78%	11.78%	13.41%	19.68%	22.65%	22.65%	23.80%	26.34%	32.98%	34.38%	36.02%
All	1100	0.28%	0.65%	0.85%	1.25%	1.66%	1.66%	1.88%	2.11%	3.41%	3.90%	4.15%	4.67%	5.87%	6.72%	7.30%	7.76%	8.89%	10.55%	10.89%	11.07%
1/1/75																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	52	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.04%	2.04%	2.04%	2.04%	2.04%	2.04%	2.04%	2.04%
Aa	103	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.05%	1.05%	1.05%	1.05%	1.05%	1.05%	2.23%	2.23%	3.45%	3.45%	3.45%	3.45%	3.45%
A	321	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.35%	0.35%	0.35%	0.73%	0.73%	1.51%	1.51%	1.93%	2.79%	3.24%	3.72%	3.72%
Baa	412	0.00%	0.00%	0.27%	0.82%	0.82%	1.44%	1.44%	2.45%	3.14%	3.85%	4.59%	4.98%	5.83%	6.26%	7.21%	9.21%	11.30%	11.30%	11.30%	11.30%
Ba	200	1.03%	2.12%	3.27%	3.89%	3.89%	3.89%	4.59%	8.44%	9.27%	9.27%	11.09%	17.01%	20.31%	21.46%	22.73%	24.06%	29.62%	31.06%	32.74%	32.74%
B	33	6.16%	6.16%	6.16%	9.51%	9.51%	9.51%	13.44%	22.10%	22.10%	22.10%	22.10%	28.59%	28.59%	28.59%	28.59%	38.79%	48.99%	48.99%	48.99%	48.99%
Caa-C	3	0.00%	0.00%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%				
IG	888	0.00%	0.00%	0.12%	0.36%	0.36%	0.62%	0.62%	1.17%	1.59%	1.88%	2.17%	2.47%	2.94%	3.58%	3.92%	4.97%	6.06%	6.24%	6.44%	6.44%
SG	236	1.74%	2.66%	4.12%	5.14%	5.14%	5.14%	6.33%	10.80%	11.50%	11.50%	13.04%	18.91%	21.69%	22.65%	23.71%	26.02%	32.04%	33.30%	34.75%	34.75%
All	1124	0.36%	0.55%	0.93%	1.32%	1.32%	1.53%	1.75%	2.99%	3.45%	3.69%	4.18%	5.32%	6.13%	6.81%	7.25%	8.46%	10.17%	10.49%	10.84%	10.84%

1/1/76																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	66	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.69%	1.69%	1.69%	1.69%	1.69%	1.69%	1.69%	1.69%	1.69%
Aa	112	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.97%	0.97%	0.97%	0.97%	0.97%	0.97%	2.06%	2.06%	3.18%	3.18%	3.18%	3.18%	3.18%	3.18%
A	367	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.61%	0.61%	0.61%	1.27%	1.27%	2.32%	2.32%	3.07%	4.24%	4.24%	4.68%	4.68%	5.16%
Baa	371	0.00%	0.29%	0.58%	0.58%	0.91%	0.91%	2.35%	3.09%	3.86%	4.65%	5.06%	5.96%	5.96%	6.98%	9.12%	10.79%	11.36%	11.36%	11.36%	11.36%
Ba	206	1.00%	2.04%	3.16%	3.16%	3.79%	4.43%	7.20%	7.96%	7.96%	9.59%	14.85%	17.80%	18.83%	19.96%	21.15%	27.30%	28.58%	30.03%	30.03%	30.03%
B	27	0.00%	0.00%	4.00%	4.00%	4.00%	8.80%	19.53%	19.53%	19.53%	19.53%	28.47%	28.47%	28.47%	28.47%	42.78%	57.08%	57.08%	57.08%	57.08%	57.08%
Caa-C	3	0.00%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%					
IG	916	0.00%	0.11%	0.23%	0.23%	0.35%	0.35%	1.00%	1.52%	1.79%	2.07%	2.50%	2.94%	3.55%	3.88%	5.04%	6.06%	6.23%	6.42%	6.42%	6.63%
SG	236	0.87%	2.25%	3.69%	3.69%	4.23%	5.36%	8.96%	9.62%	9.62%	11.06%	16.54%	19.16%	20.07%	21.07%	23.22%	29.96%	31.15%	32.50%	32.50%	32.50%
All	1152	0.18%	0.54%	0.91%	0.91%	1.11%	1.31%	2.48%	3.03%	3.26%	3.72%	4.92%	5.68%	6.33%	6.75%	8.03%	9.80%	10.10%	10.43%	10.43%	10.61%
1/1/77																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	67	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.69%	1.69%	1.69%	1.69%	1.69%	1.69%	1.69%	1.69%	1.69%	1.69%
Aa	119	0.00%	0.00%	0.00%	0.00%	0.00%	0.91%	0.91%	0.91%	0.91%	0.91%	0.91%	1.94%	1.94%	3.00%	3.00%	3.00%	3.00%	3.00%	3.00%	3.00%
A	388	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.58%	0.58%	0.58%	1.20%	1.20%	2.84%	2.84%	3.90%	5.00%	5.00%	5.42%	5.42%	5.88%	5.88%
Baa	355	0.29%	0.60%	0.60%	0.60%	0.60%	2.06%	2.81%	3.59%	4.40%	4.82%	5.73%	5.73%	6.77%	8.40%	10.09%	10.66%	10.66%	10.66%	10.66%	10.66%
Ba	195	0.54%	1.67%	1.67%	2.27%	2.92%	5.72%	6.48%	6.48%	8.11%	13.37%	16.30%	17.32%	18.44%	19.64%	25.81%	27.09%	28.55%	28.55%	28.55%	28.55%
B	31	3.23%	6.56%	6.56%	10.63%	14.88%	24.90%	24.90%	24.90%	24.90%	33.24%	33.24%	33.24%	33.24%	46.59%	59.94%	59.94%	59.94%	59.94%	59.94%	59.94%
Caa-C	3	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%						
IG	929	0.11%	0.22%	0.22%	0.22%	0.22%	0.85%	1.36%	1.62%	1.89%	2.31%	2.74%	3.63%	3.95%	5.07%	6.06%	6.23%	6.41%	6.41%	6.62%	6.62%
SG	229	1.36%	2.79%	2.79%	3.84%	4.95%	8.55%	9.21%	9.21%	10.64%	16.05%	18.63%	19.53%	20.51%	22.64%	29.34%	30.52%	31.85%	31.85%	31.85%	31.85%
All	1158	0.35%	0.72%	0.72%	0.91%	1.11%	2.25%	2.78%	3.00%	3.45%	4.63%	5.37%	6.26%	6.66%	7.91%	9.62%	9.91%	10.24%	10.24%	10.42%	10.42%
1/1/78																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	71	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.56%	1.56%	1.56%	3.15%	3.15%	3.15%	3.15%	3.15%	3.15%	3.15%	3.15%
Aa	131	0.00%	0.00%	0.00%	0.00%	0.83%	0.83%	0.83%	0.83%	0.83%	0.83%	1.74%	1.74%	1.74%	1.74%	1.74%	1.74%	1.74%	1.74%	1.74%	1.74%
A	380	0.00%	0.00%	0.00%	0.00%	0.00%	0.58%	0.58%	0.58%	1.21%	1.21%	2.55%	2.55%	4.00%	4.74%	4.74%	5.17%	5.17%	5.64%	5.64%	5.64%
Baa	341	0.00%	0.00%	0.00%	0.00%	1.42%	1.78%	2.55%	2.94%	3.35%	4.65%	5.10%	6.11%	7.68%	9.86%	10.41%	10.41%	10.41%	10.41%	10.41%	10.41%
Ba	187	1.12%	1.12%	1.12%	1.77%	4.58%	6.09%	6.09%	9.33%	15.44%	17.43%	18.44%	19.55%	21.94%	28.12%	29.41%	32.29%	32.29%	32.29%	32.29%	32.29%
B	37	5.41%	5.41%	11.71%	14.98%	22.37%	22.37%	26.94%	26.94%	32.56%	38.69%	38.69%	38.69%	47.45%	60.59%	60.59%	60.59%	60.59%	60.59%	60.59%	60.59%
Caa-C	3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	50.00%	50.00%	50.00%	50.00%	50.00%							
IG	923	0.00%	0.00%	0.00%	0.00%	0.61%	0.98%	1.24%	1.37%	1.78%	2.34%	3.21%	3.51%	4.76%	5.73%	5.89%	6.07%	6.07%	6.27%	6.27%	6.27%
SG	227	1.82%	1.82%	2.84%	3.91%	7.37%	8.61%	9.28%	12.01%	18.68%	21.16%	22.00%	22.94%	26.01%	32.61%	33.77%	36.37%	36.37%	36.37%	36.37%	36.37%
All	1150	0.35%	0.35%	0.54%	0.74%	1.84%	2.36%	2.68%	3.23%	4.60%	5.44%	6.30%	6.69%	8.18%	9.85%	10.13%	10.60%	10.60%	10.78%	10.78%	10.78%

1/1/79																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	75	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.47%	1.47%	1.47%	2.96%	2.96%	2.96%	2.96%	2.96%	2.96%	2.96%	2.96%	2.96%
Aa	132	0.00%	0.00%	0.00%	0.81%	0.81%	0.81%	0.81%	0.81%	0.81%	1.71%	1.71%	1.71%	1.71%	1.71%	1.71%	1.71%	1.71%	1.71%	1.71%	1.71%
A	376	0.00%	0.00%	0.00%	0.00%	0.57%	0.57%	0.57%	1.19%	1.19%	2.52%	2.52%	3.59%	4.32%	4.32%	4.75%	4.75%	5.21%	5.21%	5.21%	5.21%
Baa	336	0.00%	0.31%	0.31%	1.70%	2.06%	2.43%	2.81%	2.81%	4.09%	4.53%	5.51%	8.12%	10.28%	10.83%	10.83%	10.83%	10.83%	10.83%	10.83%	10.83%
Ba	207	0.51%	0.51%	1.07%	3.49%	6.07%	9.49%	12.27%	18.90%	20.55%	21.40%	22.33%	25.39%	31.70%	32.79%	35.19%	35.19%	35.19%	35.19%	35.19%	35.19%
B	34	0.00%	6.45%	9.79%	17.01%	17.01%	21.62%	26.52%	38.95%	45.74%	45.74%	45.74%	56.59%	56.59%							
Caa-C	3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	50.00%	50.00%	50.00%	50.00%	50.00%								
IG	919	0.00%	0.11%	0.11%	0.71%	1.08%	1.20%	1.33%	1.60%	2.14%	3.00%	3.30%	4.68%	5.63%	5.79%	5.96%	5.96%	6.16%	6.16%	6.16%	6.16%
SG	244	0.43%	1.34%	2.29%	5.35%	7.55%	11.06%	14.06%	21.82%	23.96%	24.70%	25.52%	29.15%	35.00%	36.03%	38.31%	38.31%	38.31%	38.31%	38.31%	38.31%
All	1163	0.09%	0.36%	0.55%	1.61%	2.31%	3.03%	3.66%	5.20%	6.00%	6.83%	7.21%	8.91%	10.52%	10.80%	11.25%	11.25%	11.42%	11.42%	11.42%	11.42%
1/1/80																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	87	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.27%	1.27%	1.27%	2.56%	2.56%	2.56%	2.56%	2.56%	2.56%	2.56%	2.56%	2.56%	2.56%
Aa	130	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.90%	1.80%	1.80%	1.80%	1.80%	1.80%	1.80%	1.80%	1.80%	1.80%	1.80%	1.80%	1.80%
A	381	0.00%	0.00%	0.28%	0.83%	0.83%	0.83%	1.73%	2.04%	3.01%	3.01%	4.06%	4.78%	4.78%	5.19%	5.19%	5.64%	5.64%	5.64%	5.64%	6.67%
Baa	336	0.00%	0.00%	1.01%	1.36%	1.73%	2.46%	2.46%	3.29%	4.14%	5.53%	7.98%	10.50%	11.53%	11.53%	11.53%	11.53%	11.53%	11.53%	11.53%	11.53%
Ba	207	0.00%	0.52%	3.89%	5.09%	8.86%	12.10%	18.21%	20.48%	21.26%	23.94%	26.81%	32.98%	35.28%	37.72%	37.72%	37.72%	37.72%	39.28%	39.28%	39.28%
B	41	5.00%	7.57%	15.72%	21.54%	27.95%	31.55%	44.78%	49.38%	49.38%	49.38%	61.43%	69.15%	69.15%	69.15%	69.15%	69.15%	69.15%	69.15%	69.15%	69.15%
Caa-C	6	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	55.56%	55.56%	55.56%	55.56%	55.56%									
IG	934	0.00%	0.00%	0.46%	0.82%	0.94%	1.18%	1.56%	2.23%	3.05%	3.48%	4.81%	5.87%	6.18%	6.35%	6.35%	6.54%	6.54%	6.54%	6.54%	6.98%
SG	254	1.63%	2.48%	6.55%	8.49%	12.59%	15.79%	23.19%	25.69%	26.33%	28.54%	32.53%	38.69%	40.72%	42.88%	42.88%	42.88%	42.88%	44.24%	44.24%	44.24%
All	1188	0.34%	0.52%	1.72%	2.38%	3.25%	4.05%	5.72%	6.70%	7.49%	8.20%	9.93%	11.72%	12.26%	12.69%	12.69%	12.85%	12.85%	13.02%	13.02%	13.40%
1/1/81																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	91	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.22%	1.22%	1.22%	2.49%	2.49%	2.49%	2.49%	2.49%	2.49%	2.49%	2.49%	2.49%	2.49%	2.49%
Aa	135	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.83%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	3.73%
A	386	0.00%	0.27%	0.27%	0.27%	0.27%	1.15%	1.46%	2.11%	2.11%	3.15%	3.88%	3.88%	4.29%	4.29%	4.74%	4.74%	4.74%	4.74%	5.76%	6.28%
Baa	339	0.00%	0.64%	1.96%	2.65%	3.37%	3.37%	3.77%	4.59%	5.91%	8.26%	10.18%	11.16%	11.16%	11.16%	11.16%	11.16%	11.16%	11.16%	11.91%	12.72%
Ba	240	0.00%	3.65%	5.13%	8.17%	11.87%	18.56%	21.07%	21.72%	24.73%	28.83%	35.16%	37.24%	39.44%	39.44%	39.44%	39.44%	40.85%	42.29%	42.29%	42.29%
B	46	4.40%	11.39%	16.32%	24.33%	27.25%	41.23%	41.23%	41.23%	41.23%	51.52%	57.58%	57.58%	57.58%	57.58%	57.58%	57.58%	57.58%	57.58%	57.58%	57.58%
Caa-C	7	0.00%	0.00%	0.00%	0.00%	0.00%	16.67%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%	33.33%
IG	951	0.00%	0.33%	0.78%	1.01%	1.25%	1.62%	2.14%	2.93%	3.35%	4.64%	5.52%	5.82%	5.99%	5.99%	6.17%	6.17%	6.17%	6.17%	6.81%	7.46%
SG	293	0.70%	4.79%	6.78%	10.51%	13.99%	21.89%	24.47%	25.01%	27.47%	32.23%	38.31%	40.09%	41.97%	41.97%	41.97%	41.97%	43.15%	44.36%	44.36%	44.36%
All	1244	0.16%	1.35%	2.14%	3.14%	4.07%	6.01%	6.93%	7.68%	8.46%	10.34%	12.04%	12.55%	12.96%	12.96%	13.11%	13.11%	13.28%	13.45%	13.98%	14.53%

1/1/82

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	94	0.00%	0.00%	0.00%	0.00%	0.00%	1.18%	1.18%	1.18%	2.41%	2.41%	2.41%	2.41%	2.41%	2.41%	2.41%	2.41%	2.41%	2.41%	2.41%	2.41%
Aa	147	0.00%	0.00%	0.00%	0.00%	0.00%	0.76%	2.28%	2.28%	2.28%	2.28%	2.28%	2.28%	2.28%	3.29%	3.29%	3.29%	3.29%	3.29%	4.46%	4.46%
A	395	0.26%	0.26%	0.26%	0.26%	1.11%	1.11%	1.74%	1.74%	3.10%	3.80%	3.80%	4.20%	4.20%	4.20%	4.20%	4.20%	4.20%	4.69%	5.19%	8.29%
Baa	325	0.33%	0.33%	1.39%	2.13%	2.51%	3.32%	4.16%	5.50%	7.87%	9.81%	10.81%	10.81%	10.81%	10.81%	10.81%	10.81%	10.81%	12.35%	13.17%	13.17%
Ba	254	2.79%	5.36%	8.05%	11.79%	18.63%	20.84%	21.42%	24.08%	28.54%	32.63%	34.53%	36.55%	36.55%	36.55%	36.55%	37.90%	39.38%	39.38%	39.38%	42.65%
B	45	2.22%	9.10%	14.08%	16.77%	29.17%	29.17%	29.17%	29.17%	34.62%	53.19%	53.19%	53.19%	53.19%	53.19%	53.19%	53.19%	53.19%	53.19%	53.19%	53.19%
Caa-C	14	21.43%	50.00%	50.00%	50.00%	58.33%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%	66.67%
IG	961	0.21%	0.21%	0.55%	0.79%	1.27%	1.77%	2.55%	2.95%	4.36%	5.22%	5.52%	5.68%	5.68%	5.86%	5.86%	5.86%	5.86%	6.48%	7.13%	8.45%
SG	313	3.54%	7.97%	10.87%	14.28%	21.92%	24.15%	24.62%	26.80%	31.11%	36.56%	38.17%	39.86%	39.86%	39.86%	39.86%	40.97%	42.18%	42.18%	42.18%	44.87%
All	1274	1.04%	2.11%	3.06%	4.03%	6.15%	7.03%	7.73%	8.49%	10.41%	12.05%	12.54%	12.94%	12.94%	13.09%	13.09%	13.25%	13.41%	13.93%	14.47%	15.94%

1/1/83

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	101	0.00%	0.00%	0.00%	0.00%	2.25%	2.25%	2.25%	3.48%	3.48%	3.48%	3.48%	3.48%	3.48%	3.48%	3.48%	3.48%	3.48%	3.48%	3.48%	3.48%
Aa	234	0.00%	0.00%	0.00%	0.00%	0.47%	1.90%	1.90%	1.90%	1.90%	1.90%	1.90%	1.90%	2.55%	2.55%	2.55%	2.55%	2.55%	2.55%	3.34%	4.17%
A	484	0.00%	0.00%	0.00%	0.23%	0.23%	0.97%	0.97%	2.31%	3.42%	4.00%	4.31%	4.31%	4.31%	4.31%	4.31%	4.31%	4.31%	4.72%	5.99%	6.43%
Baa	333	0.00%	0.96%	1.30%	2.72%	3.12%	3.53%	4.43%	6.79%	8.77%	8.77%	8.77%	8.77%	8.77%	8.77%	8.77%	8.77%	8.77%	10.29%	11.90%	13.59%	17.05%
Ba	179	1.16%	2.42%	4.42%	10.24%	11.92%	15.52%	19.80%	23.29%	26.87%	26.87%	28.53%	28.53%	28.53%	28.53%	28.53%	31.00%	33.65%	36.42%	41.94%	44.71%
B	140	2.22%	6.86%	13.30%	23.33%	26.04%	26.96%	29.18%	36.43%	47.59%	52.47%	54.97%	54.97%	54.97%	54.97%	58.72%	58.72%	58.72%	58.72%	58.72%	58.72%
Caa-C	21	45.05%	56.04%	61.54%	61.54%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%	74.36%
IG	1152	0.00%	0.27%	0.37%	0.85%	1.26%	2.01%	2.23%	3.50%	4.46%	4.71%	4.85%	4.85%	5.00%	5.00%	5.00%	5.00%	5.35%	5.89%	7.01%	8.17%
SG	340	4.27%	7.56%	11.67%	18.96%	21.80%	23.95%	27.06%	31.85%	37.79%	39.39%	41.20%	41.20%	41.20%	41.20%	42.42%	43.86%	45.38%	46.99%	50.20%	51.86%
All	1492	0.95%	1.87%	2.82%	4.73%	5.62%	6.64%	7.37%	9.20%	10.90%	11.32%	11.67%	11.67%	11.80%	11.80%	11.94%	12.08%	12.53%	13.16%	14.45%	15.62%

1/1/84

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	87	0.00%	0.00%	0.00%	1.27%	1.27%	1.27%	2.70%	2.70%	2.70%	2.70%	2.70%	2.70%	2.70%	2.70%	2.70%	2.70%	2.70%	2.70%	2.70%	2.70%
Aa	250	0.00%	0.00%	0.00%	0.86%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	1.75%	2.36%	2.36%	2.36%	2.36%	2.36%	2.36%	3.08%	3.85%	3.85%
A	506	0.00%	0.21%	0.42%	0.64%	1.57%	1.57%	3.05%	4.09%	4.62%	4.62%	4.62%	4.62%	4.62%	4.62%	4.62%	4.62%	4.99%	6.15%	6.55%	7.00%
Baa	341	0.61%	0.61%	0.93%	1.32%	1.72%	2.59%	3.53%	5.00%	5.00%	5.56%	5.56%	5.56%	5.56%	6.27%	6.27%	7.07%	8.77%	11.44%	14.19%	14.19%
Ba	196	0.52%	2.14%	9.06%	11.63%	15.11%	18.35%	24.34%	29.64%	30.72%	31.90%	31.90%	31.90%	31.90%	31.90%	31.90%	35.74%	37.75%	39.75%	43.77%	43.77%
B	159	5.17%	11.85%	20.92%	23.88%	27.01%	32.62%	40.80%	46.89%	48.79%	52.88%	52.88%	52.88%	52.88%	52.88%	52.88%	56.51%	56.51%	56.51%	56.51%	56.51%
Caa-C	11	18.18%	29.87%	29.87%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%	47.40%
IG	1184	0.17%	0.26%	0.44%	0.93%	1.65%	1.86%	2.86%	3.67%	3.91%	4.04%	4.04%	4.19%	4.19%	4.35%	4.35%	4.52%	5.06%	6.33%	7.29%	7.49%
SG	366	3.08%	7.16%	14.85%	17.86%	21.10%	25.34%	32.03%	37.33%	38.59%	40.71%	40.71%	40.71%	40.71%	40.71%	41.85%	44.28%	45.57%	46.87%	49.49%	49.49%
All	1550	0.85%	1.87%	3.78%	4.84%	6.10%	7.12%	9.18%	10.77%	11.17%	11.61%	11.61%	11.73%	11.73%	11.86%	12.00%	12.44%	13.04%	14.27%	15.40%	15.57%

1/1/85																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	92	0.00%	0.00%	0.00%	0.00%	0.00%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%
Aa	303	0.00%	0.00%	0.00%	0.77%	0.77%	0.77%	0.77%	0.77%	0.77%	0.77%	1.33%	1.33%	1.33%	1.33%	1.33%	1.33%	1.99%	2.69%	2.69%	2.69%
A	567	0.00%	0.19%	1.19%	2.27%	2.27%	3.90%	4.87%	5.12%	5.12%	5.12%	5.12%	5.12%	5.12%	5.12%	5.12%	5.49%	6.24%	6.62%	7.05%	7.05%
Baa	346	0.00%	0.98%	1.32%	1.71%	2.57%	3.02%	4.49%	5.00%	5.56%	5.56%	5.56%	5.56%	6.31%	6.31%	8.81%	10.64%	13.45%	16.41%	16.41%	16.41%
Ba	236	0.87%	5.93%	8.65%	12.09%	17.18%	23.34%	29.11%	30.72%	31.61%	31.61%	31.61%	31.61%	31.61%	31.61%	33.17%	34.88%	38.31%	41.73%	41.73%	41.73%
B	187	7.11%	15.59%	20.38%	23.59%	29.25%	37.56%	43.09%	44.81%	48.90%	48.90%	51.45%	51.45%	51.45%	54.92%	54.92%	54.92%	54.92%	54.92%	54.92%	54.92%
Caa-C	16	6.25%	12.50%	21.25%	31.09%	31.09%	31.09%	31.09%	31.09%	31.09%	31.09%	58.66%	58.66%	58.66%	58.66%	58.66%	58.66%	58.66%	58.66%	58.66%	58.66%
IG	1308	0.00%	0.33%	0.86%	1.61%	1.81%	2.74%	3.51%	3.73%	3.86%	3.86%	4.00%	4.00%	4.15%	4.15%	4.65%	5.16%	6.21%	7.13%	7.33%	7.33%
SG	439	3.72%	10.31%	14.15%	17.67%	22.86%	29.63%	35.01%	36.55%	38.28%	38.28%	40.44%	40.44%	40.44%	41.34%	42.37%	43.50%	45.76%	48.05%	48.05%	48.05%
All	1747	0.94%	2.85%	4.19%	5.61%	6.92%	9.05%	10.70%	11.16%	11.56%	11.56%	12.02%	12.02%	12.14%	12.27%	12.82%	13.40%	14.57%	15.63%	15.80%	15.80%
1/1/86																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	137	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	352	0.00%	0.00%	0.64%	0.64%	1.33%	1.33%	1.33%	1.33%	1.33%	1.79%	1.79%	1.79%	1.79%	1.79%	1.79%	2.38%	2.99%	2.99%	2.99%	2.99%
A	634	0.00%	0.17%	0.72%	0.91%	1.88%	2.48%	2.48%	2.48%	2.48%	2.48%	2.48%	2.48%	2.48%	2.48%	3.13%	3.81%	4.51%	4.91%	4.91%	5.33%
Baa	376	0.84%	1.13%	2.47%	3.55%	5.07%	6.31%	7.19%	7.67%	8.20%	8.20%	8.20%	8.84%	8.84%	10.23%	11.00%	12.55%	14.15%	14.15%	14.15%	14.15%
Ba	313	2.37%	6.94%	9.42%	13.69%	20.42%	26.55%	28.56%	30.74%	31.54%	31.54%	31.54%	32.56%	32.56%	32.56%	33.99%	39.83%	42.77%	42.77%	42.77%	44.40%
B	228	10.30%	14.54%	18.59%	23.55%	31.25%	38.68%	43.43%	46.13%	46.13%	48.28%	48.28%	48.28%	50.74%	56.54%	59.64%	59.64%	59.64%	59.64%	59.64%	59.64%
Caa-C	20	16.18%	22.62%	30.36%	30.36%	30.36%	30.36%	30.36%	30.36%	30.36%	76.79%	76.79%	76.79%	76.79%	76.79%	76.79%	76.79%	76.79%	76.79%	76.79%	76.79%
IG	1499	0.21%	0.35%	1.05%	1.37%	2.30%	2.83%	3.02%	3.12%	3.22%	3.34%	3.34%	3.46%	3.46%	3.74%	4.18%	4.92%	5.70%	5.87%	5.87%	6.05%
SG	561	6.11%	10.59%	13.90%	18.35%	25.30%	31.74%	34.63%	36.91%	37.43%	39.17%	39.17%	39.85%	40.58%	42.28%	44.09%	47.86%	49.78%	49.78%	49.78%	50.87%
All	2060	1.81%	3.10%	4.47%	5.76%	7.99%	9.72%	10.40%	10.89%	11.06%	11.43%	11.43%	11.64%	11.75%	12.21%	12.81%	13.93%	14.84%	14.98%	14.98%	15.28%
1/1/87																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	157	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	366	0.00%	0.00%	0.00%	0.32%	0.32%	0.32%	0.32%	0.32%	0.74%	0.74%	0.74%	0.74%	0.74%	0.74%	1.27%	1.83%	1.83%	1.83%	1.83%	1.83%
A	603	0.00%	0.00%	0.18%	1.13%	1.72%	1.72%	1.72%	1.72%	1.72%	1.72%	1.72%	1.72%	1.72%	2.04%	2.70%	3.39%	4.14%	4.14%	4.54%	4.96%
Baa	413	0.00%	0.85%	1.75%	3.34%	4.72%	6.16%	7.38%	7.38%	7.38%	7.38%	7.92%	8.50%	9.70%	10.36%	11.71%	12.41%	12.41%	12.41%	12.41%	12.41%
Ba	421	3.04%	4.99%	8.74%	15.97%	23.93%	26.58%	29.08%	30.78%	31.41%	32.10%	33.65%	33.65%	33.65%	34.80%	38.23%	44.15%	44.15%	44.15%	45.58%	47.09%
B	308	5.35%	11.67%	19.14%	29.19%	37.84%	42.92%	44.65%	44.65%	45.85%	45.85%	45.85%	47.40%	50.90%	56.36%	58.26%	60.34%	60.34%	60.34%	60.34%	60.34%
Caa-C	34	9.82%	16.76%	16.76%	50.06%	50.06%	50.06%	50.06%	50.06%	75.03%	75.03%	75.03%	75.03%	75.03%	75.03%	75.03%	75.03%	75.03%	75.03%	75.03%	75.03%
IG	1539	0.00%	0.21%	0.51%	1.36%	1.93%	2.26%	2.54%	2.54%	2.65%	2.65%	2.77%	2.89%	3.14%	3.41%	4.10%	4.68%	4.99%	4.99%	5.15%	5.33%
SG	763	4.28%	8.29%	13.40%	22.36%	30.35%	33.78%	35.91%	36.95%	38.52%	38.94%	39.90%	40.43%	41.68%	44.33%	47.04%	51.30%	51.30%	51.30%	52.17%	53.11%
All	2302	1.41%	2.84%	4.60%	7.71%	10.24%	11.31%	12.01%	12.23%	12.64%	12.72%	12.99%	13.19%	13.59%	14.24%	15.23%	16.37%	16.61%	16.61%	16.88%	17.16%

1/1/88																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	156	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	405	0.00%	0.52%	0.79%	0.79%	0.79%	0.79%	0.79%	1.15%	1.15%	1.15%	1.15%	1.15%	1.60%	2.06%	2.53%	2.53%	2.53%	2.53%	2.53%	2.53%
A	613	0.00%	0.17%	0.89%	1.44%	1.44%	1.44%	1.44%	1.44%	1.44%	1.44%	1.44%	1.44%	1.44%	2.08%	2.41%	3.15%	3.15%	3.54%	3.95%	3.95%
Baa	390	0.00%	0.28%	1.17%	2.44%	4.10%	5.59%	5.59%	5.59%	5.59%	6.09%	6.09%	6.66%	7.27%	9.17%	11.13%	11.13%	11.13%	11.13%	11.13%	11.13%
Ba	462	1.36%	6.47%	11.99%	20.12%	22.58%	25.00%	25.94%	26.46%	27.59%	28.85%	29.61%	30.38%	31.26%	35.85%	41.65%	41.65%	41.65%	42.92%	44.34%	47.62%
B	389	5.61%	12.63%	25.86%	34.48%	38.98%	42.04%	43.51%	46.11%	46.11%	47.15%	51.80%	54.56%	59.01%	60.59%	62.38%	62.38%	62.38%	62.38%	62.38%	62.38%
Caa-C	40	12.50%	15.74%	44.55%	44.55%	44.55%	44.55%	44.55%	72.27%	72.27%	72.27%	72.27%	72.27%	72.27%	72.27%	72.27%	72.27%	72.27%	72.27%	72.27%	72.27%
IG	1564	0.00%	0.27%	0.84%	1.36%	1.74%	2.08%	2.08%	2.18%	2.18%	2.28%	2.28%	2.40%	2.65%	3.43%	4.10%	4.38%	4.38%	4.54%	4.70%	4.70%
SG	891	3.73%	9.60%	18.86%	26.99%	30.20%	32.78%	33.88%	35.74%	36.43%	37.56%	39.74%	41.18%	43.27%	46.59%	50.70%	50.70%	50.70%	51.49%	52.34%	54.21%
All	2455	1.34%	3.50%	6.84%	9.65%	10.82%	11.73%	11.99%	12.50%	12.65%	12.97%	13.41%	13.77%	14.35%	15.55%	16.80%	17.02%	17.02%	17.27%	17.52%	17.79%
1/1/89																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	177	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	413	0.50%	0.50%	0.50%	0.50%	0.50%	0.50%	0.83%	0.83%	0.83%	0.83%	0.83%	1.26%	1.70%	2.14%	2.14%	2.14%	2.14%	2.14%	2.14%	3.35%
A	648	0.00%	0.32%	0.83%	0.83%	0.83%	0.83%	0.83%	0.83%	0.83%	0.83%	0.83%	0.83%	1.73%	2.04%	2.38%	2.38%	2.75%	3.13%	3.13%	4.40%
Baa	408	0.51%	1.58%	2.16%	3.96%	4.64%	4.64%	4.64%	4.64%	5.08%	5.08%	5.58%	6.12%	8.34%	10.06%	10.68%	10.68%	10.68%	10.68%	10.68%	11.47%
Ba	437	2.97%	9.84%	17.86%	20.13%	22.75%	23.18%	23.67%	24.77%	26.00%	28.17%	28.90%	30.55%	34.75%	41.02%	41.02%	41.02%	41.02%	42.20%	44.83%	46.37%
B	426	7.53%	21.01%	29.49%	34.59%	37.90%	40.18%	43.53%	44.31%	46.77%	49.56%	51.85%	55.67%	55.67%	58.74%	58.74%	58.74%	61.03%	61.03%	61.03%	66.23%
Caa-C	49	20.39%	49.07%	55.44%	55.44%	55.44%	55.44%	55.44%	55.44%	55.44%	55.44%	55.44%	55.44%	77.72%	77.72%	77.72%	77.72%	77.72%	77.72%	77.72%	77.72%
IG	1646	0.25%	0.64%	0.97%	1.40%	1.55%	1.55%	1.64%	1.64%	1.74%	1.74%	1.85%	2.08%	3.02%	3.64%	3.90%	3.90%	4.04%	4.19%	4.19%	5.17%
SG	912	5.90%	16.42%	24.56%	28.02%	30.86%	32.06%	33.72%	34.65%	36.35%	38.74%	40.03%	42.40%	45.35%	50.18%	50.18%	50.18%	50.86%	51.59%	53.20%	55.88%
All	2558	2.21%	5.91%	8.68%	9.98%	10.86%	11.16%	11.61%	11.82%	12.27%	12.74%	13.08%	13.70%	14.98%	16.31%	16.51%	16.51%	16.74%	16.97%	17.22%	18.38%
1/1/90																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	199	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	461	0.00%	0.00%	0.00%	0.00%	0.00%	0.28%	0.28%	0.28%	0.28%	0.28%	0.28%	0.65%	1.03%	1.03%	1.03%	1.03%	1.03%	1.03%	2.09%	3.69%
A	675	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.27%	0.55%	0.83%	1.15%	1.15%	1.49%	1.49%	1.49%	2.67%	4.28%
Baa	397	0.26%	0.83%	1.12%	1.47%	1.47%	1.47%	1.47%	1.47%	1.47%	1.97%	2.50%	5.23%	8.05%	8.65%	8.65%	8.65%	9.40%	9.40%	10.25%	12.84%
Ba	421	3.53%	12.02%	14.51%	17.33%	17.73%	18.68%	19.20%	20.37%	22.47%	23.17%	25.50%	31.14%	35.39%	35.39%	35.39%	35.39%	36.54%	39.01%	40.36%	41.72%
B	449	13.68%	23.08%	30.25%	33.38%	35.73%	38.53%	39.86%	42.85%	45.42%	47.60%	51.50%	51.50%	54.74%	54.74%	54.74%	57.01%	57.01%	57.01%	65.07%	67.98%
Caa-C	55	45.06%	49.28%	49.28%	49.28%	49.28%	49.28%	49.28%	49.28%	49.28%	74.64%	74.64%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
IG	1732	0.06%	0.18%	0.25%	0.32%	0.32%	0.40%	0.40%	0.40%	0.40%	0.50%	0.71%	1.47%	2.25%	2.49%	2.49%	2.62%	2.76%	2.76%	3.67%	5.22%
SG	925	10.43%	19.15%	23.76%	26.63%	27.93%	29.66%	30.51%	32.40%	34.65%	36.28%	38.99%	42.83%	46.43%	46.43%	46.43%	47.12%	47.85%	49.38%	52.69%	54.46%
All	2657	3.51%	6.36%	7.76%	8.61%	8.94%	9.41%	9.60%	10.01%	10.45%	10.84%	11.49%	12.76%	13.99%	14.18%	14.18%	14.39%	14.60%	14.83%	16.05%	17.55%

1/1/91

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	188	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	492	0.00%	0.00%	0.00%	0.00%	0.26%	0.26%	0.26%	0.26%	0.26%	0.26%	0.60%	0.96%	0.96%	0.96%	0.96%	0.96%	0.96%	1.95%	3.94%	3.94%
A	662	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.26%	0.52%	0.79%	1.09%	1.09%	1.41%	1.41%	1.41%	2.54%	4.08%	4.50%
Baa	419	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.66%	1.11%	3.40%	5.78%	6.28%	6.28%	6.28%	6.91%	7.56%	8.27%	9.69%	9.69%
Ba	357	3.84%	5.15%	7.05%	7.46%	7.95%	8.50%	9.72%	11.94%	12.68%	15.20%	20.44%	24.12%	24.12%	25.30%	25.30%	26.61%	26.61%	28.08%	29.79%	29.79%
B	376	13.12%	22.20%	26.79%	29.15%	32.42%	33.75%	36.06%	38.82%	41.13%	45.35%	46.92%	52.12%	52.12%	52.12%	54.64%	54.64%	57.31%	66.46%	73.17%	73.17%
Caa-C	60	15.97%	15.97%	19.79%	24.25%	29.66%	29.66%	35.52%	35.52%	46.27%	46.27%	59.70%	59.70%	59.70%	59.70%	59.70%	59.70%	59.70%	59.70%	59.70%	59.70%
IG	1761	0.06%	0.06%	0.06%	0.06%	0.13%	0.13%	0.13%	0.13%	0.22%	0.42%	1.12%	1.84%	2.07%	2.07%	2.19%	2.31%	2.45%	3.30%	4.74%	4.90%
SG	793	9.08%	14.04%	17.28%	18.77%	20.74%	21.59%	23.46%	25.75%	27.46%	30.34%	34.43%	38.23%	38.23%	38.93%	39.69%	40.48%	41.30%	44.84%	47.81%	47.81%
All	2554	2.78%	4.18%	5.03%	5.40%	5.91%	6.09%	6.48%	6.90%	7.27%	7.89%	9.11%	10.30%	10.48%	10.57%	10.78%	10.99%	11.21%	12.39%	13.96%	14.08%

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Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	174	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	498	0.00%	0.00%	0.00%	0.24%	0.24%	0.24%	0.24%	0.24%	0.24%	0.55%	0.55%	0.55%	0.55%	0.55%	0.55%	0.55%	0.55%	1.50%	2.01%	2.01%
A	762	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.22%	0.45%	0.92%	1.18%	1.18%	1.18%	1.18%	1.18%	2.83%	4.85%	5.58%	6.33%
Baa	429	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.39%	0.81%	2.95%	4.29%	4.76%	4.76%	5.30%	5.87%	6.47%	7.11%	8.40%	8.40%	9.08%
Ba	339	0.34%	1.09%	1.09%	1.59%	1.59%	2.76%	4.86%	6.29%	7.93%	12.99%	17.43%	17.43%	18.58%	19.83%	21.19%	21.19%	24.47%	26.23%	26.23%	26.23%
B	303	7.64%	14.07%	17.45%	21.43%	23.43%	25.78%	28.51%	30.84%	36.39%	37.90%	44.55%	44.55%	44.55%	44.55%	44.55%	44.55%	50.10%	59.29%	59.29%	59.29%
Caa-C	70	15.67%	21.56%	23.80%	26.62%	29.96%	33.46%	33.46%	40.85%	40.85%	49.30%	49.30%	49.30%	49.30%	49.30%	49.30%	66.20%	66.20%	66.20%	66.20%	66.20%
IG	1863	0.00%	0.00%	0.00%	0.07%	0.07%	0.07%	0.07%	0.15%	0.33%	0.96%	1.44%	1.64%	1.64%	1.75%	1.87%	1.99%	2.77%	4.10%	4.53%	4.96%
SG	712	4.93%	8.55%	10.18%	12.31%	13.40%	15.21%	17.40%	19.46%	22.26%	26.18%	30.89%	30.89%	31.55%	32.28%	33.08%	33.91%	37.66%	41.83%	41.83%	41.83%
All	2575	1.33%	2.24%	2.63%	3.15%	3.37%	3.73%	4.13%	4.54%	5.12%	6.25%	7.36%	7.53%	7.62%	7.81%	8.01%	8.22%	9.34%	10.94%	11.30%	11.68%

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Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	140	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	503	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.43%	0.91%	0.91%	0.91%
A	831	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.19%	0.59%	1.00%	1.22%	1.22%	1.22%	1.22%	1.22%	2.38%	3.87%	4.19%	4.51%	4.84%
Baa	480	0.00%	0.00%	0.25%	0.25%	0.53%	0.84%	1.47%	2.15%	3.89%	4.99%	5.77%	5.77%	6.20%	6.67%	7.17%	8.75%	10.90%	11.46%	12.64%	13.26%
Ba	353	0.62%	0.62%	1.86%	2.74%	3.69%	5.39%	7.79%	8.47%	12.69%	17.12%	17.12%	17.12%	18.10%	19.18%	19.18%	21.75%	24.40%	24.40%	24.40%	24.40%
B	291	4.35%	8.38%	13.96%	15.52%	18.09%	20.30%	23.86%	31.80%	35.12%	38.71%	39.96%	39.96%	39.96%	41.73%	43.74%	48.34%	56.58%	56.58%	56.58%	56.58%
Caa-C	73	13.95%	17.57%	22.28%	25.06%	25.06%	25.06%	31.31%	31.31%	38.94%	38.94%	38.94%	47.66%	47.66%	47.66%	47.66%	47.66%	47.66%	47.66%	47.66%	47.66%
IG	1954	0.00%	0.00%	0.06%	0.06%	0.13%	0.20%	0.35%	0.58%	1.15%	1.58%	1.85%	1.85%	1.95%	2.06%	2.17%	2.99%	4.20%	4.59%	4.98%	5.26%
SG	717	3.40%	5.43%	8.77%	10.10%	11.61%	13.41%	16.41%	19.77%	23.75%	27.56%	28.01%	28.53%	29.10%	30.33%	30.99%	34.03%	38.13%	38.13%	38.13%	38.13%
All	2671	0.89%	1.40%	2.22%	2.51%	2.88%	3.29%	3.95%	4.72%	5.86%	6.85%	7.15%	7.23%	7.40%	7.66%	7.85%	8.95%	10.49%	10.82%	11.16%	11.40%

1/1/94																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	142	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	475	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.46%	0.46%	0.46%	0.46%
A	1013	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.15%	0.45%	0.76%	1.11%	1.11%	1.11%	1.11%	1.11%	2.01%	3.41%	3.67%	4.45%	4.72%	4.72%
Baa	540	0.00%	0.20%	0.20%	0.42%	0.65%	1.87%	2.38%	3.72%	4.56%	5.15%	5.15%	5.49%	5.85%	6.23%	7.93%	10.17%	10.64%	11.62%	12.14%	12.14%
Ba	397	0.00%	0.90%	1.21%	2.24%	4.31%	7.42%	10.35%	13.48%	17.41%	18.01%	18.01%	18.81%	20.64%	20.64%	23.99%	30.10%	30.10%	30.10%	31.65%	31.65%
B	381	4.19%	8.83%	12.31%	13.96%	16.88%	21.62%	27.03%	32.88%	37.69%	41.18%	43.10%	43.10%	44.29%	45.62%	48.72%	51.98%	51.98%	51.98%	51.98%	51.98%
Caa-C	85	5.31%	11.97%	15.87%	15.87%	15.87%	25.08%	25.08%	33.40%	33.40%	33.40%	44.50%	44.50%	44.50%	44.50%	44.50%	44.50%	44.50%	44.50%	44.50%	44.50%
IG	2170	0.00%	0.05%	0.05%	0.10%	0.16%	0.47%	0.67%	1.14%	1.49%	1.79%	1.79%	1.88%	1.96%	2.06%	2.85%	3.99%	4.32%	4.89%	5.13%	5.13%
SG	863	2.34%	5.36%	7.33%	8.55%	10.87%	14.93%	18.68%	23.01%	27.08%	28.78%	29.95%	30.40%	31.89%	32.43%	35.58%	40.32%	40.32%	40.32%	41.19%	41.19%
All	3033	0.65%	1.48%	1.99%	2.33%	2.88%	3.95%	4.83%	6.02%	7.05%	7.60%	7.79%	7.93%	8.22%	8.38%	9.47%	11.04%	11.32%	11.81%	12.11%	12.11%
1/1/95																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	145	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	562	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.39%	0.81%	0.81%	0.81%	0.81%	0.81%
A	1088	0.00%	0.00%	0.00%	0.00%	0.00%	0.13%	0.40%	0.82%	1.12%	1.12%	1.12%	1.12%	1.12%	1.94%	3.00%	3.23%	3.95%	4.20%	4.20%	4.47%
Baa	556	0.00%	0.00%	0.00%	0.22%	1.34%	1.82%	3.31%	4.09%	4.36%	4.36%	4.68%	5.02%	5.39%	7.01%	9.13%	9.57%	10.49%	10.98%	10.98%	10.98%
Ba	409	0.27%	0.55%	1.76%	3.56%	7.05%	9.58%	12.71%	15.14%	16.69%	16.69%	17.41%	18.96%	18.96%	21.93%	29.67%	29.67%	29.67%	31.08%	31.08%	31.08%
B	499	3.83%	6.38%	8.82%	12.21%	16.57%	22.05%	30.82%	38.35%	42.72%	45.09%	45.80%	46.58%	47.43%	49.34%	52.32%	52.32%	52.32%	52.32%	52.32%	52.32%
Caa-C	96	11.55%	17.32%	17.32%	21.37%	31.47%	41.26%	53.74%	60.35%	60.35%	60.35%	60.35%	60.35%	60.35%	60.35%	60.35%	60.35%	60.35%	60.35%	60.35%	60.35%
IG	2351	0.00%	0.00%	0.00%	0.05%	0.32%	0.50%	0.98%	1.35%	1.56%	1.56%	1.63%	1.71%	1.80%	2.53%	3.58%	3.89%	4.42%	4.64%	4.64%	4.75%
SG	1004	3.07%	4.93%	6.65%	9.42%	13.66%	17.86%	23.90%	28.81%	31.65%	32.82%	33.50%	34.62%	35.02%	37.40%	42.50%	42.50%	42.50%	43.19%	43.19%	43.19%
All	3355	0.89%	1.42%	1.89%	2.62%	3.82%	4.88%	6.55%	7.89%	8.63%	8.86%	9.04%	9.30%	9.44%	10.41%	12.05%	12.30%	12.74%	13.02%	13.02%	13.12%
1/1/96																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	147	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	607	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.36%	0.73%	0.73%	0.73%	0.73%	0.73%	0.73%
A	1161	0.00%	0.00%	0.00%	0.00%	0.12%	0.36%	0.61%	0.89%	0.89%	0.89%	0.89%	0.89%	1.63%	2.77%	2.98%	3.62%	3.84%	3.84%	4.08%	4.08%
Baa	654	0.00%	0.00%	0.17%	0.72%	1.29%	2.68%	3.30%	3.73%	3.97%	4.22%	4.77%	4.77%	6.37%	8.06%	8.42%	9.53%	10.31%	10.31%	10.31%	11.16%
Ba	466	0.00%	0.71%	2.11%	5.67%	7.29%	10.40%	13.09%	14.76%	14.76%	15.89%	17.11%	17.80%	20.09%	25.11%	25.11%	25.11%	26.12%	26.12%	26.12%	26.12%
B	569	1.51%	4.04%	8.83%	12.16%	18.16%	25.79%	33.40%	37.45%	39.94%	40.55%	41.19%	41.92%	44.47%	49.76%	49.76%	49.76%	49.76%	49.76%	49.76%	52.48%
Caa-C	110	9.90%	14.59%	18.84%	32.36%	41.28%	51.89%	55.90%	55.90%	55.90%	55.90%	55.90%	55.90%	55.90%	55.90%	55.90%	55.90%	55.90%	55.90%	55.90%	55.90%
IG	2569	0.00%	0.00%	0.05%	0.19%	0.39%	0.86%	1.14%	1.38%	1.44%	1.51%	1.65%	1.65%	2.37%	3.38%	3.65%	4.20%	4.49%	4.49%	4.60%	4.81%
SG	1145	1.65%	3.60%	6.89%	11.01%	15.08%	20.62%	25.67%	28.37%	29.55%	30.38%	31.29%	31.96%	34.26%	39.16%	39.16%	39.16%	39.69%	39.69%	39.69%	40.91%
All	3714	0.50%	1.07%	2.02%	3.21%	4.37%	6.07%	7.46%	8.23%	8.52%	8.74%	9.02%	9.14%	10.11%	11.73%	11.95%	12.41%	12.73%	12.73%	12.81%	13.16%

1/1/97																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	143	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	686	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.31%	0.63%	0.63%	0.63%	0.63%	0.63%	0.63%	0.63%
A	1185	0.00%	0.00%	0.00%	0.10%	0.32%	0.54%	1.01%	1.01%	1.01%	1.01%	1.01%	1.99%	3.00%	3.18%	3.94%	4.33%	5.15%	5.36%	5.36%	5.36%
Baa	825	0.00%	0.13%	0.54%	0.98%	2.04%	2.68%	3.02%	3.21%	3.61%	4.03%	4.03%	5.04%	6.91%	7.19%	8.06%	9.26%	9.26%	9.26%	10.23%	10.23%
Ba	579	0.18%	1.20%	4.74%	7.65%	9.95%	13.92%	15.46%	15.46%	16.31%	17.20%	17.70%	19.33%	22.82%	22.82%	22.82%	23.51%	23.51%	24.26%	26.63%	26.63%
B	688	2.00%	6.37%	10.33%	15.91%	25.48%	31.84%	35.71%	38.23%	38.65%	40.12%	41.22%	44.35%	50.89%	50.89%	50.89%	50.89%	51.78%	51.78%	51.78%	52.72%
Caa-C	136	9.09%	15.96%	27.74%	41.67%	50.60%	53.34%	53.34%	53.34%	62.67%	62.67%	62.67%	62.67%	62.67%	62.67%	62.67%	62.67%	62.67%	62.67%	62.67%	62.67%
IG	2839	0.00%	0.04%	0.16%	0.33%	0.73%	1.02%	1.32%	1.37%	1.49%	1.61%	1.61%	2.30%	3.32%	3.55%	4.11%	4.61%	4.95%	5.04%	5.31%	5.31%
SG	1403	1.89%	5.06%	9.41%	14.29%	20.42%	25.41%	27.97%	29.17%	29.96%	31.07%	31.81%	34.02%	38.69%	38.69%	38.69%	39.05%	39.42%	39.80%	41.01%	41.42%
All	4242	0.61%	1.62%	2.99%	4.48%	6.44%	7.97%	8.83%	9.15%	9.41%	9.74%	9.89%	10.87%	12.56%	12.74%	13.18%	13.64%	13.97%	14.11%	14.54%	14.62%
1/1/98																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	130	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	738	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.23%	0.23%	0.23%	0.23%	0.23%	0.52%	0.82%	1.12%	1.12%	1.12%	1.12%	1.12%	1.12%	1.51%
A	1183	0.00%	0.00%	0.10%	0.29%	0.50%	0.94%	0.94%	0.94%	0.94%	0.94%	1.86%	2.97%	3.15%	3.69%	4.25%	5.21%	5.41%	5.41%	5.41%	5.85%
Baa	981	0.11%	0.44%	0.78%	1.75%	2.66%	2.94%	3.23%	3.55%	3.90%	3.90%	4.74%	6.53%	7.00%	7.73%	8.49%	8.76%	8.76%	9.59%	9.59%	9.59%
Ba	695	0.76%	3.23%	6.13%	9.00%	12.58%	14.44%	14.70%	15.33%	15.67%	16.06%	17.69%	21.61%	21.61%	21.61%	22.62%	22.62%	23.16%	24.85%	26.02%	26.02%
B	930	3.69%	9.46%	16.64%	26.28%	32.17%	36.94%	40.50%	41.32%	43.43%	45.16%	47.12%	50.76%	50.76%	50.76%	50.76%	51.28%	51.28%	51.85%	51.85%	52.45%
Caa-C	155	8.86%	26.50%	37.57%	51.14%	56.89%	61.06%	64.30%	67.55%	67.55%	67.55%	83.77%	83.77%	83.77%							
IG	3032	0.03%	0.14%	0.29%	0.69%	1.07%	1.34%	1.49%	1.59%	1.70%	1.70%	2.34%	3.40%	3.68%	4.20%	4.66%	5.13%	5.21%	5.46%	5.46%	5.73%
SG	1780	2.98%	8.36%	14.08%	21.16%	26.04%	29.45%	31.50%	32.31%	33.51%	34.52%	36.41%	39.98%	39.98%	39.98%	40.46%	40.70%	40.95%	42.02%	42.57%	42.85%
All	4812	1.11%	3.07%	5.12%	7.75%	9.59%	10.83%	11.53%	11.82%	12.22%	12.47%	13.39%	15.03%	15.23%	15.61%	16.06%	16.47%	16.58%	17.02%	17.14%	17.41%
1/1/99																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	114	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	738	0.00%	0.00%	0.00%	0.00%	0.00%	0.21%	0.21%	0.21%	0.21%	0.21%	0.47%	1.03%	1.30%	1.30%	1.30%	1.30%	1.30%	1.30%	1.66%	1.66%
A	1226	0.00%	0.00%	0.18%	0.28%	0.68%	0.68%	0.68%	0.68%	0.68%	1.68%	2.85%	2.85%	3.51%	4.03%	4.93%	5.11%	5.11%	5.11%	5.53%	5.53%
Baa	1124	0.09%	0.48%	1.29%	2.40%	2.86%	3.11%	3.39%	3.68%	3.68%	4.75%	6.84%	7.24%	7.87%	8.73%	8.96%	9.20%	9.93%	9.93%	9.93%	9.93%
Ba	689	1.38%	3.39%	5.85%	8.95%	10.90%	11.14%	11.72%	12.33%	12.67%	14.87%	18.37%	18.81%	18.81%	19.76%	19.76%	19.76%	20.28%	21.90%	21.90%	21.90%
B	1078	5.04%	13.64%	23.23%	31.58%	36.15%	40.01%	41.49%	43.51%	45.28%	47.38%	52.20%	52.20%	52.63%	52.63%	53.10%	53.10%	55.33%	55.89%	57.66%	58.32%
Caa-C	315	15.04%	23.96%	37.62%	41.73%	46.47%	48.08%	48.99%	48.99%	50.24%	51.66%	51.66%	51.66%	53.39%	53.39%	53.39%	53.39%	53.39%	53.39%	53.39%	53.39%
IG	3202	0.03%	0.17%	0.53%	0.95%	1.28%	1.41%	1.50%	1.60%	1.60%	2.35%	3.55%	3.81%	4.34%	4.83%	5.26%	5.41%	5.64%	5.64%	5.89%	5.89%
SG	2082	5.33%	11.77%	19.46%	25.39%	28.98%	31.21%	32.27%	33.47%	34.54%	36.52%	40.06%	40.24%	40.61%	41.01%	41.21%	41.21%	42.32%	43.23%	43.95%	44.21%
All	5284	2.09%	4.61%	7.71%	10.13%	11.56%	12.37%	12.75%	13.16%	13.45%	14.50%	16.28%	16.50%	16.98%	17.42%	17.78%	17.89%	18.33%	18.56%	18.91%	18.97%

1/1/00																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	119	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	733	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.53%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%	1.14%	1.14%	1.14%
A	1255	0.00%	0.17%	0.35%	0.73%	0.84%	0.84%	0.84%	0.84%	2.04%	3.29%	3.29%	3.91%	4.57%	5.41%	5.59%	5.77%	5.77%	6.15%	6.15%	6.15%
Baa	1199	0.35%	0.80%	1.79%	2.42%	2.64%	3.01%	3.27%	3.27%	3.88%	6.04%	6.39%	6.95%	7.52%	7.73%	7.93%	8.36%	8.36%	8.36%	8.36%	8.36%
Ba	653	1.44%	3.32%	5.76%	7.42%	7.42%	8.21%	8.50%	8.83%	11.67%	16.52%	16.93%	16.93%	17.85%	17.85%	17.85%	18.86%	20.43%	20.43%	20.43%	21.01%
B	1151	5.64%	15.57%	24.22%	29.13%	32.76%	34.59%	37.69%	39.50%	41.97%	47.90%	48.26%	48.65%	48.65%	49.10%	49.57%	51.65%	52.71%	54.40%	55.04%	57.77%
Caa-C	328	17.74%	36.17%	45.92%	53.58%	58.15%	59.25%	59.25%	59.25%	59.25%	61.03%	61.03%	63.08%	63.08%	63.08%	63.08%	63.08%	63.08%	63.08%	63.08%	63.08%
IG	3306	0.13%	0.35%	0.78%	1.16%	1.28%	1.41%	1.50%	1.50%	2.19%	3.41%	3.65%	4.15%	4.60%	5.01%	5.15%	5.36%	5.36%	5.59%	5.59%	5.59%
SG	2132	6.15%	14.72%	21.46%	25.46%	27.84%	29.18%	30.82%	31.81%	34.11%	38.96%	39.29%	39.65%	40.03%	40.22%	40.42%	41.71%	42.81%	43.51%	43.76%	45.03%
All	5438	2.46%	5.88%	8.65%	10.29%	11.17%	11.66%	12.20%	12.47%	13.55%	15.67%	15.93%	16.37%	16.79%	17.13%	17.28%	17.74%	18.01%	18.34%	18.40%	18.69%
1/1/01																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	119	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	761	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.50%	0.74%	0.74%	1.02%	1.02%	1.02%	1.02%	1.34%	1.34%	1.34%	1.34%
A	1304	0.15%	0.41%	0.67%	0.67%	0.67%	0.67%	0.67%	2.01%	3.16%	3.29%	3.86%	4.46%	5.23%	5.71%	5.87%	5.87%	6.22%	6.22%	6.22%	6.79%
Baa	1174	0.18%	1.22%	1.83%	2.26%	2.61%	2.86%	2.86%	3.30%	5.33%	5.50%	6.02%	6.75%	6.95%	6.95%	7.36%	7.57%	7.57%	7.80%	7.80%	8.78%
Ba	617	1.17%	3.20%	5.02%	5.02%	6.06%	6.36%	6.70%	9.62%	15.03%	15.46%	15.46%	16.41%	16.41%	16.41%	17.46%	18.53%	19.09%	19.09%	19.68%	20.29%
B	1091	8.72%	18.56%	23.95%	27.58%	29.40%	32.71%	34.53%	37.44%	45.36%	45.74%	46.16%	46.16%	46.63%	47.15%	49.49%	51.31%	53.26%	54.00%	56.34%	57.97%
Caa-C	326	28.27%	39.65%	53.06%	59.13%	61.15%	61.15%	61.15%	61.15%	62.83%	62.83%	64.79%	64.79%	64.79%	64.79%	64.79%	64.79%	64.79%	64.79%	66.99%	69.19%
IG	3358	0.12%	0.59%	0.90%	1.05%	1.17%	1.26%	1.26%	1.94%	3.07%	3.30%	3.76%	4.24%	4.68%	4.87%	5.07%	5.14%	5.36%	5.43%	5.43%	5.97%
SG	2034	9.31%	16.79%	21.77%	24.29%	25.79%	27.57%	28.58%	31.10%	37.08%	37.43%	37.80%	38.20%	38.40%	38.62%	40.00%	41.18%	42.17%	42.43%	43.79%	44.92%
All	5392	3.53%	6.52%	8.42%	9.33%	9.85%	10.39%	10.65%	11.76%	14.01%	14.25%	14.68%	15.13%	15.50%	15.69%	16.14%	16.44%	16.82%	16.93%	17.21%	17.86%
1/1/02																					
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	128	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	773	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.46%	0.69%	0.69%	0.95%	0.95%	0.95%	0.95%	1.25%	1.25%	1.25%	1.25%	1.25%
A	1308	0.16%	0.32%	0.32%	0.32%	0.32%	0.32%	1.57%	2.52%	2.77%	3.30%	3.99%	4.70%	5.15%	5.30%	5.30%	5.63%	5.63%	5.63%	5.98%	5.98%
Baa	1258	0.84%	1.28%	1.37%	1.58%	1.69%	1.69%	2.07%	3.88%	3.88%	4.03%	4.69%	4.86%	4.86%	5.23%	5.42%	5.42%	5.62%	5.83%	6.49%	6.49%
Ba	598	1.77%	4.06%	4.93%	5.88%	6.70%	7.02%	9.79%	14.92%	15.31%	15.75%	17.10%	17.10%	17.10%	18.57%	19.08%	19.61%	19.61%	20.18%	21.95%	21.95%
B	895	4.57%	9.41%	12.25%	14.60%	17.88%	19.53%	23.80%	33.58%	35.15%	36.47%	36.47%	36.96%	37.51%	39.35%	41.90%	44.64%	45.41%	47.82%	50.32%	50.32%
Caa-C	370	26.40%	40.96%	48.67%	50.00%	52.75%	53.83%	55.32%	59.84%	59.84%	61.67%	61.67%	61.67%	61.67%	61.67%	61.67%	61.67%	61.67%	64.06%	66.46%	66.46%
IG	3467	0.37%	0.59%	0.62%	0.70%	0.74%	0.74%	1.36%	2.35%	2.56%	2.87%	3.37%	3.77%	3.95%	4.13%	4.20%	4.39%	4.46%	4.53%	4.89%	4.89%
SG	1863	7.79%	13.25%	16.03%	17.67%	19.82%	20.80%	24.00%	30.93%	31.78%	32.71%	33.31%	33.51%	33.73%	35.11%	36.30%	37.55%	37.82%	39.21%	41.22%	41.22%
All	5330	2.92%	4.85%	5.72%	6.24%	6.83%	7.07%	8.28%	10.61%	10.96%	11.40%	11.91%	12.26%	12.44%	12.87%	13.16%	13.56%	13.67%	14.00%	14.67%	14.67%

1/1/03

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	125	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	704	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.47%	0.70%	0.70%	0.96%	0.96%	0.96%	0.96%	1.27%	1.27%	1.27%	1.27%	1.27%	1.27%
A	1299	0.00%	0.00%	0.00%	0.00%	0.00%	1.20%	2.00%	2.37%	3.01%	3.54%	4.23%	4.66%	4.81%	4.81%	5.13%	5.13%	5.13%	5.48%	5.48%	5.48%
Baa	1218	0.00%	0.00%	0.20%	0.30%	0.41%	0.53%	1.68%	1.68%	1.68%	2.27%	2.43%	2.43%	2.77%	2.94%	2.94%	3.13%	3.32%	3.93%	3.93%	4.36%
Ba	584	0.89%	1.29%	1.95%	2.44%	2.74%	5.58%	10.73%	11.48%	11.48%	12.78%	12.78%	12.78%	13.74%	14.71%	15.23%	15.23%	15.78%	17.50%	17.50%	18.74%
B	877	2.56%	4.21%	6.40%	9.15%	10.52%	14.98%	24.43%	26.03%	27.11%	28.68%	29.48%	29.92%	31.85%	33.85%	35.95%	37.66%	40.64%	42.51%	42.51%	45.80%
Caa-C	366	20.45%	31.59%	34.24%	37.86%	38.68%	39.77%	48.70%	48.70%	53.42%	53.42%	53.42%	53.42%	53.42%	53.42%	53.42%	53.42%	56.33%	59.24%	59.24%	59.24%
IG	3346	0.00%	0.00%	0.07%	0.11%	0.15%	0.66%	1.39%	1.63%	1.93%	2.35%	2.74%	2.91%	3.09%	3.15%	3.34%	3.40%	3.47%	3.82%	3.82%	3.97%
SG	1827	5.32%	8.05%	9.74%	11.71%	12.58%	15.90%	23.38%	24.43%	25.43%	26.69%	27.05%	27.25%	28.50%	29.80%	30.94%	31.67%	33.41%	35.23%	35.23%	37.18%
All	5173	1.83%	2.72%	3.28%	3.86%	4.12%	5.32%	7.70%	8.14%	8.59%	9.20%	9.58%	9.75%	10.16%	10.49%	10.88%	11.08%	11.50%	12.14%	12.14%	12.66%

1/1/04

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Aaa	144	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	668	0.00%	0.00%	0.00%	0.00%	0.20%	0.20%	0.66%	0.89%	0.89%	1.16%	1.16%	1.16%	1.16%	1.47%	1.47%	1.47%	1.47%	1.47%	1.47%	1.47%
A	1304	0.00%	0.00%	0.00%	0.00%	1.12%	1.55%	1.89%	2.49%	2.86%	3.65%	4.05%	4.19%	4.19%	4.50%	4.50%	4.50%	4.66%	4.66%	4.66%	4.84%
Baa	1284	0.00%	0.18%	0.18%	0.18%	0.28%	1.89%	1.89%	1.89%	2.55%	2.83%	2.83%	3.13%	3.28%	3.28%	3.45%	3.62%	4.34%	4.34%	4.91%	5.11%
Ba	562	0.38%	0.58%	1.30%	1.85%	4.60%	9.99%	10.35%	10.35%	11.21%	11.21%	11.21%	12.68%	13.68%	14.22%	14.22%	15.35%	17.13%	17.13%	18.37%	18.37%
B	957	0.68%	1.97%	3.95%	4.82%	8.82%	19.24%	20.51%	21.36%	22.95%	23.60%	23.96%	25.18%	26.48%	29.25%	30.72%	32.74%	34.33%	34.33%	38.35%	38.35%
Caa-C	328	10.90%	17.15%	21.87%	24.80%	28.34%	42.17%	43.24%	50.07%	52.93%	52.93%	52.93%	52.93%	55.07%	55.07%	55.07%	58.06%	61.06%	61.06%	61.06%	61.06%
IG	3400	0.00%	0.07%	0.07%	0.07%	0.58%	1.34%	1.56%	1.84%	2.23%	2.70%	2.86%	3.02%	3.08%	3.26%	3.32%	3.38%	3.70%	3.70%	3.91%	4.06%
SG	1847	2.30%	4.00%	5.93%	6.95%	10.41%	19.43%	20.34%	21.52%	22.95%	23.28%	23.46%	24.63%	25.85%	27.38%	28.07%	29.73%	31.46%	31.46%	33.86%	33.86%
All	5247	0.78%	1.37%	1.95%	2.24%	3.52%	6.41%	6.80%	7.30%	7.93%	8.36%	8.52%	8.90%	9.21%	9.67%	9.86%	10.25%	10.86%	10.86%	11.51%	11.62%

1/1/05

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Aaa	135	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	683	0.00%	0.00%	0.00%	0.18%	0.18%	0.82%	1.03%	1.03%	1.27%	1.27%	1.27%	1.27%	1.55%	1.55%	1.55%	1.55%	1.55%	1.55%	1.55%
A	1355	0.00%	0.00%	0.00%	0.75%	1.24%	1.56%	2.11%	2.47%	3.09%	3.47%	3.60%	3.60%	3.89%	3.89%	3.89%	4.05%	4.05%	4.05%	4.22%
Baa	1305	0.16%	0.16%	0.16%	0.66%	2.06%	2.06%	2.06%	2.67%	3.06%	3.06%	3.34%	3.48%	3.48%	3.64%	3.96%	4.47%	4.47%	5.20%	5.39%
Ba	555	0.00%	0.65%	0.91%	3.68%	8.98%	9.61%	9.61%	10.74%	10.74%	10.74%	12.45%	12.89%	13.82%	13.82%	14.82%	16.91%	16.91%	18.60%	18.60%
B	1040	0.71%	2.44%	3.62%	8.35%	19.74%	20.97%	21.92%	23.53%	24.65%	24.97%	26.39%	28.28%	30.76%	32.11%	33.96%	36.35%	36.35%	41.64%	41.64%
Caa-C	366	7.27%	11.73%	14.80%	22.23%	38.81%	41.85%	50.53%	52.83%	54.11%	55.69%	57.46%	59.40%	59.40%	59.40%	62.10%	64.81%	64.81%	64.81%	64.81%
IG	3478	0.06%	0.06%	0.06%	0.57%	1.28%	1.53%	1.79%	2.15%	2.59%	2.74%	2.89%	2.94%	3.11%	3.17%	3.29%	3.53%	3.53%	3.80%	3.93%
SG	1961	1.72%	3.61%	4.82%	9.33%	19.51%	20.78%	22.29%	23.78%	24.48%	24.79%	26.30%	27.54%	29.07%	29.68%	31.16%	33.35%	33.35%	36.45%	36.45%
All	5439	0.64%	1.26%	1.63%	3.29%	6.70%	7.23%	7.81%	8.46%	8.95%	9.14%	9.61%	9.93%	10.40%	10.58%	10.99%	11.65%	11.65%	12.51%	12.61%

1/1/06

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Aaa	131	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	702	0.00%	0.00%	0.17%	0.17%	0.76%	0.96%	0.96%	1.18%	1.41%	1.41%	1.41%	1.67%	1.67%	1.67%	1.67%	1.67%	1.67%	1.67%
A	1413	0.00%	0.00%	0.68%	1.04%	1.33%	1.83%	2.15%	2.94%	3.17%	3.29%	3.29%	3.56%	3.56%	3.56%	3.70%	3.70%	4.01%	4.17%
Baa	1288	0.00%	0.00%	0.56%	1.57%	1.57%	1.57%	2.05%	2.42%	2.42%	2.55%	2.70%	2.70%	2.85%	3.32%	3.81%	3.81%	4.86%	5.05%
Ba	562	0.20%	0.20%	3.12%	9.49%	9.49%	9.49%	10.45%	10.79%	11.15%	13.36%	13.74%	15.38%	15.79%	16.69%	19.02%	19.02%	22.17%	22.17%
B	1101	1.09%	2.05%	8.30%	18.73%	21.06%	22.68%	24.96%	25.95%	26.50%	28.96%	31.68%	34.64%	36.24%	38.71%	41.71%	42.18%	46.60%	46.60%
Caa-C	400	5.50%	9.65%	18.75%	37.60%	42.58%	48.53%	50.09%	52.04%	53.24%	54.57%	59.27%	59.27%	59.27%	61.67%	64.22%	64.22%	64.22%	64.22%
IG	3534	0.00%	0.00%	0.51%	1.02%	1.25%	1.49%	1.79%	2.28%	2.42%	2.51%	2.57%	2.72%	2.78%	2.95%	3.18%	3.18%	3.67%	3.80%
SG	2063	1.67%	2.92%	8.64%	19.24%	21.24%	22.90%	24.59%	25.44%	25.97%	28.13%	29.98%	31.99%	32.87%	34.55%	37.09%	37.30%	40.60%	40.60%
All	5597	0.59%	1.00%	3.15%	6.83%	7.60%	8.23%	8.91%	9.49%	9.73%	10.33%	10.82%	11.41%	11.65%	12.16%	12.90%	12.95%	14.03%	14.13%

1/1/07

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Aaa	141	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	751	0.00%	0.15%	0.15%	0.67%	0.85%	0.85%	1.05%	1.25%	1.25%	1.25%	1.48%	1.48%	1.48%	1.48%	1.48%	1.48%	1.48%
A	1405	0.00%	0.71%	1.05%	1.32%	1.79%	2.19%	3.14%	3.35%	3.47%	3.47%	3.72%	3.72%	3.72%	3.86%	3.86%	4.58%	4.73%
Baa	1284	0.00%	0.70%	1.74%	1.74%	1.74%	2.19%	2.54%	2.54%	2.67%	2.80%	2.94%	3.08%	3.52%	4.14%	4.14%	4.81%	4.98%
Ba	626	0.00%	1.79%	7.25%	7.96%	7.96%	9.30%	10.13%	10.43%	12.65%	13.96%	15.39%	16.14%	18.15%	20.21%	20.21%	23.01%	23.01%
B	1130	0.10%	6.52%	17.92%	20.87%	23.05%	25.76%	27.09%	28.09%	30.85%	33.90%	35.93%	37.40%	38.94%	41.81%	42.27%	47.94%	47.94%
Caa-C	442	4.78%	15.88%	34.57%	39.19%	43.12%	45.68%	48.83%	49.76%	50.83%	57.92%	60.92%	60.92%	62.88%	65.20%	65.20%	65.20%	65.20%
IG	3581	0.00%	0.56%	1.07%	1.28%	1.50%	1.82%	2.36%	2.49%	2.58%	2.63%	2.82%	2.87%	3.03%	3.30%	3.30%	3.82%	3.94%
SG	2198	0.99%	6.93%	17.95%	20.51%	22.20%	24.36%	25.70%	26.39%	28.65%	31.21%	32.98%	33.93%	35.62%	37.93%	38.12%	41.77%	41.77%
All	5779	0.36%	2.81%	6.99%	7.99%	8.67%	9.52%	10.28%	10.56%	11.21%	11.88%	12.46%	12.72%	13.22%	13.95%	13.99%	15.16%	15.25%

1/1/08

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Aaa	176	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	835	0.50%	0.50%	0.93%	1.39%	1.55%	1.88%	2.22%	2.40%	2.40%	2.79%	2.79%	2.79%	2.79%	2.79%	2.79%	2.79%
A	1321	0.40%	0.91%	1.18%	1.46%	1.97%	3.06%	3.17%	3.17%	3.17%	3.30%	3.30%	3.44%	3.58%	3.58%	4.33%	4.64%
Baa	1256	1.01%	1.82%	1.82%	1.82%	2.03%	2.14%	2.14%	2.14%	2.38%	2.51%	2.78%	3.05%	3.63%	3.63%	4.25%	4.25%
Ba	633	2.33%	7.35%	7.55%	7.55%	8.69%	9.40%	9.91%	12.08%	13.50%	15.05%	15.69%	17.07%	18.87%	18.87%	22.78%	22.78%
B	1009	4.04%	14.61%	17.65%	19.75%	22.15%	23.23%	24.23%	27.51%	29.99%	32.42%	33.96%	36.45%	40.43%	40.89%	46.43%	46.43%
Caa-C	663	10.70%	28.92%	34.23%	37.59%	40.96%	43.91%	46.57%	49.33%	55.03%	56.40%	57.22%	59.05%	62.20%	63.46%	63.46%	66.50%
IG	3588	0.62%	1.09%	1.29%	1.50%	1.80%	2.30%	2.42%	2.47%	2.55%	2.74%	2.83%	2.98%	3.24%	3.24%	3.73%	3.84%
SG	2305	5.47%	16.66%	19.53%	21.28%	23.47%	24.82%	25.95%	28.58%	31.09%	32.85%	33.80%	35.57%	38.24%	38.58%	42.28%	42.67%
All	5893	2.50%	7.12%	8.32%	9.05%	9.96%	10.73%	11.14%	11.91%	12.65%	13.24%	13.55%	14.08%	14.90%	14.98%	16.15%	16.32%

1/1/09

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Aaa	150	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	750	0.00%	0.15%	0.46%	0.63%	0.97%	1.15%	1.52%	1.52%	1.93%	1.93%	1.93%	1.93%	1.93%	1.93%	1.93%
A	1310	0.24%	0.49%	0.76%	1.14%	2.16%	2.37%	2.37%	2.37%	2.50%	2.50%	2.63%	2.63%	2.63%	3.35%	3.65%
Baa	1243	0.92%	0.92%	1.21%	1.31%	1.41%	1.52%	1.52%	1.75%	1.87%	2.12%	2.37%	3.32%	3.47%	4.20%	4.20%
Ba	601	1.77%	2.16%	2.16%	3.52%	4.23%	5.00%	6.64%	7.79%	9.34%	10.29%	11.65%	13.08%	13.08%	16.51%	16.97%
B	856	6.80%	8.00%	9.55%	11.64%	12.53%	13.82%	18.39%	20.94%	23.49%	24.73%	27.44%	31.75%	32.25%	40.66%	40.66%
Caa-C	763	26.39%	33.72%	37.17%	41.08%	43.99%	46.42%	48.51%	53.92%	55.48%	56.08%	57.37%	60.19%	61.07%	61.07%	63.07%
IG	3453	0.42%	0.55%	0.82%	1.04%	1.53%	1.69%	1.77%	1.86%	2.04%	2.13%	2.27%	2.62%	2.67%	3.21%	3.32%
SG	2220	12.11%	15.13%	16.78%	19.14%	20.50%	21.87%	24.59%	27.12%	28.93%	29.84%	31.52%	34.07%	34.40%	38.61%	39.17%
All	5673	4.98%	6.19%	6.94%	7.86%	8.62%	9.14%	9.97%	10.72%	11.33%	11.63%	12.15%	13.03%	13.14%	14.48%	14.69%

1/1/10

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Aaa	99	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	591	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.23%	0.23%	0.23%	0.23%	0.23%	0.23%	0.23%
A	1253	0.17%	0.51%	0.98%	1.76%	2.07%	2.29%	2.29%	2.53%	2.53%	2.65%	2.65%	2.65%	2.79%	3.07%
Baa	1352	0.07%	0.31%	0.40%	0.84%	1.03%	1.03%	1.33%	1.44%	1.87%	2.10%	2.82%	2.82%	3.87%	3.87%
Ba	595	0.00%	0.00%	0.84%	1.71%	2.19%	3.95%	5.01%	6.45%	7.33%	8.59%	10.24%	10.60%	14.91%	15.33%
B	818	0.39%	1.85%	3.78%	4.61%	6.79%	11.36%	14.08%	16.19%	17.38%	19.06%	22.64%	23.11%	31.28%	31.83%
Caa-C	728	8.63%	13.24%	18.99%	23.19%	26.35%	29.37%	34.65%	37.11%	38.55%	41.18%	45.69%	46.34%	47.01%	47.75%
IG	3295	0.09%	0.32%	0.53%	1.01%	1.20%	1.28%	1.40%	1.58%	1.76%	1.90%	2.19%	2.19%	2.66%	2.77%
SG	2141	3.07%	5.11%	7.92%	9.76%	11.61%	14.66%	17.40%	19.30%	20.39%	22.10%	25.02%	25.47%	30.03%	30.55%
All	5436	1.24%	2.13%	3.24%	4.16%	4.87%	5.84%	6.72%	7.37%	7.79%	8.34%	9.30%	9.41%	10.84%	11.04%

1/1/11

Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12	13
Aaa	87	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	552	0.19%	0.19%	0.19%	0.19%	0.19%	0.19%	0.43%	0.43%	0.43%	0.43%	0.43%	0.43%	0.43%
A	1247	0.00%	0.09%	0.74%	0.84%	0.94%	0.94%	1.17%	1.17%	1.29%	1.29%	1.29%	1.42%	1.82%
Baa	1442	0.36%	0.43%	0.91%	1.25%	1.25%	1.61%	1.71%	2.10%	2.31%	3.08%	3.08%	4.16%	4.16%
Ba	643	0.16%	1.39%	2.14%	2.95%	4.67%	6.04%	7.28%	8.30%	9.43%	10.91%	11.23%	15.46%	15.83%
B	962	0.34%	1.89%	3.21%	5.29%	9.75%	13.45%	15.41%	16.53%	18.08%	22.39%	23.45%	30.90%	32.09%
Caa-C	739	6.22%	13.53%	18.35%	22.24%	26.37%	31.84%	34.36%	36.40%	39.23%	44.92%	45.67%	47.26%	49.11%
IG	3328	0.19%	0.25%	0.70%	0.88%	0.92%	1.08%	1.24%	1.41%	1.54%	1.87%	1.87%	2.37%	2.52%
SG	2344	2.08%	5.22%	7.30%	9.42%	12.78%	15.98%	17.76%	19.00%	20.60%	23.92%	24.60%	29.48%	30.40%
All	5672	0.94%	2.19%	3.24%	4.09%	5.22%	6.33%	6.98%	7.46%	8.00%	9.13%	9.31%	10.90%	11.24%

1/1/12													
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11	12
Aaa	85	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	444	0.00%	0.00%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%	0.25%
A	1213	0.00%	0.09%	0.09%	0.09%	0.09%	0.20%	0.20%	0.31%	0.31%	0.31%	0.44%	0.83%
Baa	1483	0.07%	0.43%	0.51%	0.51%	0.76%	0.93%	1.29%	1.48%	2.38%	2.38%	3.37%	3.37%
Ba	718	0.14%	1.67%	2.17%	3.45%	5.18%	6.66%	7.33%	8.30%	9.58%	9.85%	14.43%	14.75%
B	998	0.54%	1.66%	3.95%	7.94%	11.51%	14.16%	15.08%	16.64%	21.15%	22.06%	28.72%	30.09%
Caa-C	830	8.01%	13.29%	17.65%	22.95%	28.55%	32.15%	35.73%	37.97%	45.00%	45.67%	47.87%	50.43%
IG	3225	0.03%	0.23%	0.30%	0.30%	0.42%	0.54%	0.70%	0.83%	1.23%	1.23%	1.71%	1.86%
SG	2546	2.81%	5.31%	7.63%	11.01%	14.42%	16.83%	18.20%	19.64%	23.19%	23.78%	28.61%	29.67%
All	5771	1.24%	2.40%	3.35%	4.60%	5.88%	6.76%	7.31%	7.84%	9.16%	9.33%	10.99%	11.39%

1/1/13												
Rating	n(0)	1	2	3	4	5	6	7	8	9	10	11
Aaa	62	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	303	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
A	1181	0.09%	0.18%	0.18%	0.18%	0.18%	0.18%	0.28%	0.39%	0.39%	0.51%	0.87%
Baa	1673	0.12%	0.18%	0.18%	0.32%	0.40%	0.70%	0.86%	1.54%	1.54%	2.58%	2.58%
Ba	694	0.60%	1.09%	1.83%	3.36%	5.47%	5.70%	6.70%	8.02%	8.30%	12.32%	12.65%
B	1059	1.01%	2.63%	7.05%	11.36%	13.32%	14.60%	15.63%	19.73%	20.75%	26.40%	28.12%
Caa-C	935	6.29%	10.37%	17.27%	23.07%	27.66%	31.73%	34.71%	43.10%	43.63%	47.74%	49.08%
IG	3219	0.09%	0.16%	0.16%	0.23%	0.27%	0.42%	0.55%	0.93%	0.93%	1.47%	1.62%
SG	2688	2.69%	4.81%	8.91%	12.79%	15.46%	17.03%	18.47%	22.40%	23.02%	27.61%	28.69%
All	5907	1.25%	2.18%	3.81%	5.32%	6.30%	6.93%	7.47%	8.94%	9.12%	10.80%	11.20%

1/1/14											
Rating	n(0)	1	2	3	4	5	6	7	8	9	10
Aaa	58	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	297	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
A	1172	0.09%	0.09%	0.09%	0.09%	0.09%	0.19%	0.30%	0.30%	0.30%	0.65%
Baa	1736	0.06%	0.06%	0.19%	0.19%	0.47%	0.61%	1.24%	1.24%	2.46%	2.46%
Ba	723	0.15%	0.48%	1.67%	3.19%	3.39%	4.75%	6.66%	6.91%	10.87%	11.47%
B	1097	0.50%	3.89%	7.73%	10.44%	12.23%	13.33%	17.08%	18.71%	24.70%	26.53%
Caa-C	1142	4.76%	11.88%	18.51%	24.62%	29.02%	32.34%	40.57%	41.25%	45.23%	46.18%
IG	3263	0.06%	0.06%	0.13%	0.13%	0.28%	0.39%	0.75%	0.75%	1.36%	1.49%
SG	2962	2.03%	5.98%	10.06%	13.50%	15.55%	17.33%	21.58%	22.46%	27.10%	28.24%
All	6225	0.97%	2.70%	4.45%	5.80%	6.64%	7.33%	8.98%	9.26%	11.08%	11.51%

1/1/15										
Rating	n(0)	1	2	3	4	5	6	7	8	9
Aaa	57	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	307	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
A	1248	0.00%	0.00%	0.00%	0.00%	0.09%	0.19%	0.19%	0.19%	0.51%
Baa	1744	0.00%	0.00%	0.00%	0.27%	0.41%	1.16%	1.16%	2.17%	2.26%
Ba	719	0.30%	1.56%	2.77%	2.95%	3.37%	4.70%	4.70%	9.23%	9.80%
B	1076	2.42%	6.13%	7.89%	10.12%	12.35%	16.94%	18.86%	25.33%	26.78%
Caa-C	1367	6.62%	13.03%	19.22%	23.47%	28.32%	37.83%	38.83%	42.33%	43.36%
IG	3356	0.00%	0.00%	0.00%	0.14%	0.24%	0.66%	0.66%	1.15%	1.32%
SG	3162	3.71%	7.94%	11.25%	13.63%	16.19%	21.28%	22.27%	27.09%	28.10%
All	6518	1.75%	3.67%	5.07%	6.10%	7.11%	9.16%	9.49%	11.35%	11.77%

1/1/16									
Rating	n(0)	1	2	3	4	5	6	7	8
Aaa	53	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	311	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
A	1305	0.00%	0.00%	0.00%	0.09%	0.09%	0.09%	0.09%	0.29%
Baa	1730	0.00%	0.00%	0.19%	0.26%	0.99%	1.06%	1.47%	1.64%
Ba	734	0.14%	0.77%	0.93%	1.50%	3.31%	3.31%	8.48%	9.28%
B	1019	1.68%	2.99%	4.83%	6.80%	11.92%	13.73%	20.77%	21.93%
Caa-C	1434	9.10%	15.83%	19.99%	24.72%	34.19%	35.58%	39.28%	41.20%
IG	3399	0.00%	0.00%	0.10%	0.17%	0.53%	0.56%	0.76%	0.92%
SG	3187	4.63%	8.06%	10.37%	12.95%	18.55%	19.63%	24.84%	26.09%
All	6586	2.20%	3.73%	4.76%	5.80%	8.09%	8.49%	10.37%	10.87%

1/1/17								
Rating	n(0)	1	2	3	4	5	6	7
Aaa	54	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	305	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
A	1276	0.00%	0.00%	0.09%	0.09%	0.09%	0.09%	0.28%
Baa	1712	0.00%	0.00%	0.06%	0.34%	0.41%	0.73%	0.89%
Ba	795	0.53%	0.53%	0.53%	1.58%	1.78%	6.91%	7.61%
B	979	0.45%	1.36%	3.31%	10.54%	12.17%	19.62%	21.08%
Caa-C	1442	7.55%	13.16%	18.98%	29.13%	30.71%	33.97%	36.48%
IG	3347	0.00%	0.00%	0.07%	0.21%	0.24%	0.39%	0.55%
SG	3216	3.59%	6.23%	9.15%	15.62%	16.78%	21.96%	23.48%
All	6563	1.71%	2.90%	4.17%	6.86%	7.32%	9.29%	9.93%

1/1/18							
Rating	n(0)	1	2	3	4	5	6
Aaa	56	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	258	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
A	1305	0.00%	0.16%	0.16%	0.16%	0.16%	0.34%
Baa	1761	0.00%	0.00%	0.32%	0.39%	0.82%	0.97%
Ba	830	0.00%	0.00%	1.08%	1.26%	7.13%	7.96%
B	976	0.68%	2.06%	8.96%	11.11%	17.41%	20.36%
Caa-C	1424	5.17%	11.43%	21.67%	23.43%	26.21%	30.34%
IG	3380	0.00%	0.06%	0.23%	0.26%	0.48%	0.62%
SG	3230	2.45%	5.48%	12.09%	13.51%	18.34%	20.99%
All	6610	1.17%	2.57%	5.52%	6.12%	8.11%	9.17%

1/1/19						
Rating	n(0)	1	2	3	4	5
Aaa	54	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	294	0.00%	0.00%	0.00%	0.00%	0.00%
A	1304	0.00%	0.00%	0.00%	0.00%	0.18%
Baa	1818	0.11%	0.23%	0.29%	1.09%	1.23%
Ba	799	0.00%	0.72%	1.04%	5.51%	6.69%
B	1003	1.20%	7.37%	9.33%	15.79%	18.37%
Caa-C	1414	6.52%	17.58%	19.41%	22.53%	28.67%
IG	3470	0.06%	0.12%	0.15%	0.55%	0.69%
SG	3216	3.20%	10.03%	11.50%	16.07%	19.55%
All	6686	1.53%	4.65%	5.30%	7.39%	8.82%

1/1/20					
Rating	n(0)	1	2	3	4
Aaa	54	0.00%	0.00%	0.00%	0.00%
Aa	333	0.00%	0.00%	0.00%	0.00%
A	1307	0.00%	0.00%	0.00%	0.17%
Baa	1839	0.11%	0.11%	1.05%	1.18%
Ba	771	0.13%	0.43%	5.71%	6.61%
B	962	3.98%	5.43%	11.07%	13.75%
Caa-C	1396	12.83%	15.21%	19.13%	25.81%
IG	3533	0.06%	0.06%	0.53%	0.66%
SG	3129	6.92%	8.47%	13.29%	17.00%
All	6662	3.23%	3.91%	6.20%	7.76%

1/1/21				
Rating	n(0)	1	2	3
Aaa	50	0.00%	0.00%	0.00%
Aa	330	0.00%	0.00%	0.00%
A	1289	0.00%	0.00%	0.17%
Baa	1863	0.00%	0.87%	1.12%
Ba	780	0.00%	4.79%	5.28%
B	891	1.33%	8.25%	10.31%
Caa-C	1460	2.96%	6.57%	15.19%
IG	3532	0.00%	0.45%	0.64%
SG	3131	1.75%	6.64%	11.11%
All	6663	0.80%	3.18%	5.15%

1/1/22			
Rating	n(0)	1	2
Aaa	50	0.00%	0.00%
Aa	317	0.00%	0.00%
A	1321	0.00%	0.16%
Baa	1927	0.91%	1.14%
Ba	772	4.17%	4.60%
B	955	5.14%	6.84%
Caa-C	1589	3.84%	11.75%
IG	3615	0.48%	0.66%
SG	3316	4.29%	8.57%
All	6931	2.28%	4.33%

1/1/23		
Rating	n(0)	1
Aaa	50	0.00%
Aa	319	0.00%
A	1329	0.15%
Baa	1860	0.11%
Ba	699	0.73%
B	866	0.72%
Caa-C	1525	9.16%
IG	3558	0.11%
SG	3090	4.83%
All	6648	2.28%

Source: Moody's Investors Service

Endnotes

1 Russian defaults accounted for 65 defaults in the 2022 tally and zero in that of 2023.

2 Dollar-weighted bond default rates only include defaulted bonds from US and European issuers.

3 Average recovery rates of senior unsecured bonds and other debts can be based on different defaulters because, for example, some defaulters may have senior unsecured bonds and no other debts.

4 The analysis on ultimate recovery is provided by Julia Chursin, a corporate finance analyst at Moody's. The data is from Moody's Ultimate Recovery Database, which includes robust detailed recovery information for more than 6400 loans and bonds from 1,337 US corporate nonfinancial defaulters since 1987.

5 Glossary of Moody's Ratings Performance Metrics for details on data methodology.

6 These forecasts were made in January 2024. We plan to update them towards the end of February.

7 For details about the EDF model, see Public Firm Expected Default Frequency (EDFTM) Credit Measures: Methodology, Performance, and Model Extensions, June 2012.

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